

72nd BPSC PRELIMS

प्रहार

Test Series

Test- 02

World Geogarphy

Solution

English



Khan Global Studies

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BPSC PRELIMS TEST - 2 (SOLUTION)

Q 1. Ans. A

Explanation:

- Ruhr region:
 - ✓ The Ruhr region in Germany is primarily known for iron and steel production.
 - ✓ The Ruhr region is located in North Rhine–Westphalia, Germany.
 - ✓ The region developed mainly due to abundant coal deposits in the Ruhr basin.
 - ✓ Iron ore was imported through the Rhine River, which provided an efficient transport route.
- The Chhota Nagpur Plateau is known as the “Ruhr of India”.
 - ✓ This region contains:
 - Rich coal deposits (Gondwana coalfields)
 - Large iron ore reserves (hematite)
 - This made it India’s major iron–steel production region.
 - ✓ Major industrial establishments:
 - Bokaro Steel Plant: established with Soviet collaboration
 - Heavy Engineering Corporation (HEC), Ranchi
- Other Examples:
 - ✓ Petrochemical Industry
 - Located near oil fields or major ports.
 - Examples:
 - ◇ Jamnagar (India) — major refinery hub
 - ✓ Fertiliser Industry:
 - Depends mainly on natural gas, not coal.
 - In India, fertiliser plants are linked to the HVJ (Hazira–Vijaipur–Jagdishpur) gas pipeline.
- ✓ Burning produces ash, which temporarily increases soil fertility.
- ✓ After 3–5 years, fertility declines and farmers shift to a new plot.
- ✓ This method is also called Slash-and-Burn Agriculture.
- Global names of shifting cultivation:
 - ✓ Milpa — Mexico & Central America
 - ✓ Taungya — Myanmar
 - ✓ Chena — Sri Lanka
 - ✓ Roca (Roka) — Brazil
 - ✓ Ladang — Indonesia & Malaysia
 - ✓ Masole — Central Africa (Congo Basin)
 - ✓ Conuco — Venezuela
 - ✓ Ray — Vietnam
- Indian names of shifting cultivation:
 - ✓ Jhumming — Assam, Meghalaya, Mizoram
 - ✓ Pamlou — Manipur
 - ✓ Dipa — Bastar (Chhattisgarh), Andaman & Nicobar
 - ✓ Bewar / Dahiya — Madhya Pradesh
 - ✓ Podu / Penda — Andhra Pradesh
 - ✓ Pama Dabi / Koman / Bringa — Odisha
 - ✓ Kumari — Western Ghats
 - ✓ Kuruwa — Jharkhand
 - ✓ In the Rajmahal Hills region (Santhal Pargana area), shifting cultivation is locally called Kuruwa.
 - ✓ It has been practised by tribal groups such as:
 - Sauria Paharia
 - Mal Paharia

Q 2. Ans. C

Explanation:

- Shifting Cultivation:
 - ✓ Shifting cultivation is a primitive subsistence farming method.
 - ✓ Forest land is cleared by cutting and burning vegetation.
- Out of the total 100 per cent insolation received at the top of the atmosphere, about 35 per cent is reflected back to space before reaching the Earth’s surface which is known as the albedo of the Earth.
- This includes 27 per cent reflected by clouds and 2 per cent reflected by snow and ice covered areas, with the remaining part reflected by other atmospheric components.
- The remaining 65 per cent is absorbed by the atmosphere and the Earth’s surface.

Q 3. Ans. B

Explanation:

- Out of the total 100 per cent insolation received at the top of the atmosphere, about 35 per cent is reflected back to space before reaching the Earth’s surface which is known as the albedo of the Earth.
- This includes 27 per cent reflected by clouds and 2 per cent reflected by snow and ice covered areas, with the remaining part reflected by other atmospheric components.
- The remaining 65 per cent is absorbed by the atmosphere and the Earth’s surface.

- ✓ 14 units are absorbed by the atmosphere.
- ✓ 51 units are absorbed by the Earth's surface.
- The Earth radiates back 51 units as terrestrial radiation.
 - ✓ Out of these, 17 units escape directly to space.
- The remaining 34 units are absorbed by the atmosphere.
 - ✓ Out of 34 units absorbed by atmosphere:
 - ✓ 6 units absorbed directly by atmosphere.
 - ✓ 9 units transferred through convection and turbulence.
 - ✓ 19 units transferred through latent heat of condensation.
- Total radiation absorbed by atmosphere becomes:
 - ✓ 48 units = 14 units (insolation) + 34 units (terrestrial radiation).
 - ✓ The atmosphere radiates 48 units back to space.
- Thus, total radiation returned to space equals:
 - ✓ 65 units = 17 units (Earth) + 48 units (Atmosphere).
- This balance between incoming and outgoing radiation is called the Heat Budget or Heat Balance of the Earth.
- Surface albedo variations: Albedo is highly dependent on surface color and texture. Fresh snow has the highest albedo (up to 90%), while dark ocean water has a very low albedo (less than 10%), meaning it absorbs almost all incoming heat.
- ✓ However, due to dominant vertical air movement (convection), horizontal surface winds are generally weak or calm.
- Horse Latitudes (Sub-Tropical High-Pressure Belt)
 - ✓ Horse Latitudes occur around 30° N and 30° S.
 - ✓ They are characterised by descending air, which suppresses cloud formation.
 - ✓ These regions experience dry and arid conditions.
 - ✓ Most major deserts occur near this belt, including:
 - Sahara Desert, Kalahari Desert, Arabian Desert, and Thar Desert.
- Mid-Latitude Westerlies
 - ✓ Mid-latitude Westerlies blow between 35° and 60° latitude in both hemispheres.
 - ✓ They blow from the Sub-Tropical High-Pressure Belt towards the Sub-Polar Low-Pressure Belt.
 - ✓ In the Southern Hemisphere, these winds are very strong due to the absence of large landmasses.
 - ✓ They are known as:
 - Roaring Forties, Furious Fifties, and Screaming Sixties.
- Polar Easterlies
 - ✓ Polar Easterlies blow from the Polar High-Pressure areas (90°) towards the Sub-Polar Low-Pressure Belt (60°).
 - ✓ These winds are very cold and dry.
 - ✓ Where Polar Easterlies meet Westerlies, a Polar Front forms.
 - ✓ The Polar Front leads to the formation of extra-tropical cyclones.

Q 4. Ans. B

Explanation:

- The Doldrums refer to the Intertropical Convergence Zone (ITCZ) or the Equatorial Low Pressure Belt, located along the equator roughly between 5°N and 5°S latitude.
- This region receives intense, nearly perpendicular solar radiation, causing surface air to heat up, expand, and rise.
 - ✓ These ascending convective currents create persistent low pressure at the surface and lead to frequent thunderstorms and heavy rainfall as the rising air cools and condenses at higher altitudes.
- It is also a zone of wind convergence where the Northeast and Southeast Trade Winds meet.

Q 5. Ans. D

Explanation:

- Photosynthesis is the primary mechanism of carbon fixation. Plants absorb CO₂ and convert it into glucose (organic matter) using sunlight, releasing O₂ as a by-product. This is the main entry point of carbon into the biosphere.
- Oceans are among the largest carbon reservoirs. Carbon is stored mainly as:
 - ✓ Dissolved CO₂
 - ✓ Bicarbonate (HCO₃⁻)
 - ✓ Carbonate (CO₃²⁻)

- ✓ This buffering role makes oceans critical regulators of atmospheric CO₂.
 - Decomposition by microorganisms (bacteria and fungi) breaks down dead organic matter and releases CO₂ and methane back into the atmosphere, closing the biological carbon loop.
 - Respiration does not remove CO₂ permanently. Instead:
 - ✓ Organisms release CO₂ during respiration.
 - ✓ It is a source, not a sink, of atmospheric carbon.
 - Major Carbon Sources:
 - ✓ Respiration
 - ✓ Decomposition
 - ✓ Volcanic eruptions
 - ✓ Fossil fuel combustion
 - Major Carbon Sinks:
 - ✓ Oceans
 - ✓ Forests
 - ✓ Soil organic matter
 - ✓ Carbonate rocks
 - Types of Carbon Cycle:
 - ✓ Biological Carbon Cycle — Atmosphere ↔ Living organisms
 - ✓ Geological Carbon Cycle — Rocks, fossil fuels, volcanism
 - ✓ Oceanic Carbon Cycle — Dissolution and deep-sea storage
 - Human Disruption Factors:
 - ✓ Burning fossil fuels
 - ✓ Deforestation
 - ✓ Land-use change.
- Q 6. Ans. D**
- Explanation:**
- **Statement 1 is incorrect:** Fog reduces horizontal visibility to less than 1,000 meters (1 km), whereas mist limits visibility between 1,000 meters and 5,000 meters (1–5 km), not merely up to 2 km.
 - ✓ Fog:
 - Reduces horizontal visibility to less than 1 km.
 - Forms when temperature reaches the dew point and condensation nuclei are present.
 - Common in plains and industrial regions with abundant dust, smoke, and ash particles.
 - ✓ Mist:
 - Less dense than fog.
 - Reduces visibility between 1 km and 5 km.
 - More common in mountain regions, where rising air cools rapidly.
 - **Statement 2 is incorrect:** Mist does not contain more total moisture than fog; rather, each condensation nucleus in mist carries a thicker moisture layer, making droplets fewer but larger, whereas fog contains numerous finer droplets, making it denser and more effective in reducing visibility.
 - ✓ Condensation nuclei are microscopic particles, such as dust, smoke, and salt, on which water vapour condenses to form fog or mist.
 - ✓ Fog generally contains more numerous fine droplets, producing dense atmospheric obscuration.
 - Fog is essentially a cloud at or near the ground, formed by condensation of water vapour on dust, smoke, or salt particles, commonly under cold conditions when warm moist air comes into contact with colder surfaces or currents.
 - Progression of Atmospheric Obscuration (Very Important):
 - ✓ Haze:
 - Caused by dry particles such as dust and smoke.
 - No water condensation occurs.
 - ✓ Mist:
 - Formed by water condensation.
 - Visibility between 1–5 km.
 - ✓ Fog:
 - Formed by dense water condensation.
 - Visibility less than 1 km.
 - ✓ Smog:
 - Combination of smoke + fog.
 - Contains toxic pollutants and is common in winter urban areas.
 - Agricultural Impact:
 - ✓ Winter fog affects Rabi crops such as wheat and mustard.
 - ✓ Short-duration fog can be beneficial as it reduces transpiration and conserves soil moisture.
 - ✓ Persistent dense fog reduces Photosynthetically Active Radiation (PAR) reaching crops.
 - ✓ This slows crop growth and promotes fungal diseases, especially blight.

Q 7. Ans. C

Explanation:

- **Statement 1 is correct:** Temperature inversion is the reversal of the normal lapse rate, where temperature increases with altitude instead of decreasing.
 - ✓ Under normal conditions in the troposphere, temperature decreases with height.
 - ✓ The Normal Lapse Rate averages 6.5°C per 1,000 meters.
 - ✓ During inversion, cold air remains near the surface, while warmer air overlies it.
 - ✓ The warm upper layer acts as a lid, preventing vertical air mixing.
- **Statement 2 is correct:** Temperature inversion commonly develops during clear nights with calm air, when the Earth loses heat rapidly by radiation, causing the surface to cool faster than the air above.
- The most common type is Radiation (Surface) Inversion.
- It develops under the following conditions:
 - ✓ Long winter nights allow maximum heat loss.
 - ✓ Clear skies permit rapid radiation of heat.
 - ✓ Calm winds prevent mixing of air layers.
 - This results in cold air near the ground and warmer air above, producing stable atmospheric conditions.
 - Such stability leads to trapping of pollutants and formation of fog, especially radiation fog.
- Additional points:
 - ✓ Valley Inversion:
 - Occurs in mountainous and hilly regions.
 - Cold air flows downslope at night due to katabatic winds.
 - Cold air accumulates in valley bottoms, while warmer air remains above.
 - Hence, orchards and plantations are often located on hill slopes rather than valley floors.
 - ✓ Advection Inversion:
 - Occurs when warm air moves over a colder surface, such as cold ocean water or snow-covered land.
 - The lower air layer cools rapidly, forming an inversion.

Q 8. Ans. C

Explanation:

- Bab-el-Mandeb Strait:
 - ✓ Bab-el-Mandeb means “Gate of Tears”.
 - ✓ It separates Yemen (Arabian Peninsula) from Djibouti and Eritrea (Horn of Africa).
 - ✓ It connects the Red Sea with the Gulf of Aden, leading to the Arabian Sea and the Indian Ocean.
 - ✓ Width:
 - At its narrowest point, the width is about 26–29 km.
 - ✓ Physical Structure:
 - The strait is divided into two channels by Perim (Mayyun) Island.
 - The eastern channel is narrow (about 3 km).
 - The western channel is wider (about 25 km) and used for major navigation.
 - ✓ Strategic Importance
 - Bab-el-Mandeb is the southern gateway to the Suez Canal route.
 - About 10% of the global seaborne petroleum trade passes through this strait.
 - It is a major route linking Europe with the Indo-Pacific region.
 - Disruptions in this strait force ships to reroute around the Cape of Good Hope in Africa.
- Bering Strait:
 - ✓ The Bering Strait connects the Bering Sea (Pacific Ocean) with the Chukchi Sea (Arctic Ocean).
 - ✓ It separates Russia (Asia) from Alaska, USA (North America).
 - ✓ Width:
 - At its narrowest, the width is about 82 km.
 - ✓ Depth:
 - Average depth is about 50 meters, making it relatively shallow.
 - ✓ The International Date Line (180° longitude) passes through the Bering Strait, between the Diomed Islands.
 - Big Diomed belongs to Russia.
 - Little Diomed belongs to the USA.
 - ✓ During the Ice Age, low sea levels exposed the Bering Land Bridge (Beringia).
 - ✓ This land bridge enabled human migration from Asia to North America.

Q 9. Ans. A

Explanation:

- Megacity:
 - ✓ A megacity is an urban agglomeration with a population exceeding 10 million (1 crore).
 - ✓ The term is defined by population size, not density or area.
 - ✓ First Megacity:
 - New York became the world's first megacity in 1950.
 - ✓ Largest Megacity:
 - Tokyo is the world's largest megacity.
 - ✓ Megacities in India:
 - Greater Mumbai
 - Delhi
 - Kolkata
 - Related terms:
 - ✓ Metropolitan City (Metropolis):
 - Population 1 million (10 lakh) to 10 million.
 - ✓ Megacity:
 - Population more than 10 million.
 - ✓ Megalopolis:
 - A continuous urban region formed by the merging of multiple metropolitan cities.
 - Example:
 - ◇ BosWash Corridor — Boston to Washington, D.C., USA.
 - ✓ Conurbation:
 - Merging of nearby towns into a continuous urban area.
 - Patna is classified as a Million-Plus City (Metropolis).
- ◇ Poor sanitation and healthcare
 - ◇ Uncertain and irregular food supply
 - ◇ High infant mortality
 - Birth rates remain high to compensate for frequent deaths.
 - Natural increase is very low, but not completely zero, so population growth is slow rather than absent.
 - Present Status:
 - ◇ No country today is fully in Stage 1.
 - ◇ Only some isolated tribal populations display Stage 1 characteristics.
 - **Statement 2 is correct:** At this stage, the population pyramid is broad at the base due to high birth rates, but narrows sharply toward the top because mortality rates are high across all age groups, particularly among children.
 - ✓ Population pyramid in stage 1:
 - Shape: Triangular (Expanding Type).
 - ◇ Very wide base: Indicates a high birth rate.
 - ◇ Very narrow top: Indicates low life expectancy and high mortality.
 - ✓ Population pyramid types:
 - Expanding (Triangular):
 - ◇ Wide base and narrow top.
 - ◇ Typical of Stage 1 and Stage 2.
 - ◇ Examples: Nigeria, Bangladesh.
 - Stable (Bell-shaped):
 - ◇ Nearly equal base and middle sections.
 - ◇ Typical of Stage 3.
 - ◇ Examples: USA, Australia.
 - Declining (Urn-shaped):
 - ◇ Narrow base and wider middle.
 - ◇ Typical of Stage 4 and Stage 5.
 - ◇ Examples: Japan, Germany.

Q 10. Ans. A

Explanation:

- **Statement 1 is incorrect:** Stage 1 is characterised by high birth rates and high death rates, resulting in negligible or very low population growth.
 - ✓ Demographic transition theory: Stage 1 (high stationary stage):
 - Represents a pre-industrial agrarian society.
 - Both Crude Birth Rate (CBR) and Crude Death Rate (CDR) are high.
 - High death rates occur due to:
 - ◇ Epidemics and infectious diseases

Q 11. Ans. B

Explanation:

- Entrepôt:
 - ✓ An entrepôt is a trading port where goods are imported, stored, and re-exported usually without paying import duty.
 - ✓ Goods are stored in bonded warehouses before re-export.
 - ✓ It acts as a break-of-bulk point, where large shipments are divided into smaller ones.
 - ✓ Key Global Examples:

- Singapore — a major entrepôt of Southeast Asia
- Rotterdam — gateway to Europe (Rhine hinterland)
- Copenhagen — Baltic region entrepôt
- Hong Kong — entrepôt for China trade
- ✓ Entrepôt ports focus on storage, sorting, grading, repackaging, and logistics, not manufacturing.
- Example from Bihar:
 - ✓ Bihar has no maritime entrepôt because it is landlocked.
 - ✓ Bihta (Patna):
 - ✓ Functions as an Inland Container Depot (ICD) or dry port.
 - ✓ Goods are consolidated, cleared, and transported to seaports like Haldia or Visakhapatnam.
 - ✓ Fatuha (Patna):
 - ✓ Proposed Multi-Modal Logistics Park (MMLP) to function as a regional distribution hub.
- Related terms:
 - ✓ Export Processing Zone (EPZ) / SEZ:
 - An area where goods are manufactured and exported.
 - Example: Kandla (India's first EPZ, 1965).
 - ✓ Commercial Centre (Mandi):
 - Centre for domestic trade only.
 - Example: Gulabbagh (Purnia) grain market.
 - ✓ Outport:
 - A subsidiary coastal port serving an inland port.
 - Examples:
 - Haldia: outport of Kolkata
 - Piraeus: outport of Athens
 - Bremerhaven: outport of Bremen
- ✓ WMO: Coordinates global weather and climate data.
- ✓ ITU: Allocates radio spectrum and satellite orbits.
- ✓ UNHCR: Established in 1950 for refugee protection.
- ✓ WEF: Organises annual Davos meeting.
- ✓ ICRC: Oversees the Geneva Conventions.
- New York (USA):
 - ✓ UN Headquarters: Houses 5 principal UN organs.
 - ✓ UNICEF: Publishes State of the World's Children report.
 - ✓ UNDP: Publishes Human Development Index (HDI).
- Washington, D.C. (USA):
 - ✓ World Bank Group: Includes IBRD, IDA, IFC, MIGA, and ICSID.
 - ✓ IMF: Issues Special Drawing Rights (SDRs).
- Paris (France):
 - ✓ UNESCO: Declares World Heritage Sites.
 - ✓ OECD: Economic policy and tax standards body.
 - ✓ FATF: Maintains Grey and Black lists.
- Lyon (France):
 - ✓ INTERPOL: Issues international Red Notices.
- Vienna (Austria):
 - ✓ IAEA: Nuclear monitoring; Atoms for Peace.
 - ✓ OPEC: Global oil cartel headquarters.
- Rome (Italy):
 - ✓ FAO: Celebrates World Food Day on October 16.
 - ✓ WFP: Largest global food aid agency.
- The Hague (Netherlands):
 - ✓ ICJ: The only principal UN organ NOT located in New York.
 - ✓ ICC: Tries genocide and war crimes cases.
- Nairobi (Kenya):
 - ✓ UNEP: Major UN body headquartered in Africa.

Q 12. Ans. B

Explanation:

- Geneva (Switzerland):
 - ✓ WTO: Replaced GATT in 1995; oversees TRIPS.
 - ✓ WHO: Declares Public Health Emergencies (PHEIC).
 - ✓ ILO: Oldest UN agency; founded in 1919.
 - ✓ WIPO: Administers Patent Cooperation Treaty (PCT).
- London (UK):
 - ✓ IMO: Regulates shipping safety and marine pollution.
- Montreal (Canada):
 - ✓ ICAO: Sets international aviation standards.
- Gurugram (India):
 - ✓ ISA: First treaty-based global organization headquartered in India.

- New Delhi (India):
 - ✓ CDRI: Launched in 2019 for disaster-resilient infrastructure.
- Kathmandu (Nepal):
 - ✓ SAARC: South Asian regional organization.
- Dhaka (Bangladesh):
 - ✓ BIMSTEC: India-focused Bay of Bengal grouping.
- Jakarta (Indonesia):
 - ✓ ASEAN: Southeast Asian regional bloc.
- Beijing (China):
 - ✓ SCO: Eurasian political and security organization.
 - ✓ AIIB: China-led multilateral development bank.
- Shanghai (China):
 - ✓ NDB: BRICS development bank.

Q 13. Ans. A

Explanation:

- Trans-Siberian Railway:
 - ✓ Trans-Siberian Railway is the longest continuous railway line in the world.
 - ✓ Constructed between 1891 and 1916 under the Russian Empire.
 - ✓ Length: About 9,289 km.
 - ✓ Route:
 - Moscow → Ural Mountains → Siberian region → Vladivostok.
 - Start Point: Moscow (European Russia).
 - End Point: Vladivostok on the Sea of Japan (Pacific Ocean).
 - ✓ Crosses the Ural Mountains, which form the boundary between Europe and Asia.
 - ✓ Crosses 8 time zones.
 - ✓ Built to connect western Russia with resource-rich Siberia.
 - ✓ Helped in the population settlement and economic development of Siberia.

Q 14. Ans. B

Explanation:

- Assertion (A) is correct:
 - ✓ Quaternary activities involve research, development, and information-based services such as IT services, R&D, financial analysis, and data processing.
 - ✓ The internet enabled instant transmission

of data across the world, removing geographical barriers.

- ✓ This allowed global expansion of quaternary services through outsourcing, such as BPO and KPO.
- ✓ Example:
 - IT services from India serving foreign companies.
 - Medical data and financial services are handled remotely across countries.
- Reason (R) is correct:
 - ✓ The spread of internet technology has accelerated automation in primary and secondary sectors.
 - ✓ Primary Sector:
 - ✓ Use of precision agriculture, drones, and sensors.
 - ✓ Digital monitoring in mining operations.
 - ✓ Secondary Sector:
 - Development of Industry 4.0, smart factories, and robotic assembly lines.
 - Internet-connected machines reduce human intervention.
- Why (R) is not the correct explanation of (A):
 - ✓ The expansion of quaternary services occurred mainly because the internet allows the rapid transfer of information globally.
 - ✓ Automation in agriculture and manufacturing is a separate effect of internet technology.
 - ✓ It does not explain the global spread of knowledge-based services.

Economic Activity Sectors:

- Primary Sector:
 - ✓ Extraction of natural resources.
 - ✓ Examples: Agriculture, Mining, Fishing.
- Secondary Sector:
 - ✓ Processing raw materials into finished goods.
 - ✓ Examples: Manufacturing, Construction.
- Tertiary Sector:
 - ✓ Service-based activities.
 - ✓ Examples: Transport, Banking, Tourism, Trade.
- Quaternary Sector:
 - ✓ Knowledge-based services.
 - ✓ Examples: IT, R&D, Education, Data services.
- Quinary Sector:
 - ✓ High-level decision-making services.
 - ✓ Examples: Top government officials, CEOs, policymakers.

Q 15. Ans. A

Explanation:

- Statement 1 is correct:
 - ✓ Population concentrates in moderate climates with reliable rainfall.
 - ✓ Such climates support agriculture and longer growing seasons.
 - ✓ Examples include the Monsoon and Mediterranean regions.
- Statement 2 is incorrect:
 - ✓ High altitude and extremely cold regions have low population density.
 - ✓ Reasons include:
 - Low atmospheric pressure (hypoxia)
 - Rugged terrain limits transport
 - Presence of permafrost, restricting agriculture
- Statement 3 is correct:
 - ✓ Fertile soils and river valleys support dense population clusters.
 - ✓ Alluvial and loamy soils allow intensive agriculture with high carrying capacity.
 - ✓ Examples:
 - Nile Valley
 - Indus Valley
 - Ganga Valley
 - Bihar lies on the fertile alluvial plains of the Middle Ganga Valley.
 - Despite low industrialisation, Bihar is the most densely populated state in India (1,106 persons per sq. km, Census 2011).
- Statement 4 is incorrect:
 - ✓ Deserts have low population density mainly due to water scarcity, not because of reduced chemical weathering.
 - ✓ Primary limiting factors:
 - Lack of rainfall
 - Scarcity of drinking water
 - Extreme temperature range
 - Although chemical weathering is low in deserts, this is not the main reason for the sparse population.

Q 16. Ans. D

Explanation:

- Maasai — Kenya:
 - ✓ Semi-nomadic pastoralists of the East

African Rift Valley in Kenya and Tanzania.

- ✓ Depend mainly on cattle rearing.
- ✓ Known for conflicts with wildlife conservation areas such as the Maasai Mara National Reserve, which restricted traditional grazing lands.
- Navajo — USA:
 - ✓ The largest federally recognised Native American tribe in the USA.
 - ✓ Located mainly in Arizona and New Mexico.
 - ✓ Live in the arid desert regions of the southwestern USA.
- Wodaabe — Chad:
 - ✓ Subgroup of the Fulani people.
 - ✓ Highly nomadic cattle herders and traders.
 - ✓ Found across the semi-arid Sahel region, including Chad, Niger, and Cameroon.
- Achuar — Ecuador:
 - ✓ Indigenous hunter-gatherers of the Amazon Basin.
 - ✓ Located along the Pastaza River on the Ecuador-Peru border.
 - ✓ Similar to the Tharu tribe of West Champaran (Bihar), whose livelihood is closely linked to the forest ecology of the Shivalik foothills.
- Bihar's tribal population is about 1.3% (Census 2011), mainly concentrated in West Champaran (Tharu) and Katihar-Purnia (Santhal and Oraon) regions.

Q 17. Ans. D

Explanation:

- Expansive Pyramid — Nigeria:
 - ✓ Triangular shape with a very wide base and a narrow top.
 - ✓ Indicates a high birth rate and low life expectancy.
 - ✓ Population dominated by youth.
 - ✓ Standard examples include Nigeria, Bangladesh, and Mexico.
- Stationary Pyramid — United States:
 - ✓ Shape is column or bell-shaped, tapering only at older ages.
 - ✓ Indicates birth rate nearly equal to death rate.
 - ✓ Population growth is stable or slow.
 - ✓ Standard examples include Australia; the USA fits due to near-replacement fertility and support from immigration.

- Constrictive Pyramid — Germany:
 - ✓ Shape is urn-shaped, with a narrow base and a wider middle.
 - ✓ Indicates a very low birth rate and an ageing population.
 - ✓ Typical of advanced countries in Stage 4 or 5 of demographic transition.
 - ✓ Standard examples include Japan, Germany, and Latvia.
 - ✓ Leads to a high old-age dependency ratio, increasing the need for pensions and elderly care.

Q 18. Ans. A

Explanation:

- Aeolian (Wind) Erosional Landforms
 - ✓ Rock Pedestals (Mushroom Rocks):
 - Formed by wind abrasion near ground level (0.5 to 1.5 m).
 - Stronger erosion at the base creates a narrow stem and wider top.
 - ✓ Yardangs:
 - Formed where hard and soft rocks lie vertically and parallel to prevailing winds.
 - Wind removes soft rock, leaving long, streamlined ridges of hard rock.
 - ✓ Zeugen:
 - Formed where hard rock overlies soft rock (horizontal layers).
 - Moisture widens joints in hard caprock.
 - Wind erodes underlying soft rock, forming flat-topped ridges.

Q 19. Ans. A

Explanation:

- Assertion (A) is correct:
 - ✓ Walker Circulation is the east–west (zonal) atmospheric circulation over the equatorial Pacific Ocean.
 - ✓ It operates between the western Pacific (Indonesia–Australia) and eastern Pacific (Peru–Ecuador) regions.
- Reason (R) is correct:
 - ✓ Walker Circulation is driven by temperature and pressure differences across the Pacific Ocean.
 - ✓ Western Pacific:
 - Warm ocean waters create low pressure.
 - Warm air rises, producing heavy rainfall in Indonesia and Australia.

- ✓ Eastern Pacific:
 - Cold waters due to Peru (Humboldt) Current and upwelling create high pressure.
 - Air sinks, producing dry conditions, such as the Atacama Desert region.
 - Surface winds called Trade Winds blow from east to west, while upper-level winds return from west to east, completing the Walker Circulation loop.

- Why (R) is the correct explanation of (A):

- ✓ Walker Circulation exists because of the east–west contrast in ocean temperature and pressure across the Pacific.
- ✓ The warm western Pacific causes rising air, and the cold eastern Pacific causes sinking air.
- ✓ This pressure difference directly drives the zonal circulation pattern, which explains the formation of the Walker Circulation.

Q 20. Ans. B

Explanation:

- **Statement 1 is correct:** Ocean salinity mainly depends on the balance between evaporation and precipitation. More evaporation increases salinity, while more rainfall decreases salinity.
- **Statement 2 is correct:** Salinity decreases near coasts due to large freshwater inflow from rivers, which dilutes seawater.
- **Statement 3 is incorrect:** Highest ocean salinity is not at the equator, but in the subtropics (20° to 30° N and S).
 - ✓ At the equator, high evaporation is balanced by heavy convectional rainfall, reducing salinity.
 - ✓ In subtropics, dry descending air and very low rainfall cause very high evaporation, leading to maximum salinity.
- **Statement 4 is correct:** Ocean currents and winds redistribute saline water. Example: Gulf Stream carries saline tropical water to the North Atlantic.
- **Global Salinity extremes:**
 - ✓ Lake Van (Turkey) — 330‰

- ✓ Dead Sea (Israel–Jordan) — 238%
- ✓ Great Salt Lake (USA) — 220%
- ✓ Average ocean salinity — 35%

● Indian Context:

- ✓ Arabian Sea has higher salinity than the Bay of Bengal.
- ✓ Bay of Bengal receives large freshwater input from the Ganga, Brahmaputra, Godavari, and Krishna rivers.
- ✓ The Arabian Sea has high evaporation and fewer major rivers (Narmada, Tapi), so salinity remains higher.

Q 21. Ans. A

Explanation:

- Troposphere — Weather phenomena and decreasing temperature:
 - ✓ Lowest atmospheric layer, with an average height of about 13 km.
 - ✓ All weather phenomena, such as clouds, rain, and storms, occur here.
 - ✓ Temperature decreases with height at the Normal Lapse Rate of 6.5°C per 1,000 m.
- Stratosphere — Presence of ozone layer:
 - ✓ Extends up to about 50 km.
 - ✓ Contains the ozone layer, which absorbs harmful UV radiation.
 - ✓ Temperature increases with height due to absorption of UV rays.
 - ✓ Generally free from weather disturbances, so jet aircraft fly in this layer.
- Mesosphere — Temperature decreases to about –100°C:
 - ✓ Extends up to about 80 km.
 - ✓ Coldest atmospheric layer.
 - ✓ Most meteors burn up in this layer due to friction.
- Thermosphere — Contains the ionosphere and rising temperature:
 - ✓ Extends from about 80 km to 400 km.
 - ✓ Contains the ionosphere, which reflects radio waves and supports long-distance communication.
 - ✓ Temperature rises sharply due to absorption of high-energy solar radiation (X-rays and UV).

Q 22. Ans. D

Explanation:

- **Statement 1 is correct:** Cirques are amphitheatre-shaped depressions formed at the head of glacial valleys due to intense erosion by moving ice.
 - ✓ Cirque:
 - An armchair-shaped or amphitheatre-like depression.
 - Characterised by:
 - ◇ Steep back wall
 - ◇ Concave basin
 - Formed by glacial erosion, especially plucking and abrasion at the head of glaciers.
 - Among the most common glacial erosional landforms.
 - ✓ Alternate Names:
 - Corrie — Scotland
 - Cwm — Wales
 - ✓ Related Feature — Tarn Lake:
 - ✓ A tarn is a small glacial lake formed when a cirque fills with water after glacier melting.
 - ✓ Typically found at high altitudes.
- **Statement 2 is correct:** Horns are sharp, pointed pyramidal peaks formed when three or more cirques erode a mountain from different sides.
 - ✓ Horn (Pyramidal Peak):
 - Formed due to headward erosion by multiple cirques.
 - Produces a sharp, pyramidal summit.
 - ✓ Examples:
 - Matterhorn — classic global example.
 - Mount Everest
 - Kangchenjunga
- **Statement 3 is correct:** Arêtes are narrow, knife-edge ridges formed between adjacent cirques due to headward erosion.
 - ✓ Arête:
 - A sharp, serrated ridge separates two neighbouring cirques.
 - Forms when two glaciers erode toward each other, leaving a narrow ridge between them.

Q 23. Ans. B

Explanation:

- Abyssal Plains:
 - ✓ Abyssal plains are extensive, flat deep-sea plains of the ocean floor.

- ✓ They occur between the continental rise and mid-ocean ridges.
- ✓ Depth Range:
 - Typically found between 3,000 m and 6,000 m depth.
- ✓ Morphology:
 - They are the flattest and smoothest regions on Earth.
 - Flatness results from long-term deposition of fine sediments over volcanic rock.
- ✓ Sediment Types:
 - Terrigenous sediments: clay and silt from land.
 - Pelagic ooze: microscopic shells of marine organisms.
- ✓ Abyssal Plains contains:
 - These nodules contain important minerals such as:
 - ◊ Manganese
 - ◊ Nickel
 - ◊ Copper
 - ◊ Cobalt
 - These minerals are critical for electric vehicle batteries and renewable energy technologies.
- Deflation (wind erosion):
 - ✓ Deflation is the removal of fine particles such as sand and silt by wind.
 - ✓ Deflation leads to the formation of desert pavement.
 - ✓ It also forms shallow depressions called deflation hollows or blowouts.
 - ✓ Example:
 - Deflation is highly active in the **Thar Desert** of Rajasthan.
- Abrasion (mechanical erosion):
 - ✓ Abrasion is the wearing away of rock surfaces by friction of particles carried by wind, glaciers, or rivers.
 - ✓ Landforms produced by abrasion:
 - Wind forms ventifacts.
 - Glaciers form glacial striations on bedrock.
 - Rivers form potholes in riverbeds.
- Corrasion:
 - ✓ Corrasion is another term for abrasion in geomorphology.
 - ✓ It refers to the wearing away of rock by rock fragments striking against the surface.
 - ✓ Distinction:
 - Attrition refers to rock particles breaking against each other.
 - Corrasion/Abrasion refers to rock particles wearing away the bed or surface.

Q 24. Ans. B

Explanation:

- Solifluction:
 - ✓ Slow movement of water-saturated soil over permafrost in cold regions.
 - ✓ Occurs in the active layer that thaws in summer while ground below remains frozen.
 - ✓ Frozen layer blocks percolation –upper soil becomes saturated and moves slowly under gravity.
 - ✓ Typical of periglacial environments; movement is creep-like in nature.
 - ✓ Produces solifluction lobes and solifluction sheets on slopes.
 - ✓ Distribution in India:
 - Solifluction occurs in periglacial regions of the Himalayas.
 - Important areas include Ladakh, Spiti Valley (Himachal Pradesh), Sikkim, and Arunachal Pradesh, generally above 4,000 m altitude.

Q 25. Ans. D

Explanation:

- **Statement 1 is incorrect:** Pothole formation is driven by **Eddy Currents** (circular, swirling water).
 - ✓ When gravel or pebbles get trapped in a shallow depression on the bedrock, these eddies spin the rocks at high speeds, acting like a drill bit.
- This is an example of Corrasion (Abrasion).
 - ✓ Nighoj Potholes, on the Kukdi River (Ahmednagar, Maharashtra), form one of the largest pothole fields in Asia.
 - ✓ They are carved in basalt rock.
- **Statement 2 is incorrect:** Plunge pools are large deep depressions formed at the base of waterfalls due to the intense impact of falling water and the rotational abrasion of rock fragments.

- ✓ Both features represent important forms of river erosion especially in youthful stages of river development.
- ✓ Examples:
 - Kakolat Waterfall (Nawada District): Water falls from about 150–160 feet, forming a deep plunge pool.
 - Karkatgarh Waterfall (Kaimur District):
 - ◇ Located on the Karmanasa River, a tributary of the Ganga.
 - ◇ Forms a large plunge pool in Vindhyan rock formations.

• Waterfall retreat:

- ✓ Waterfall retreat occurs when erosion of the plunge pool undercuts the rock ledge, causing the waterfall to move upstream over time.

• V-Shaped valleys:

- ✓ V-shaped valleys are formed by vertical downcutting of rivers.
- ✓ They are typical of the upper course of rivers.

• Gorge:

- ✓ A gorge is a deep, narrow valley with steep sides.
- ✓ It usually forms in hard rock regions.
- ✓ Example:
 - Indus Gorge in the Himalayas.

• Canyon:

- ✓ A canyon is a deep valley with step-like side slopes.
- ✓ It usually forms in horizontally bedded sedimentary rocks.
- ✓ Example:
 - Grand Canyon (USA).

• River terraces:

- ✓ River terraces are step-like flat surfaces along river valleys.
- ✓ They represent abandoned floodplains formed due to river rejuvenation.

- ✓ Characterised by serrated crests with peaks and saddles.
- ✓ Example:
 - Common in the Thar Desert, especially in the Barmer and Jaisalmer districts of Rajasthan.

• Barchan Dunes (crescent-shaped dunes):

- ✓ Barchan dunes are crescent-shaped dunes formed by unidirectional winds.
- ✓ Their horns point in the direction of the wind (downwind).
- ✓ They form where sand supply is limited, and the surface is hard.

• Parabolic Dunes (U-shaped Dunes):

- ✓ Parabolic dunes are U-shaped dunes formed where vegetation stabilises sand.
- ✓ Their horns point against the wind (upwind).
- ✓ They form in semi-arid regions.
- ✓ Examples:
 - Common in the semi-arid margins of the Thar Desert.

• Star Dunes:

- ✓ Star dunes form due to multidirectional winds blowing from three or more directions.
- ✓ They have a pyramidal shape with multiple radiating arms.
- ✓ Star dunes grow vertically and become the tallest dunes in desert regions.

Q 27. Ans. B

Explanation:

- **Assertion (A) is correct:** Ten Percent Law (Raymond Lindeman, 1942):

- ✓ During the transfer of food energy from one trophic level to the next, only about 10% of energy is stored as biomass (flesh) and becomes available to the next level.
- ✓ The remaining ~90% energy is not lost mysteriously but is expended in respiration, digestion, movement, reproduction, and released as metabolic heat or eliminated as unassimilated waste (faeces/detritus).
- ✓ This reduction in usable energy at successive trophic levels is explained by the Second Law of Thermodynamics, which states that energy transfer increases entropy and dissipates energy as heat.

Q 26. Ans. B

Explanation:

• Seif Dunes (longitudinal dunes):

- ✓ Long, narrow sand ridges formed by bimodal winds (two dominant wind directions).
- ✓ Aligned parallel to the prevailing wind direction.
- ✓ Formed by dominant winds elongating dunes and cross winds increasing height and width.
- ✓ Occur as parallel ridges, often extending over long distances.

- ✓ Due to this rapid depletion of energy, grazing food chains (GFC) in nature rarely exceed 3–4 trophic levels.
- **Reason (R) is also correct:** Unidirectional Flow of Energy:
 - ✓ Energy flow in an ecosystem is strictly linear and non-cyclic.
 - ✓ Solar energy captured by producers (autotrophs) moves to herbivores, carnivores, and decomposers, and ultimately dissipates as heat into the environment.
 - ✓ This energy never returns to the Sun or to previous trophic levels.
 - ✓ In contrast, nutrients (Carbon, Nitrogen, Phosphorus) move in cyclic biogeochemical cycles, repeatedly circulating between biotic and abiotic components.
- Why (R) does NOT explain (A):
 - ✓ The 10% transfer limit occurs due to energy loss during metabolic processes and thermodynamic inefficiency, not because energy flow is unidirectional.
 - ✓ Unidirectional flow explains the direction of movement, whereas thermodynamic loss explains the reduced quantity of transferable energy.
 - ✓ Applied Ecological Illustration (Example-type Understanding):
 - ✓ In a lake ecosystem, phytoplankton → zooplankton → small fish → apex predators, energy moves stepwise upward.
 - ✓ Because only ~10% energy passes at each step, a very large biomass of producers is required to support a small number of apex predators.
 - ✓ This is why top predators always have the smallest population size in a stable ecosystem.
- ✓ It is a major global maritime chokepoint, carrying nearly one-third of global shipping, including energy supplies to Japan and South Korea.
- ✓ **Maritime Trade Link:**
 - Cargo from eastern Indian ports such as Kolkata and Haldia moves through the Strait of Malacca into the South China Sea, and then through the Taiwan Strait to reach East Asian markets.
- Palk Strait
 - ✓ The Palk Strait lies between India and Sri Lanka.
 - ✓ It connects Palk Bay (Bay of Bengal) with the Gulf of Mannar (Laccadive Sea).
 - ✓ Width ranges from 53 km to 80 km.
 - ✓ It contains Rama Setu (Adam's Bridge), a chain of limestone shoals.
- Turkish Straits System
 - ✓ The Turkish Straits connect the Black Sea with the Mediterranean Sea.
- Bosphorus Strait:
 - ✓ Connects the Black Sea with the Sea of Marmara.
- Dardanelles Strait:
 - ✓ Connects the Sea of Marmara with the Aegean Sea (Mediterranean).
 - ✓ These straits provide the only maritime outlet for Russia and Ukraine from the Black Sea.
- Bab-el-Mandeb Strait
 - ✓ The Bab-el-Mandeb Strait connects the Red Sea with the Gulf of Aden and then the Arabian Sea.
 - ✓ Its narrowest width is about 26 km.
 - ✓ It is a major global oil transport chokepoint, linking Europe and Asia via the Suez Canal route.

Q 28. Ans. A

Explanation:

- Taiwan Strait:
 - ✓ The Taiwan Strait separates mainland China (Fujian Province) from the island of Taiwan.
 - ✓ It connects the South China Sea (southwest) with the East China Sea (northeast).
 - ✓ The average width of the Taiwan Strait is about 180 km.

Q 29. Ans. D

Explanation:

- Saudi Arabia:
 - ✓ Saudi Arabia is the largest country on the Arabian Peninsula.
 - ✓ It is landlocked from the Gulf of Oman.
 - ✓ Maritime Borders:
 - ✓ Saudi Arabia borders the Red Sea to the west.
 - ✓ Saudi Arabia borders the Persian Gulf to the east.

- ✓ Major Ports:
 - Jeddah lies on the Red Sea coast.
 - Dammam lies on the Persian Gulf coast.
 - ✓ Oil Transport Bypass:
 - Saudi Arabia uses the East–West Pipeline to transport oil from the Persian Gulf coast to the Red Sea coast.
 - This pipeline helps bypass the Strait of Hormuz, reducing geopolitical risk.
 - Gulf of Oman:
 - ✓ Location & Nature
 - ✓ The Gulf of Oman is the northwestern arm of the Arabian Sea.
 - ✓ It connects the Arabian Sea to the Strait of Hormuz, which further connects to the Persian Gulf.
 - ✓ Countries Bordering the Gulf of Oman
 - ✓ The Gulf of Oman is bordered by:
 - Iran and Pakistan to the north.
 - Oman and the United Arab Emirates (UAE) to the south.
 - ✓ Oceanographic Feature:
 - The Gulf of Oman contains one of the largest hypoxic zones (Dead Zones) in the world.
 - Dead Zones are areas with very low oxygen, unsuitable for marine life.
 - ✓ INSTC (International North–South Transport Corridor)
 - The Gulf of Oman is a key maritime route in the International North–South Transport Corridor (INSTC).
 - Goods from India move to Iran’s Chabahar Port, located on the Gulf of Oman.
 - From Chabahar, cargo moves to Afghanistan, Central Asia, and Russia, bypassing Pakistan.
 - ✓ Key Strategic Ports on the Gulf of Oman:
 - United Arab Emirates (UAE)
 - The Emirate of Fujairah lies on the Gulf of Oman.
 - Fujairah Port allows oil exports bypassing the Strait of Hormuz.
 - Iran:
 - ✓ Chabahar Port lies on Iran’s Gulf of Oman coast.
 - ✓ It is India’s major overseas port investment.
 - Pakistan
 - ✓ Gwadar Port lies on Pakistan’s Makran Coast along the Gulf of Oman.
 - ✓ Gwadar Port is part of the China–Pakistan Economic Corridor (CPEC).
- Q 30. Ans. C**
- Explanation:**
- **Statement 1 is correct:** Warm ocean currents such as the Gulf Stream and North Atlantic Drift raise temperatures over adjoining ocean regions, causing isotherms to bend northward (poleward) over oceans during winter.
 - ✓ In January (Northern hemisphere winter):
 - Land cools faster than oceans due to differential heating.
 - Land becomes colder than oceans at the same latitude.
 - Therefore:
 - ◇ Isotherms bend northward over oceans (warmer conditions).
 - ◇ Isotherms bend southward over continents (colder conditions).
 - ✓ Examples:
 - Strong northward bending occurs in the North Atlantic due to the Gulf Stream and North Atlantic Drift.
 - Similar northward bending occurs in the North Pacific due to the Kuroshio Current.
 - Applied Example:
 - ◇ Murmansk (Russia) remains ice-free in winter due to the warming influence of the North Atlantic Drift (warm current).
 - **Statement 2 is correct:** Land surfaces heat and cool more rapidly than oceans, leading to greater seasonal variation and deviation of isotherms over continents.
 - ✓ Continentiality principle:
 - Water has approximately five times higher specific heat capacity than land.
 - Oceans heat and cool slowly, producing moderated temperature patterns.
 - Land heats and cools rapidly, producing extreme seasonal temperature variations.
 - ✓ July (Northern hemisphere summer) pattern:
 - Land heats faster than oceans.
 - Therefore:
 - ◇ Isotherms bend northward over continents.
 - ◇ Isotherms bend southward over oceans.
 - This pattern is the reverse of January.

Q 31. Ans. B**Explanation:**

- Statement 1 is correct:
 - ✓ Industries locate where costs are minimum, and profits are maximum.
 - ✓ Key factors include cheap land, labour, power, raw materials, and markets.
 - ✓ Based on Alfred Weber's Industrial Location Theory.
- Statement 2 is incorrect:
 - ✓ Agglomeration means clustering of industries to share labour, transport, and services.
 - ✓ It does not always lead to dispersal.
 - ✓ Only excessive clustering may later cause diseconomies like high rent, congestion, and pollution.
- Statement 3 is correct:
 - ✓ Governments influence industrial location through subsidies, tax benefits, cheap land, and power.
 - ✓ This helps industries develop in backward regions.
- Key Exception (Footloose Industries):
 - ✓ Some industries are not tied to raw materials or markets.
 - ✓ Examples include software, diamond cutting, and watchmaking.
 - ✓ They depend mainly on skilled labour and transport connectivity.

Q 32. Ans. B**Explanation:**

- ASEAN — Jakarta, Indonesia:
 - ✓ ASEAN stands for the Association of Southeast Asian Nations.
 - ✓ Founded in 1967 by the Bangkok Declaration.
 - ✓ Has 10 member countries.
 - ✓ Headquarters located in Jakarta, Indonesia.
- EU — Brussels, Belgium:
 - ✓ EU stands for European Union.
 - ✓ Brussels is the de facto capital of the EU.
 - ✓ Hosts major institutions such as the European Commission and Council of the European Union.
- SAARC — Kathmandu, Nepal:
 - ✓ SAARC stands for South Asian Association for Regional Cooperation.
 - ✓ Founded in 1985 at Dhaka.

- ✓ Secretariat was established in Kathmandu in 1987.

● OPEC — Vienna, Austria:

- ✓ OPEC stands for the Organisation of Petroleum Exporting Countries.
- ✓ Headquarters located in Vienna, Austria.
- ✓ Austria is not an OPEC member and produces very little oil.
- ✓ Vienna hosts OPEC due to its historic neutrality.

Q 33. Ans. D**Explanation:**

- Statement 1 is correct:
 - ✓ Defines the Most-Favoured-Nation (MFN) principle of WTO.
 - ✓ A country must treat all WTO members equally in trade.
 - ✓ If a tariff concession is given to one country, it must be extended to all WTO members.
- Statement 2 is correct:
 - ✓ WTO promotes predictability and transparency in trade.
 - ✓ Countries fix Bound Tariffs, which cannot be raised arbitrarily.
 - ✓ Transparency maintained through Trade Policy Reviews (TPRM).
- Statement 3 is correct:
 - ✓ WTO allows Special and Differential Treatment (S&DT) for developing and least developed countries.
 - ✓ They receive preferential access, flexibility, and a longer time to implement WTO rules.
- Key WTO Non-Discrimination Principles:
 - ✓ MFN: Equal treatment to all trading partners.
 - ✓ National Treatment: Imported goods are treated the same as domestic goods after entry.
- India Context:
 - ✓ India supports S&DT to continue MSP and the Public Distribution System (PDS).
 - ✓ The Peace Clause (WTO Bali Ministerial, 2013) protects agricultural subsidies important for farmers, including those in states like Bihar.

Q 34. Ans. B**Explanation:**

- Human Development Index:

- ✓ HDI (Human Development Index) was developed by Mahbub ul Haq with contributions from Amartya Sen and is published by UNDP.
- ✓ HDI measures human development using three dimensions and four indicators:
- Health Dimension:
 - ✓ Measured by Life expectancy at birth.
- ✓ Education Dimension:
 - ✓ Measured by Mean years of schooling (adults 25+ years).
 - ✓ Measured by expected years of schooling (for children entering school).
- Standard of Living Dimension:
 - ✓ Measured by GNI per capita (PPP adjusted).
 - ✓ Infant Mortality Rate (IMR) is an important health indicator, but not included in HDI calculation.
 - ✓ It is used in other indices such as the Multidimensional Poverty Index (MPI), where child mortality is included.
- Other UNDP Indices:
 - ✓ IHDI: Adjusts HDI for inequality.
 - ✓ GDI: Measures gender-based HDI differences.
 - ✓ MPI: Uses indicators, including child mortality.
- India ranks 134th globally (HDI 0.644, Medium Human Development category).

Q 35. Ans. B

Explanation:

- Assertion (A) is correct:
 - ✓ Aluminium smelting is a highly electricity-intensive industry.
 - ✓ Electricity accounts for about 30% to 40% of total production cost, making it the largest cost component in aluminium production.
- Reason (R) is correct:
 - ✓ Aluminium is extracted from alumina by the Hall-Héroult process, which uses electrolysis.
 - ✓ Alumina is dissolved in molten cryolite, and a strong continuous electric current is passed to separate aluminium from oxygen.
 - ✓ This process requires a very large and continuous electricity supply.
- Why (R) is the correct explanation of (A):
 - ✓ Aluminium production depends on electrolysis, which consumes massive

amounts of electricity.

- ✓ Therefore, the high electricity requirement directly explains why aluminium smelting is electricity-intensive.

● Industrial Location:

- ✓ Aluminium industries are usually located near cheap and abundant power sources, as electricity cannot be stored easily and transmission losses occur over long distances.

● Examples:

- ✓ HINDALCO (Renukoot, Uttar Pradesh): Located near Rihand Dam and Singrauli coalfields for power supply.
- ✓ BALCO (Korba, Chhattisgarh): Located near the Korba coalfields.
- ✓ NALCO (Angul, Odisha): Located near Talcher coalfields; bauxite supplied from Panchpatmali mines.

Q 36. Ans. C

Explanation:

- Maginot Line — France and Germany:
 - ✓ A line of defensive fortifications built by France in the 1930s.
 - ✓ Constructed to prevent German invasion.
 - ✓ During World War II, Germany bypassed it through Belgium.
- Blue Line — Lebanon and Israel:
 - ✓ A UN-demarcated boundary was created in 2000.
 - ✓ Formed to verify Israel's withdrawal from Lebanon.
 - ✓ Remains a highly sensitive and conflict-prone border.
- 38th Parallel — North Korea and South Korea:
 - ✓ Latitude line used to divide Korea after World War II by the USA and USSR.
 - ✓ Today, the Demilitarised Zone (DMZ) lies along this boundary.
- 49th Parallel — USA and Canada:
 - ✓ Forms the major boundary between the USA and Canada.
 - ✓ Known as the longest undefended international border in the world.
- Other important Boundary Lines:
 - ✓ Durand Line – Boundary between Pakistan and Afghanistan
 - ✓ Hindenburg Line – German defensive line (WWI)

- ✓ Mason–Dixon Line – Boundary between states in the USA
- ✓ Mannerheim (Marginal) Line – Fortification on Russia–Finland border
- ✓ McMahon Line – Boundary between India and China
- ✓ Medicine Line – Border between USA and Canada
- ✓ Radcliffe Line – Boundary between India and Pakistan
- ✓ Siegfried Line – German fortification on border with France
- ✓ Oder–Neisse Line – Boundary between Germany and Poland
- ✓ 17th Parallel – Boundary between North and South Vietnam (before unification)
- ✓ Used by flat-bottomed barges due to shallow depth and siltation.
- ✓ Example: Kolkata Port (Hooghly River) — India's major river port.
- Fishing Port — Used for landing and distribution of fish:
 - ✓ Specialized for fish landing, storage, and processing.
 - ✓ Facilities include cold storage and auction centers.
- Dry Port — Connected to sea port by road or rail for cargo storage:
 - ✓ Also called Inland Container Depot (ICD).
 - ✓ Handles custom clearance, cargo storage, and distribution inland.
 - ✓ Example: Tughlakabad ICD (Delhi) — major inland dry port.

Q 37. Ans. D

Explanation:

- Djibouti — Port of Ghoubet:
 - ✓ Port of Ghoubet is a specialized mineral port in Djibouti.
 - ✓ Developed mainly for export of salt from Lake Assal, one of the most saline lakes.
 - ✓ Djibouti's main commercial port is Port of Djibouti (Doraleh).
- Eritrea — Massawa Port:
 - ✓ Massawa is a major deep-water port of Eritrea on the Red Sea coast.
 - ✓ Along with Assab, it serves as one of Eritrea's primary seaports.
- Ethiopia — Mogadishu Port (Incorrect Pair):
 - ✓ Mogadishu is the capital and main port of Somalia, not Ethiopia.
 - ✓ Ethiopia is a landlocked country and has no seaport of its own.

Q 38. Ans. D

Explanation:

- Sea Port — Used to handle ocean-going vessels:
 - ✓ Located on the coast with deep water.
 - ✓ Designed for large cargo ships and oil tankers.
 - ✓ Example: JNPT (Mumbai) — India's largest container sea port.
- River Port — Handles shallow draft vessels and barges:
 - ✓ Located on navigable rivers inland.

Q 39. Ans. A

Explanation:

- Pampas — South America:
 - ✓ Located mainly in Argentina and Uruguay (also parts of southern Brazil).
 - ✓ A temperate grassland known for fertile soils and large-scale wheat farming and cattle ranching.
- Savanna — Africa:
 - ✓ Typical tropical grassland, especially in sub-Saharan Africa.
 - ✓ Characterized by tall grasses and scattered trees with wet and dry seasons.
- Downs — Australia:
 - ✓ Temperate grasslands are found mainly in the Murray–Darling Basin region.
 - ✓ Famous for commercial sheep rearing, especially Merino wool production.
- Steppes — Eurasia:
 - ✓ Vast temperate grassland belt stretching from Eastern Europe (Ukraine) through Russia to Central Asia.
 - ✓ Dominated by short grasses and is widely used for wheat cultivation and grazing.
- Key Classification:
 - ✓ Tropical Grasslands:
 - Savanna — Africa
 - Campos — Brazil
 - Llanos — Venezuela and Colombia
 - ✓ Temperate Grasslands:
 - Pampas — South America

- Downs — Australia
- Steppes — Eurasia
- Prairies — North America
- Veld — South Africa
- Economic Significance:
 - ✓ Temperate grasslands are among the most productive agricultural regions due to fertile soils and mechanised farming.
 - ✓ Prairies — known as the granaries of the World.
 - ✓ Downs — famous for sheep rearing.
 - ✓ Pampas — known for wheat cultivation and cattle ranching.
 - ✓ Steppes — known as the Breadbasket regions, especially in Ukraine.

Q 40. Ans. A

Explanation:

- Somalia:
 - ✓ Forms the easternmost tip of Africa and is a core country of the Horn.
 - ✓ Has the longest coastline in mainland Africa (about 3,025 km).
 - ✓ Borders both the Indian Ocean and the Gulf of Aden.
- Ethiopia:
 - ✓ Largest and most populous country in the Horn region.
 - ✓ Most populous landlocked country in the world.
 - ✓ Contains Lake Tana, the main source of the Blue Nile.
 - ✓ Hosts headquarters of the African Union at Addis Ababa.
- Eritrea:
 - ✓ Located along the Red Sea coast, forming the northern part of the Horn.
 - ✓ Became independent from Ethiopia in 1993, making Ethiopia landlocked.
 - ✓ Controls part of the Red Sea route toward the Suez Canal.
- Djibouti:
 - ✓ The fourth country completing the Horn of Africa.
 - ✓ Controls the strategic Bab el-Mandeb Strait, a major global shipping route.
- Kenya (Not part of the Horn of Africa):
 - ✓ Located south of Somalia and Ethiopia but belongs to East Africa, not the Horn.
- ✓ The Equator passes through Kenya.
- ✓ Shares Lake Victoria with Tanzania and Uganda.
- Other Important Regional Groupings:
 - ✓ Maghreb (North-West Africa):
 - Includes Morocco, Algeria, Tunisia (often Libya and Mauritania).
 - Located along the Mediterranean and Atlantic coasts.
 - ✓ Sahel Region:
 - Semi-arid belt south of the Sahara Desert.
 - Extends from Senegal to Sudan across Africa.
 - ✓ Baltic States:
 - Countries along the Baltic Sea.
 - Includes Estonia, Latvia, and Lithuania.
 - ✓ Nordic Countries:
 - Northern European region.
 - Includes Denmark, Finland, Iceland, Norway, and Sweden.
 - ✓ Caucasus (Transcaucasia):
 - Region between the Black Sea and the Caspian Sea.
 - Includes Georgia, Armenia, and Azerbaijan.

Q 41. Ans. B

Explanation:

- **Statement 1 is Incorrect:** The Suez Canal cuts through the Isthmus of Suez located in northeastern Egypt, not northwestern. It forms the physical boundary between the African continent and Asia.
- **Statement 2 is correct:** The canal geographically separates mainland Egypt (which is on the African continent) from the Sinai Peninsula (which is geographically part of the Asian continent).
- **Statement 3 is correct:** The canal serves as a direct maritime link. Its northern terminus on the Mediterranean Sea is Port Said, and its southern terminus on the Gulf of Suez (an arm of the Red Sea) is the city of Suez (Port Tawfiq).
- **Additional information about Suez canal:**
 - ✓ Constructed in 1869.
 - It is a sea-level canal without locks (unlike Panama Canal).
 - Lakes crossed (North to South order):

- Lake Manzala → Lake Timsah → Great Bitter Lake → Little Bitter Lake
- ✓ Location:
 - Cuts across the Isthmus of Suez, connecting the Mediterranean Sea to the Red Sea.
- ✓ Significance:
 - Shortens sea route between Europe and Asia.
 - Avoids long journey around the Cape of Good Hope.
 - Major global trade and shipping route.
- ✓ Sovereignty:
 - Entirely under Egypt's control.
 - Operated by the Suez Canal Authority (SCA).
- ✓ Historical fact:
 - Nationalised in 1956 by Gamal Abdel Nasser, leading to the Suez Crisis.

Q 42. Ans. D

Explanation:

- Statement 1 is incorrect:
 - ✓ Primary activities depend directly on natural resources such as land, water, vegetation, and minerals.
 - ✓ Most primary activities like subsistence farming, fishing, herding, and gathering are labour-intensive and small-scale.
 - ✓ Large-scale, capital-intensive production is typical of the secondary sector, not the primary sector.
- Statement 2 is incorrect:
 - ✓ Primary activities involve direct extraction of natural resources without processing.
 - ✓ Secondary activities involve processing raw materials into finished goods, adding value.
- Sector Classification:
 - ✓ Primary Activities — Red-collar workers:
 - Extraction of natural resources.
 - Examples: Agriculture, fishing, forestry, mining, pastoralism.
 - ✓ Secondary Activities — Blue-collar workers:
 - Manufacturing and construction.
 - Examples: Steel production, textile manufacturing, and infrastructure building.
 - ✓ Tertiary Activities — White-collar workers:

- Service-based activities.
- Examples: Trade, transport, banking, education, and healthcare.
- ✓ Quaternary Activities — Knowledge workers:
 - Information and research-based services.
 - Examples: IT services, R&D, data analysis.
- ✓ Quinary Activities — Gold-collar workers:
 - Top-level decision making and policy roles.
 - Examples: Senior executives, government policymakers, top scientists.

Q 43. Ans. A

Explanation:

- Chagos Archipelago:
 - ✓ Located in the central Indian Ocean.
 - ✓ It lies about 1,600 km south of India and about 500 km south of the Maldives.
 - ✓ Geographically positioned between India (north) and Mauritius (south).
 - ✓ Hence, describing it as located between India and Mauritius is most accurate.
- Strategic Importance:
 - ✓ Diego Garcia:
 - The largest island of the Chagos Archipelago.
 - Hosts a major US–UK military base developed during the Cold War.
 - Used in operations such as the Persian Gulf War, Afghanistan, and the Iraq War.
 - Located near key oil shipping routes in the Indian Ocean.
- Geographical Facts:
 - ✓ Located on the Chagos–Laccadive Ridge (submarine ridge in the Indian Ocean).
 - ✓ Diego Garcia is a coral atoll.
 - ✓ A strategic position allows control over sea routes linking:
 - Persian Gulf — East Africa — South and Southeast Asia.
- Historical and Political Facts:
 - ✓ Originally controlled by France, transferred to the United Kingdom in 1814 (Treaty of Paris).

- ✓ British Indian Ocean Territory (BIOT) was created in 1965 by separating Chagos from Mauritius.
- ✓ Native Chagossians were displaced in the 1970s during base establishment.
- Recent developments:
 - ✓ 2025: UK–Mauritius agreement signed to transfer sovereignty to Mauritius, with Diego Garcia leased to the UK for 99 years.
 - ✓ India supported the agreement as part of decolonization process.
- India's Strategic Interest:
 - ✓ Chagos lies along India's major Sea Lines of Communication (SLOCs) used for energy imports from the Gulf.
 - ✓ Strategic control in this region affects India's Indian Ocean security strategy and SAGAR doctrine (Security and Growth for All in the Region).

Q 44. Ans. B

Explanation:

- The **Essequibo region** forms about two-thirds of Guyana but is claimed by Venezuela. It is rich in oil, gold, and forests, making it geopolitically important.
- **Additional facts:**
 - ✓ 1899 Arbitral Award: Fixed the Guyana–Venezuela border; Guyana considers it final, Venezuela disputes it.
 - ✓ Oil discovery (2015): Revived tensions and made Guyana one of the fastest-growing economies.
 - ✓ 2023 Venezuelan referendum: Supported creation of a Venezuelan state in Essequibo, escalating tensions.
 - ✓ The dispute is currently under the International Court of Justice (ICJ) for final settlement.

Q 45. Ans. C

Explanation:

- Nomadic herding is practiced mainly in **arid and semi-arid regions** (deserts and dry steppes) where rainfall is low and crop farming is difficult.
- **Equatorial rainforest:** Dense forests, heavy rainfall, and tsetse fly make livestock rearing difficult.

- **Mediterranean:** Favors commercial orchard farming (grapes, olives, citrus), not nomadism.
- **Tundra:** Reindeer herding exists but is limited, not the main global zone.
- Additional information:
 - ✓ Major belts: North Africa → Arabian Peninsula → Central Asia → Mongolia (largest).
 - ✓ Nomadism: Continuous movement for pasture.
 - ✓ Transhumance: Seasonal movement between fixed summer and winter pastures.

Q 46. Ans. D

Explanation:

- **Statement 1 is correct:** Stalactites are icicle-like depositional features that hang from cave ceilings and taper downward due to the deposition of calcium carbonate from dripping water.
 - ✓ Stalactites:
 - Depositional features that hang from the ceilings of caves.
 - Form when calcium carbonate is deposited from mineral-rich dripping water.
 - Grow downward from the roof of the caves.
 - The deposited mineral is calcite, also known as travertine.
- **Statement 2 is correct:** Stalagmites are upward-growing depositional features on cave floors, formed by the accumulation of mineral deposits from water dripping from above.
 - ✓ Stalagmites:
 - Depositional features that rise from cave floors.
 - Formed from calcium carbonate deposits left by dripping water.
 - Grow upward from the ground.
- **Statement 3 is correct:** When stalactites and stalagmites continue to grow and eventually join, they form vertical columns or pillars, creating a continuous structure from ceiling to floor.
- **Formation Process Karst Topography:**
 - ✓ These features develop mainly in limestone and dolomite regions.
 - ✓ Rainwater mixes with carbon dioxide (CO₂) to form carbonic acid.

- ✓ Carbonic acid dissolves limestone, creating underground cavities.
- ✓ When mineral-rich water drips inside caves, CO₂ escapes, leading to calcium carbonate deposition.
- ✓ Continuous deposition forms speleothems such as stalactites, stalagmites, and pillars.
- Classification of Karst landforms:
 - ✓ Depositional Karst Features (by calcium carbonate deposition):
 - Stalactites
 - Stalagmites
 - Pillars (Columns)
 - Cave Drapes
 - ✓ Erosional Karst Features (by limestone dissolution and collapse):
 - Sinkholes
 - Swallow holes
 - Dolines
 - Uvalas
 - Lapies
 - Blind valleys
- Example:
 - ✓ Kaimur Plateau:
 - Extends across the Rohtas and Kaimur districts.
 - Forms part of the Vindhyan Range.
 - Contains extensive limestone formations, favourable for cave development.
 - ✓ Gupteshwar Cave (Guptadham), Rohtas:
 - A major limestone cave is located in the Kaimur Hills.
 - The Shiva Lingam inside the cave is a natural stalagmite formed by long-term mineral deposition.
- ✓ Result:
 - Upper soil becomes ash-grey due to the dominance of silica.
 - Podzolic soils become highly acidic and naturally infertile.
- **Statement 2 is correct:** Laterisation occurs in hot and wet tropical regions, where heavy leaching removes soluble minerals and leaves behind iron and aluminium oxides.
 - ✓ Occurs in hot, wet tropical climates with alternating wet and dry seasons.
 - ✓ The process involves desilication, where silica and soluble bases are leached downward.
 - ✓ Result:
 - Upper soil accumulates Iron and Aluminum oxides (sesquioxides).
 - Soil becomes reddish in colour.
 - Laterite soil becomes hard on drying and is often used as a building material.
 - These soils are poor in Nitrogen, Potash, and organic matter.
- Other Soil-Forming Processes:
 - ✓ Calcification:
 - Occurs in arid and semi-arid climates.
 - Calcium Carbonate accumulates in the upper layers due to evaporation.
 - Forms calcareous nodules known as Kankar.
 - Common in older alluvial plains (Bangar).
 - ✓ Gleisation:
 - Occurs in waterlogged and anaerobic conditions.
 - Iron compounds are reduced, producing blue-grey or greenish clay soil, known as Gley soil.
 - Common in boggy and marshy areas.
 - ✓ Salinisation:
 - Occurs in dry regions with poor drainage or excessive irrigation.
 - Salts accumulate on the surface, forming white salt crusts.
 - Leads to alkaline or saline soils, known as Usar soils.
 - Common in parts of Punjab and Haryana.
- Laterite soil distribution in Bihar:
 - ✓ Laterite soils in Bihar are mainly found in the southern plateau margins bordering Jharkhand.
 - ✓ Important districts include:

Q 47. Ans. C

Explanation:

- **Statement 1 is correct:** Podzolisation involves intense leaching in cool, humid climates, leading to the formation of an ash-grey upper soil layer.
 - ✓ Occurs mainly in cool and humid climates, especially under coniferous forests (Taiga).
 - ✓ The process involves intense leaching caused by acidic forest litter, such as pine needles.
 - ✓ Calcium, Iron, and Aluminum are leached from the upper soil layer.
- **Statement 2 is correct:** Laterisation occurs in hot and wet tropical regions, where heavy leaching removes soluble minerals and leaves behind iron and aluminium oxides.
 - ✓ Occurs in hot, wet tropical climates with alternating wet and dry seasons.
 - ✓ The process involves desilication, where silica and soluble bases are leached downward.
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 - Leads to alkaline or saline soils, known as Usar soils.
 - Common in parts of Punjab and Haryana.
- Laterite soil distribution in Bihar:
 - ✓ Laterite soils in Bihar are mainly found in the southern plateau margins bordering Jharkhand.
 - ✓ Important districts include:

- Rohtas
- Aurangabad
- Gaya
- Munger (Kharagpur Hills)
- Jamui
- ✓ These soils are acidic and less fertile for food crops without the application of fertilisers.
- ✓ Widely used as Murrum, especially in rural road construction, due to its hardening property.

Q 48. Ans. C

Explanation:

- Ecological Niche:
 - ✓ Functional role and position of a species in an ecosystem (trophic role + interactions + tolerance range).
 - ✓ Includes physical space, environmental conditions, and interactions with biotic and abiotic factors.
 - ✓ Habitat: Place of living; Niche: Functional role of the organism.
 - ✓ Gause's Competitive Exclusion Principle:
 - Two species cannot occupy the same niche in the same habitat for long → one outcompetes the other.
 - ✓ Types of Niche:
 - Fundamental Niche: Full potential range of conditions without competition.
 - Realised Niche: Actual range occupied under competition and predation.
 - ✓ Gangetic Dolphin (Susu):
 - Found in Vikramshila Gangetic Dolphin Sanctuary (Bhagalpur).
 - It is a blind aquatic predator using echolocation in turbid Ganga waters.
 - ✓ Kabartal Wetland (Begusarai):
 - Bihar's first Ramsar Site.
 - Supports over 50 species of migratory birds.
 - Different bird species occupy different ecological niches, reducing competition.
- Habitat:
 - ✓ A habitat is the natural physical environment where an organism lives and reproduces.
 - ✓ Many species can share the same habitat, but not the same ecological niche.

- ✓ Components of Habitat:
 - Abiotic factors include temperature, water, light, and soil.
 - Biotic factors include predators, parasites, and pathogens.

- Ecosystem:
 - ✓ The term Ecosystem was coined by Sir A.G. Tansley (1935).
 - ✓ An ecosystem includes biotic components (living organisms) and abiotic components (physical environment) interacting together.
- Structure of Ecosystem:
 - ✓ Species show stratification, such as trees → shrubs → herbs.
- Major Functions of Ecosystem:
 - ✓ Productivity
 - ✓ Decomposition
 - ✓ Energy Flow
 - ✓ Nutrient Cycling
 - ✓ Energy flow follows the 10% Law, proposed by Lindeman.
- Biosphere:
 - ✓ The biosphere is the largest ecological unit, consisting of all ecosystems on Earth.
 - ✓ It is the zone where lithosphere, hydrosphere, and atmosphere interact to support life.
 - ✓ Vertical Limits of Biosphere:
 - Extends from about 11,000 m below sea level to about 15,000 m above sea level.
 - Most life occurs within –200 m (oceans) to +3,000 m (land).

Q 49. Ans. C

Explanation:

- **Statement 1 is correct:** Convectional rainfall is caused by intense surface heating, which makes warm air rise rapidly.
 - ✓ It is associated with Cumulonimbus clouds.
 - ✓ Rainfall is heavy, short-duration, and usually accompanied by thunder and lightning.
 - ✓ It is common in Equatorial regions.
 - ✓ Examples:
 - Mango Showers occur in Kerala and Karnataka during April–May.
 - Cherry Blossom Showers occur in Karnataka during April–May.

- Nor'westers (Kalbaisakhi) occur in April–May due to strong convection over the Chota Nagpur Plateau.
 - They affect eastern Bihar districts such as Purnea and Kishanganj.
- ✓ These rains are important for jute and summer paddy (Garma) crops.
- **Statement 2 is correct:** Orographic rainfall occurs when moist winds rise over mountains, cool, and produce rainfall on the windward side.
 - ✓ The leeward side receives less rainfall and forms a rain shadow region.
 - ✓ Examples:
 - Western Ghats show clear windward and rain shadow effects.
 - Mahabaleshwar receives over 6,000 mm of rainfall annually.
 - Pune, located in the rain shadow region, receives about 700 mm of rainfall annually.
 - Mawsynram and Cherrapunji receive very high rainfall due to uplift over the Khasi Hills.
 - West Champaran district touches the Shiwalik (Someshwar Hills).
 - Orographic lifting of monsoon winds makes West Champaran the wettest district in Bihar.
- Cyclonic (Frontal) rainfall:
 - ✓ Cyclonic rainfall occurs when warm and cold air masses meet, forcing warm air to rise.
 - ✓ It produces widespread, steady rainfall.
 - ✓ Examples:
 - Tropical cyclones form over the Bay of Bengal, affecting Odisha and Andhra Pradesh.
 - Western Disturbances cause winter rainfall during January–February.
 - These systems originate in the Mediterranean region.
 - Winter rainfall from Western Disturbances is important for Rabi crops, especially wheat.
- its maximum holding capacity at a given temperature.
- ✓ Relative Humidity is inversely related to temperature when the moisture content remains constant.
- ✓ Increase in temperature decreases RH.
- ✓ Decrease in temperature increases RH.
- When Relative Humidity reaches 100%, air becomes saturated.
 - ✓ At 100% RH, air temperature equals dew point, and further cooling causes condensation forming clouds, fog, or dew.
- Absolute Humidity
 - ✓ Absolute Humidity is the mass of water vapour present in a unit volume of air.
 - ✓ Unit of measurement is grams per cubic meter (g/m^3).
 - ✓ Absolute humidity changes when air volume changes.
- Specific Humidity
 - ✓ Specific Humidity is the mass of water vapour present in a unit mass of air.
 - ✓ Unit of measurement is grams per kilogram (g/kg).
 - ✓ Specific humidity remains constant when air expands or contracts.
 - ✓ It is widely used in weather and climate studies.
- Vapour Pressure
 - ✓ Vapour Pressure is the partial pressure exerted by water vapour in air.
 - ✓ Unit of measurement is millibar (mb) or hectopascal (hPa).
- Saturation Vapour Pressure
 - ✓ Saturation Vapour Pressure is the maximum vapour pressure possible at a given temperature.
 - ✓ When actual vapour pressure equals saturation vapour pressure, Relative Humidity becomes 100%.

Q 51. Ans. C

Explanation:

- Methane (CH_4): 12 years
 - ✓ Methane is a short-lived climate pollutant (SLCP).
 - ✓ It remains in the atmosphere for about 12 years, mainly broken down by reactions with hydroxyl radicals (OH).

Q 50. Ans. B

Explanation:

- Relative Humidity (RH)
 - ✓ Relative Humidity is the percentage of moisture present in the air compared to

- Nitrous oxide (N₂O): 109 years
 - ✓ Nitrous oxide persists for about 109 years.
 - ✓ Major sources include agricultural soils, fertilizers, and biomass burning.
- Sulfur hexafluoride (SF₆): Over 1000 years
 - ✓ SF₆ is an extremely long-lived synthetic greenhouse gas, lasting over 3,000 years.
 - ✓ It is widely used as an electrical insulator in high-voltage systems.
- CFC-12: 102 years
 - ✓ CFC-12 has an atmospheric lifetime of about 102 years.
 - ✓ Though phased out, previously emitted CFC-12 remains in the atmosphere for decades.
- Additional information:
 - ✓ A gas with a short lifetime can still be dangerous if its heat-trapping ability is high.
 - ✓ Methane: Short-lived but very powerful warming gas
 - ✓ SF → Extremely long-lived and considered a “near-permanent” gas
 - ✓ Kyoto Protocol: Targeted Greenhouse Gases (“Kyoto Six”):
 - Carbon dioxide (CO₂)
 - Methane (CH₄)
 - Nitrous oxide (N₂O)
 - Hydrofluorocarbons (HFCs)
 - Perfluorocarbons (PFCs)
 - Sulfur hexafluoride (SF₆)
 - ✓ Kigali Amendment (2016):
 - Focused on phasing down HFCs, linking ozone protection with climate change mitigation.
- moves outward, so crust farther from the ridge is older.
- Statement 3 is incorrect:
 - ✓ Sediments are not thickest at the ridge.
 - ✓ At mid-ocean ridges, crust is new, so sediment cover is very thin or absent.
 - ✓ Sediment thickness increases with distance from the ridge toward continental margins.
- Statement 4 is incorrect:
 - ✓ Subduction does not occur at mid-ocean ridges.
 - ✓ Subduction happens at convergent boundaries, such as deep-sea trenches.
 - ✓ Mid-ocean ridges produce shallow-focus earthquakes, not deep-focus ones.
- Additional facts:
 - ✓ Key Proof of Seafloor Spreading:
 - Paleomagnetism:
 - ◇ Cooling basalt records Earth’s magnetic polarity.
 - ◇ Alternating magnetic stripes on both sides of ridges confirm crust movement.
 - ✓ Oceanic Crust Age Rule:
 - Youngest crust: At mid-ocean ridges
 - Oldest crust: Near continental margins
 - Maximum oceanic crust age: About 200 million years
 - ✓ Example:
 - Mid-Atlantic Ridge:
 - ◇ Runs through the Atlantic Ocean.
 - ◇ Rises above sea level at Iceland, where volcanic activity is clearly visible.

Q 52. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ Mid-oceanic ridges are the longest mountain chains on Earth, located underwater.
 - ✓ They occur at divergent plate boundaries, where plates move apart and magma rises to form new crust.
- Statement 2 is correct:
 - ✓ This describes seafloor spreading, proposed by Harry Hess (1960).
 - ✓ New oceanic crust forms at the ridge and

Q 53. Ans. C

Explanation:

- Tropical Monsoon: Am
- Subtropical Desert: BWh
- Marine West Coast: Cfb
- Humid Subtropical: Cfa
- Meaning of Each Köppen Code:
 - ✓ Am — Tropical Monsoon:
 - ✓ A: Tropical climate (all months above 18°C)
 - ✓ m: Short dry season but heavy monsoon rainfall
 - ✓ Typical Regions:
 - Western coastal India
 - Bangladesh

- Myanmar
- ✓ BWh — Hot Desert Climate:
 - B: Dry climate
 - W: Desert (very low rainfall)
 - h: Hot (mean annual temperature above 18°C)
 - Typical Regions:
 - Thar Desert (India)
 - Sahara Desert
 - Arabian Desert
- ✓ Cfb — Marine West Coast Climate:
 - C: Temperate climate
 - f: No dry season
 - b: Warm (not hot) summer
 - Typical Regions:
 - United Kingdom
 - Western Europe
 - New Zealand
- ✓ Cfa — Humid Subtropical Climate:
 - C: Temperate climate
 - f: Fully humid
 - a: Hot summer
 - Typical Regions:
 - Southeastern United States
 - Eastern China
 - Japan
- Rule:
 - ✓ A → Tropical
 - ✓ B → Dry
 - ✓ C → Temperate
 - ✓ D → Continental
 - ✓ E → Polar
- India-Specific Mapping:
 - ✓ Am → Western Ghats, Malabar Coast
 - ✓ Aw → Peninsular India (Savanna type)
 - ✓ BWh → Thar Desert
 - ✓ Cwg → Indo-Gangetic Plains
- Statement B is correct:
 - ✓ The Crimean Peninsula extends into the northern part of the Black Sea.
 - ✓ It separates the Black Sea from the Sea of Azov to the northeast.
- Statement C is correct:
 - ✓ The Black Sea connects to the Mediterranean Sea through a chain of waterways:
 - ✓ Bosphorus Strait → Sea of Marmara → Dardanelles Strait → Aegean Sea → Mediterranean Sea
- Statement D is correct:
 - ✓ The Black Sea lies between Europe and Asia, making it a transcontinental water body.
- Additional facts:
 - ✓ Countries Bordering the Black Sea:
 - Bulgaria
 - Romania
 - Ukraine
 - Russia
 - Georgia
 - Turkey
 - ✓ Strategic Treaty:
 - Montreux Convention (1936):
 - Gives Turkey control over the Bosphorus and Dardanelles straits.
 - Turkey can regulate or restrict passage of warships during conflicts.
 - ✓ Kerch Strait:
 - Connects the Black Sea to the Sea of Azov.
 - Separates the Crimean Peninsula from mainland Russia.
 - ✓ Unique Physical Feature:
 - The Black Sea is the largest permanently anoxic sea in the world.
 - Deep waters lack oxygen below roughly 200 meters, preserving ancient shipwrecks.

Q 54. Ans. A

Explanation:

- Statement A is incorrect:
 - ✓ Russia does not have the longest coastline along the Black Sea.
 - ✓ Countries with longer Black Sea coastlines include Turkey (southern coast) and Ukraine (northern coast, depending on measurement methods).
 - ✓ Russia's coastline along the Black Sea is comparatively shorter.

Q 55. Ans. A

Explanation:

- **Statement 1 is correct:** Sinkholes and lapies are typical erosional features of karst topography, formed due to the chemical dissolution of limestone or dolomite rich in calcium carbonate (CaCO_3).
 - ✓ Chemical basis — Carbonation Process:

- Karst topography develops due to chemical weathering, not mechanical erosion.
- Rainwater absorbs carbon dioxide (CO_2) to form carbonic acid (H_2CO_3).
- Carbonic acid reacts with calcium carbonate (CaCO_3) in limestone.
- This reaction produces calcium bicarbonate, which is soluble and removed by water.
- This chemical process is known as carbonation.
- ✓ Karst Erosional Features:
 - Lapiés:
 - ◊ Sharp, grooved limestone surfaces formed by dissolution along joints and cracks.
 - ◊ Produce a rugged and uneven rocky surface.
 - Sinkholes:
 - ◊ Funnel-shaped or cylindrical depressions formed due to limestone dissolution.
 - ◊ Represent the smallest karst depressions.
- **Statement 2 is correct:** Stalactites and stalagmites are depositional landforms formed inside limestone caves when water carrying dissolved calcium carbonate loses carbon dioxide or evaporates, causing precipitation of calcite.
 - ✓ Karst Depositional Features:
 - Form when calcium carbonate (calcite) is deposited inside caves.
 - Stalactites:
 - ◊ Hang from cave ceilings.
 - ◊ Form by calcite deposition from dripping water.
 - Stalagmites:
 - ◊ Grow upward from cave floors.
 - ◊ Form when calcite accumulates from falling droplets.
 - When stalactites and stalagmites join together, they form pillars (columns).
- **Statement 3 is incorrect:** Karst topography does not develop due to mechanical erosion, as the dominant processes involved are chemical solution and precipitation, not physical removal of material.
 - ✓ Dominant role of chemical weathering:
 - Karst formation depends primarily on chemical dissolution of soluble rocks.
 - Groundwater percolates through joints and cracks in limestone.
 - Dissolution enlarges cavities, forming karst depressions and underground features.
 - Subsequent deposition inside caves creates speleothems such as stalactites and stalagmites.
- Additional points:
 - ✓ Karst depressions increase in size in the following order:
 - Sinkhole / Swallow Hole → Doline → Uvala → Polje
 - ✓ Doline:
 - Formed when multiple sinkholes merge.
 - ✓ Uvala:
 - Formed by the merging of several dolines.
 - Usually large and elongated.
 - ✓ Polje:
 - The largest karst depression.
 - A flat-bottomed enclosed valley.
 - May contain residual limestone hills, known as Hums.

Q 56. Ans. D

Explanation:

- **Statement 1 is incorrect:** Nitrogen fixation is carried out only by prokaryotes, not by fungi or higher plants.
 - ✓ Nitrogen Fixation:
 - The conversion of atmospheric nitrogen (N_2) into ammonia (NH_3).
 - Performed exclusively by prokaryotic organisms, called diazotrophs.
 - Higher plants cannot fix nitrogen directly; they depend on symbiotic bacteria.
 - ✓ Nitrogen-Fixing Organisms:
 - Free-living Bacteria:
 - ◊ Azotobacter — aerobic
 - ◊ Beijerinckia
 - ◊ Rhodospirillum — anaerobic
 - Cyanobacteria:
 - ◊ Anabaena
 - ◊ Nostoc

- Symbiotic Bacteria:
 - ◊ Rhizobium — associated with leguminous plants
 - ◊ Frankia — associated with non-leguminous plants
- ✓ Fact:
 - Cyanobacteria are common in paddy fields and wetlands, where they enhance soil fertility through nitrogen fixation.
- **Statement 2 is incorrect:** The enzyme nitrogenase does not function in the presence of oxygen, as it is highly oxygen-sensitive and operates under low-oxygen conditions.
 - ✓ Key features of Nitrogenase:
 - A Mo-Fe (Molybdenum-Iron) protein enzyme.
 - Converts Nitrogen (N_2) into Ammonia (NH_3).
 - Requires about 16 ATP molecules per N-molecule fixed.
 - Functions only under low-oxygen conditions.
 - ✓ Root Nodule Adaptation:
 - Root nodules contain leg-haemoglobin, which:
 - ◊ Regulates oxygen supply.
 - ◊ Protects nitrogenase from oxygen damage.
 - The pink colour of root nodules is due to leg-haemoglobin.
- **Statement 3 is incorrect:** Lightning does not convert nitrogen into ammonia; instead, it converts atmospheric nitrogen into nitrogen oxides.
 - ✓ Atmospheric Nitrogen Fixation (Lightning):
 - Lightning converts Nitrogen (N_2) into:
 - ◊ Nitric oxide (NO)
 - ◊ Nitrogen dioxide (NO_2)
 - These gases dissolve in rainwater to form nitrates, which enter the soil.
- Types of Nitrogen Fixation:
 - ✓ Biological Fixation:
 - Performed by nitrogen-fixing bacteria and cyanobacteria.
 - ✓ Atmospheric Fixation:
 - Occurs due to lightning, producing nitrogen oxides.
 - ✓ Industrial Fixation:
 - Occurs during fertiliser production, such as the Haber Process.
- Nitrogen Cycle:
 - ✓ Nitrification Bacteria:
 - Nitrosomonas:
 - ◊ Converts Ammonia (NH_3) → Nitrite (NO_2^-)
 - Nitrobacter:
 - ◊ Converts Nitrite (NO_2^-) → Nitrate (NO_3^-)
 - ✓ Denitrification Bacteria:
 - Pseudomonas
 - Thiobacillus
 - These bacteria convert nitrates (NO_3^-) back into nitrogen gas (N_2), returning nitrogen to the atmosphere.

Q 57. Ans. C

Explanation:

- **Statement 1 is correct:** The Dardanelles Strait (Hellespont) forms the southwestern link connecting the Aegean Sea to the Sea of Marmara and serves as an important maritime passage in northwestern Turkey.
 - ✓ Dardanelles Strait (Hellespont):
 - Connects the Aegean Sea (an arm of the Mediterranean Sea) to the Sea of Marmara.
 - Approximately 61 km long.
 - Separates:
 - ◊ European Turkey — Gallipoli Peninsula
 - ◊ Asian Turkey — Anatolia
 - Historically known as the Hellespont.
 - Forms the southwestern segment of the Turkish Straits System.
- **Statement 2 is correct:** The Bosphorus Strait connects the Sea of Marmara to the Black Sea, forming the northeastern segment of the Turkish Straits System.
 - ✓ Bosphorus Strait:
 - Connects the Sea of Marmara to the Black Sea.
 - Approximately 31 km long.
 - At its narrowest point, it is about 700 meters wide, making it one of the narrowest international navigation straits.
 - The city of Istanbul lies on both sides of the Bosphorus, linking Europe and Asia.
- Turkish Straits System:

- ✓ The Turkish Straits form the only maritime link between the Mediterranean Sea and the Black Sea.
- ✓ Complete Route Sequence:
 - Black Sea → Bosphorus → Sea of Marmara → Dardanelles → Aegean Sea → Mediterranean Sea
- Montreux Convention (1936):
 - ✓ The Montreux Convention Regarding the Regime of the Straits (1936) regulates the control and navigation of the Turkish Straits.
 - ✓ Key Provisions:
 - Civilian ships enjoy free passage during peacetime.
 - Turkey has the authority to restrict warship passage during wartime.
- Economic Importance:
 - ✓ The Black Sea region (notably Russia and Ukraine) is a major global exporter of:
 - Sunflower oil
 - Fertilisers
 - Wheat and other grains
 - ✓ All ships moving between the Black Sea and the Mediterranean Sea must pass through the Bosphorus and Dardanelles.
 - ✓ Any disruption in this route significantly affects global food security and fertiliser supply chains.
- ✓ Oyashio Current (Cold Current):
 - A cold ocean current in the North Pacific Ocean.
 - Originates from Arctic waters.
 - Flows southward through the Bering Sea and the Sea of Okhotsk toward Japan.
 - The term “Oyashio” means “Parental Tide” in Japanese.
 - Carries cold, nutrient-rich water, supporting rich marine ecosystems.
- Additional facts:
 - ✓ The warm Kuroshio Current meets the cold Oyashio Current near the eastern coast of Japan, especially around Hokkaido.
 - ✓ Major Effects:
 - Rich Fishing Grounds:
 - ◇ Mixing of warm and cold waters promotes plankton growth.
 - ◇ Creates one of the world’s richest fishing regions.
- Similar example:
 - ✓ A similar rich fishing zone occurs at the Grand Banks of Newfoundland, where the warm Gulf Stream meets the cold Labrador Current.

Q 59. Ans. C

Explanation:

- Q 58. Ans. D**
- Explanation:**
- **Statement 1 is incorrect:** The Kuroshio Current is a warm ocean current, and it originates near the east coast of the Philippines, not directly from the North Equatorial Current.
 - ✓ Kuroshio Current (Warm Current):
 - A warm ocean current in the North Pacific Ocean.
 - Serves as the western boundary current of the North Pacific Subtropical Gyre.
 - Originates near the east coast of the Philippines.
 - Flows northward past Taiwan and Japan.
 - The term “Kuroshio” means “Black Tide” in Japanese.
 - **Statement 2 is incorrect:** The Oyashio is a cold current which originates from Arctic waters, not from the Bering Sea and the Sea of Okhotsk.
 - **Statement 1 is correct:** Fjords are deep glacial valleys that have been submerged under seawater, forming narrow sea inlets characterised by steep sides and great depth.
 - ✓ Fjords:
 - Formed when U-shaped glacial valleys are flooded due to rising sea levels after glacier melting.
 - Characterised by:
 - ◇ Deep, narrow sea inlets
 - ◇ Steep vertical sides
 - ◇ Often, a shallow mouth (sill) is formed by terminal moraine deposits.
 - Common along high-latitude mountainous coasts.
 - ✓ Examples:
 - Norway — classic fjord coastline (e.g., Sognefjord).
 - Southern Chile

- New Zealand — Milford Sound
- Alaska
- **Statement 2 is correct:** Hanging valleys are formed due to differential glacial erosion.
 - ✓ Hanging valleys:
 - When a main glacier erodes its valley deeply, while tributary glaciers erode less deeply.
 - After glacier melting, tributary valleys remain elevated above the main valley.
 - Characteristic Feature:
 - ◇ Waterfalls commonly form where streams descend from the hanging valley into the main valley.
- Glacial landforms classification:
 - ✓ Erosional glacial landforms:
 - Cirque: An armchair-shaped depression formed at the head of a glacier.
 - Horn: Pyramidal peak formed by erosion from multiple sides (example: Matterhorn).
 - Arête: Sharp ridge formed between adjacent glacial valleys.
 - U-shaped valley: Broad valley with steep sides carved by glaciers.
 - Hanging valley: Elevated tributary valley above the main valley.
 - ✓ Depositional glacial landforms:
 - Moraines: Deposits of glacial debris (types: terminal, lateral, medial).
 - Eskers: Long, winding ridges of sand and gravel deposited by glacial meltwater.
 - Drumlins: Spoon-shaped hills indicating the direction of ice movement.
 - Outwash plains: Extensive plains formed by deposits of meltwater streams.
- Depth generally ranges between 120 m and 370 m.
- ✓ Geographical Extremes:
 - Continental shelves are very narrow or absent near subduction zones (example: Chile coast, West Sumatra).
 - The Siberian Shelf (Arctic Ocean) is the widest in the world, extending up to 1,500 km.
- Continental shelves are the richest fishing grounds due to abundant plankton growth.
- Important fishing grounds include the Grand Banks and the Dogger Bank.
- Continental shelves contain about 20% of global oil and gas reserves.
- They also contain placer deposits, including diamonds off South Africa.
- Indian Continental Shelf:
 - ✓ India's Western Continental Shelf is much wider than the Eastern Shelf.
 - ✓ The Western Shelf reaches about 350 km in width near the Narmada–Tapi estuary and the Gulf of Cambay.
 - ✓ The Eastern Continental Shelf has an average width of about 50 km.
 - ✓ Bombay High, India's largest offshore oilfield, lies on the Western Continental Shelf.
 - ✓ It produces more than half of India's crude oil.

Deep oceanic trenches:

- Oceanic trenches are the deepest parts of the ocean floor.
- They form at convergent plate boundaries where subduction occurs.
- Depth Range:
 - ✓ Typically 5,500 m to over 11,000 m.
- Global Examples:
 - ✓ The Mariana Trench is the deepest ocean trench, about 11,034 m deep at Challenger Deep.
 - ✓ Other examples include the Puerto Rico Trench and the Sunda (Java) Trench.
- Significance:
 - ✓ Ocean trenches are major zones of earthquakes and tsunami generation.

Mid-Ocean Ridge:

- A Mid-Ocean Ridge is a long submarine

Q 60. Ans. B

Explanation:

- Continental shelf (submerged continental margin):
 - ✓ The continental shelf is the submerged extension of a continent covered by shallow seas.
 - ✓ Key Dimensions:
 - The average width of the continental shelf is about 80 km.
 - Shelf slope is very gentle, usually 1° or less.

mountain chain formed at divergent plate boundaries.

- It is the longest mountain system on Earth, extending about 65,000 km.
- New oceanic crust is formed here by seafloor spreading.
- Example:
 - ✓ The Mid-Atlantic Ridge is a major mid-ocean ridge.
 - ✓ Iceland is an exposed part of the Mid-Atlantic Ridge above sea level.

Q 61. Ans. B

Explanation:

- Neap Tide:
 - ✓ Neap tides occur when the Earth, Moon, and Sun form a 90° angle.
 - ✓ This alignment is called Quadrature.
 - ✓ Neap tides occur during the First Quarter and Third (Last) Quarter moon phases.
 - ✓ They produce the lowest tidal range.
 - ✓ During Neap tides, high tides are lower and low tides are higher than normal.
 - ✓ Reduced tidal range during Neap tides affects navigation at Syama Prasad Mookerjee Port (Kolkata) on the Hooghly River, limiting the entry of large ships.
- Spring Tide
 - ✓ Spring tides occur when the Sun, Moon, and Earth lie in a straight line.
 - ✓ This alignment is called Syzygy.
 - ✓ Spring tides occur during the New Moon and Full Moon phases.
 - ✓ They produce the highest tidal range, with very high high tides and very low low tides.
- Tidal Bore
 - ✓ A tidal bore is a strong tidal wave moving upstream in a river or estuary.
 - ✓ Examples:
 - Hooghly River (West Bengal)
 - Mahi River (Gulf of Khambhat, Gujarat)
 - ✓ Tidal bores are strongest during Spring tides.
- Perigean Tide
 - ✓ Perigean tides occur when the Moon is closest to Earth (Perigee).
 - ✓ They produce a higher-than-normal tidal range.
 - ✓ When a Perigean tide coincides with a

Spring tide, it produces an extremely large tide called a King Tide or Perigean Spring Tide.

Q 62. Ans. B

Explanation:

- Upwelling:
 - ✓ Upwelling is the rising of deep, cold ocean water to the surface.
 - ✓ It occurs due to Ekman Transport, where surface water moves 90° to the wind direction due to the Coriolis force.
 - ✓ When surface water moves away from the coast, deep water rises to replace it.
 - ✓ Upwelling brings nutrient-rich water (nitrates and phosphates) to the surface.
 - ✓ It leads to phytoplankton growth and supports rich fishing grounds.
 - ✓ Examples:
 - Major upwelling zones occur along the Peru Coast and the California Coast.
 - Arabian Sea Upwelling
 - ◇ Seasonal upwelling occurs during the Southwest Monsoon.
 - ◇ Strong monsoon winds along the Somali and Arabian coasts drive upwelling.
 - Upwelling also occurs along the Southwest Coast of India (Kerala).
 - It supports major fisheries, especially the Indian Oil Sardine and Mackerel.
- Downwelling:
 - ✓ Downwelling is the sinking of surface water to deeper ocean layers.
 - ✓ It occurs when winds push surface water towards the coast.
 - ✓ Downwelling zones have low nutrient supply and generally low fish productivity.
- Convection (thermohaline circulation):
 - ✓ Thermohaline circulation is driven by density differences caused by temperature and salinity.
 - ✓ Cold, salty water sinks in polar regions, forming deep ocean currents.
 - ✓ This circulation forms the Global Ocean Conveyor Belt.
- Tides:
 - ✓ Tides are the periodic rise and fall of the sea level.

- ✓ They are caused by the gravitational pull of the Moon and Sun and Earth's rotation.
- ✓ Tides mainly cause horizontal water movement near coasts.

Q 63. Ans. C

Explanation:

- Hadley Cell
 - ✓ The Hadley Cell operates between the Equator and 30° N/S latitudes.
 - ✓ It is a thermally direct circulation cell.
 - ✓ Hot air rises at the Equator and reaches the tropopause (~18 km).
 - ✓ The air then moves poleward and sinks near 30° N/S, forming subtropical high-pressure belts.
 - ✓ Poleward-moving air is deflected by the Coriolis force, forming the Subtropical Jet Stream near 30° latitude.
 - ✓ Hadley Cell & Monsoon:
 - The Indian Monsoon is linked to the seasonal shifting of the Hadley Cell.
 - During summer, intense heating of the Tibetan Plateau shifts the ITCZ northward into India.
 - This pulls Southeast Trade Winds across the Equator, which deflect to become the Southwest Monsoon winds.
- Ferrel Cell
 - ✓ The Ferrel Cell operates between 30° and 60° N/S latitudes.
 - ✓ It is a thermally indirect circulation cell.
 - ✓ It is mechanically influenced by the Hadley Cell and Polar Cell.
 - ✓ The Ferrel Cell produces the Prevailing Westerlies.
- Polar Cell
 - ✓ The Polar Cell operates between 60° and 90° latitudes.
 - ✓ It is a thermally direct circulation cell.
 - ✓ Cold, dense air sinks at the poles, forming Polar High Pressure.
 - ✓ Surface winds move equatorward as Polar Easterlies.
 - ✓ At about 60° latitude, Polar Easterlies meet Westerlies, forming the Sub-Polar Low Pressure Belt.
- Walker Cell (Walker Circulation) — Very Important Distinction

- ✓ The Walker Cell operates along the Equator in an East–West (zonal) direction.
- ✓ Other circulation cells (Hadley, Ferrel, Polar) operate in a north–south (meridional) direction.
- ✓ The Walker Cell is driven by Sea Surface Temperature (SST) differences across the Pacific Ocean.
- ✓ Normally, air rises over the Western Pacific (Indonesia–Australia) and sinks over the Eastern Pacific (Peru region).

● ENSO:

- ✓ El Niño (Weak Walker Cell)
 - Walker circulation weakens or reverses.
 - High pressure develops over India, reducing rainfall.
 - This often causes weak monsoon and drought conditions, including in Bihar.
- ✓ La Niña (Strong Walker Cell)
 - Walker's circulation strengthens.
 - More moisture reaches India.
 - This causes strong monsoon rainfall.
 - In North Bihar, strong monsoons often lead to flooding, especially in the Kosi and Gandak river basins.

Q 64. Ans. B

Explanation:

- Statement 1 is incorrect:
 - ✓ The Most Favoured Nation (MFN) principle is not limited to bilateral trade.
 - ✓ It is a core rule of multilateral trade under the World Trade Organization (WTO).
 - ✓ MFN Principle meaning:
 - If a country grants a trade benefit (such as lower tariffs) to one WTO member, it must extend the same benefit to all WTO members.
 - This ensures non-discrimination in international trade.
- Statement 2 is correct:
 - ✓ Trade diversion commonly occurs under bilateral or regional trade agreements.
 - ✓ When tariffs are reduced only for selected partners, countries may shift imports to less efficient producers, simply because they receive tariff advantages.
- Additional facts:
 - ✓ Multilateral Trade System:

- Operates under uniform rules for many countries.
- Promotes trade creation, where goods are sourced from the most efficient producers globally.
- ✓ Bilateral/Regional Agreements:
 - Involve only two or a few countries.
 - May lead to trade diversion, where cheaper global suppliers are replaced by favored but less efficient partners.
- ✓ Exceptions to MFN Rule:
 - Allowed under WTO in specific cases such as:
 - Free Trade Agreements (FTAs)
 - Customs Unions
 - Special treatment for developing countries
 - National security exceptions

Q 65. Ans. B

Explanation:

- Statement 1 is incorrect:
 - ✓ Plantation agriculture is not polyculture.
 - ✓ It is a monoculture system, meaning a single crop is grown over large areas for commercial export.
 - ✓ Typical plantation crops include:
 - Tea
 - Coffee
 - Rubber
 - Sugarcane
 - Cocoa
 - Banana
 - ✓ Plantation farming often involves large estates, heavy capital investment, and export-oriented production.
- Statement 2 is correct:
 - ✓ Mediterranean agriculture is a specialized commercial farming system.
 - ✓ It produces high-value crops suited to its climate of dry summers and wet winters.
 - ✓ Typical Mediterranean crops include:
 - Grapes (viticulture)
 - Olives
 - Citrus fruits
 - Figs
- Additional facts:
 - ✓ Plantation Agriculture Features:
 - Large landholdings
 - Single cash crop cultivation

- Dependence on hired labor
- Export-oriented production
- Common in tropical regions

- **Fazendas:** Large coffee plantations in Brazil..

Q 66. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ The Gaza Strip is a narrow territory about 41 km long.
 - ✓ It shares borders with:
 - ✓ Israel to the east and north
 - ✓ Egypt (Sinai Peninsula) to the southwest (about 11 km border)
- Statement 2 is correct:
 - ✓ The western boundary of the Gaza Strip lies along the Mediterranean Sea, with a coastline of roughly 40 km.
- Additional facts:
 - ✓ Gaza Strip:
 - Coastal territory
 - Borders Israel, Egypt, and Mediterranean Sea
 - ✓ West Bank:
 - Landlocked territory
 - Borders Israel on three sides
 - Borders Jordan to the east along the Jordan River and Dead Sea
 - ✓ Rafah Crossing:
 - Located on the Egypt–Gaza border
 - Main crossing not directly controlled by Israel
 - ✓ Erez Crossing:
 - Located on the northern Gaza–Israel border
 - Used mainly for movement of people
 - ✓ Kerem Shalom Crossing:
 - Located near the tri-junction of Israel, Gaza, and Egypt
 - Primarily used for goods and supplies
 - ✓ Philadelphi Corridor:
 - Narrow strip along the Gaza–Egypt border
 - Important for controlling tunnel smuggling routes

Q 67. Ans. C

Explanation:

- Statement 1 is incorrect:

- ✓ Nagorno-Karabakh is not related to the Israel–Iran conflict.
- ✓ It is a disputed mountainous region between Armenia and Azerbaijan, internationally recognized as part of Azerbaijan.
- Statement 2 is incorrect:
 - ✓ Armenia never officially recognized Nagorno-Karabakh as an independent country.
 - ✓ Although Armenia supported ethnic Armenians there, formal recognition was deliberately avoided to prevent wider international conflict.
- Additional facts:
 - ✓ Nagorno-Karabakh:
 - Located in the South Caucasus region.
 - Internationally recognized as Azerbaijan's territory.
 - Historically populated by ethnic Armenians, leading to prolonged conflict.
 - ✓ Recent Developments:
 - 2020 War:
 - ◇ Azerbaijan regained significant territory using advanced drone warfare.
 - ◇ Turkey provided strong support to Azerbaijan.
 - 2023 Military Operation:
 - ◇ Azerbaijan launched a rapid offensive.
 - ◇ Armenian forces surrendered.
 - ◇ The self-declared Republic of Artsakh dissolved by January 1, 2024.
 - Lachin Corridor:
 - ◇ Previously the only land link between Armenia and Nagorno-Karabakh.
 - ◇ Its blockade played a major role in weakening Armenian resistance before the 2023 takeover.

Q 68. Ans. C

Explanation:

- Estonia: About 59° N
- Latvia: About 57° N
- Denmark: About 56° N
- Lithuania: About 55° N
- Additional facts:
 - ✓ Baltic States Identity:
 - Estonia, Latvia, and Lithuania are

collectively called the Baltic States.

- All three were formerly part of the Soviet Union and regained independence in 1990–1991.
- They are now members of NATO and the European Union.

● Denmark's strategic importance:

- ✓ Denmark controls the Danish Straits, which connect the Baltic Sea to the North Sea.
- ✓ These waterways are critical for naval and commercial movement from ports such as St. Petersburg and Kaliningrad.

Q 69. Ans. D

Explanation:

- Africa currently has the highest population growth rate in the world, averaging about 2.4 percent annually.
- This is mainly due to high birth rates along with declining death rates across much of the continent.
- Asia:
 - ✓ Asia has the largest population, but growth rates are declining rapidly.
 - ✓ Countries like China now show negative population growth, and India's fertility rate has fallen below replacement level (2.1).
- North America:
 - ✓ Population growth is low naturally.
 - ✓ Most growth comes from immigration, not high birth rates.
- South America:
 - ✓ Many countries have low fertility rates and aging populations, indicating advanced stages of demographic transition.
- Population trend:
 - ✓ A large share of global population growth up to 2050 is expected to occur in African countries such as:
 - Nigeria
 - Ethiopia
 - Democratic Republic of the Congo
 - Tanzania
 - Egypt

Q 70. Ans. D

Explanation:

- Statement 1 is correct:

- ✓ Footloose industries are generally clean or non-polluting industries.
- ✓ They differ from heavy industries, which often produce smoke, waste, and pollution.
- Statement 2 is correct:
 - ✓ Footloose industries require a small but highly skilled labor force.
 - ✓ They depend more on technical expertise than on large numbers of workers.
- Statement 3 is correct:
 - ✓ Diamond cutting and semiconductor manufacturing are classic examples.
 - ✓ Their raw materials are lightweight but highly valuable, so transport cost is not a major factor.
- Core Characteristics of Footloose Industries:
 - ✓ Low transport cost sensitivity
 - ✓ Use lightweight, high-value raw materials
 - ✓ Require skilled labor
 - ✓ Produce minimal pollution
 - ✓ Can locate almost anywhere
 - ✓ Often rely on component parts rather than bulky raw materials
- Location factors:
 - ✓ Since raw material location is not critical, these industries prefer:
 - ✓ Good transport connectivity (especially airports)
 - ✓ Availability of skilled labor
 - ✓ Presence of research centers and universities
 - ✓ High-quality urban infrastructure
- Examples:
 - ✓ Surat (India):
 - Global hub for diamond cutting and polishing.
 - ✓ Bengaluru (India):
 - Major center for IT and software industries, the purest example of footloose activity.
 - ✓ Silicon Valley (USA):
 - World-famous hub for semiconductor and technology industries.

Q 71. Ans. C

Explanation:

- **Statement 1 is correct:** Quaternary activities involve some of the following: the collection, production and of information or even the production of information. Quaternary activities

centre around research, development and may be seen as an advanced form of services involving specialised knowledge and technical skills.

- **Statement 2 is correct:** Quinary activities involve high-level services like creating and interpreting ideas, data analysis, and evaluating new technologies, performed by highly skilled professionals such as executives, scientists, and consultants in advanced economies.

Q 72. Ans. B

Explanation:

The Trans-Canadian railway line is considered the economic artery of Canada due to the following reasons:

- This railway line covers a distance of 7,050 km running from Halifax in the East to Vancouver on the Pacific coast.
- It connects important cities of Montreal, Ottawa, Winnipeg and Calgary.
- It connects the Quebec-Montreal industrial region with the wheat belt of the Prairie region, thus gaining economic significance.
- This line also connects the coniferous forest region in the North to Quebec-Montreal and the Prairies. All these regions have become complementary to each other.

Q 73. Ans. D

Explanation:

- **Physiological density:**
 - ✓ Formula: $\text{Total Population} \div \text{Net Cultivated Area}$
 - ✓ Meaning: Measures the pressure of total population on food-producing land.
- **Agricultural density:**
 - ✓ Formula: $\text{Agricultural Population} \div \text{Net Cultivable Area}$
 - ✓ Meaning: Measures the number of farmers per unit of cultivable land (labor pressure on land).
- **Arithmetic density:**
 - ✓ Formula: $\text{Total Population} \div \text{Total Land Area}$
- **Additional facts:**
 - ✓ Example:
 - Egypt has low Arithmetic Density because much land is desert.

- But it has very high Physiological Density, as most people depend on the narrow Nile Valley farmland.
- ✓ Developed vs Developing Pattern:
 - Developed Countries (USA, Canada):
 - ◇ Low Agricultural Density
 - ◇ Highly mechanized farming
 - ◇ Few farmers manage large land areas
- ✓ Developing Countries (India, Bangladesh):
 - High Agricultural Density
 - Labor-intensive agriculture
 - Small fragmented landholdings
- ✓ India's Situation:
 - India has high Physiological and Agricultural Density, indicating:
 - ◇ Heavy pressure on cultivable land
 - ◇ Disguised unemployment in agriculture
 - ◇ Need for modernisation and diversification.
- ✓ More suitable for cattle ranching than orchard farming.
- Additional facts:
 - ✓ Geographical distribution:
 - Mediterranean climate occurs between 30° and 45° latitudes on western continental margins.
 - ✓ Major Global Regions:
 - California region (USA)
 - Central Chile (South America)
 - Western Cape region (South Africa)
 - Southern and Western Australia
 - Mediterranean Basin (Southern Europe and North Africa)
 - ✓ Seasonal Wind Mechanism:
 - Summer: Dominated by dry trade winds, causing drought conditions.
 - Winter: Influenced by westerlies, bringing rainfall.

Q 74. Ans. C

Explanation:

- The **Mediterranean climate** is known as the “Orchard of the World” and is globally famous for viticulture (grape cultivation) and orchard farming.
 - ✓ Key Climatic Features:
 - Hot, dry summers
 - Mild, wet winters
 - Moderate rainfall concentrated in winter
 - Bright sunshine during growing season
 - These conditions create controlled water stress, ideal for producing high-quality grapes, olives, citrus fruits, and figs.
- Monsoon Climate:
 - ✓ Characterized by heavy summer rainfall.
 - ✓ Excess moisture damages grapes and promotes fungal diseases.
 - ✓ Main crops: Rice, sugarcane, jute.
- Steppe Climate:
 - ✓ Semi-arid grasslands with low rainfall.
 - ✓ Better suited for commercial grain farming, especially wheat.
 - ✓ Known as “granaries of the world.”
- Savanna Climate:
 - ✓ Tropical wet-and-dry climate.
 - ✓ Dominated by grasslands and livestock grazing.

Q 75. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ Stage 2 of the Demographic Transition Model (DTM) is known as the Population Explosion stage.
 - ✓ Population grows rapidly because death rates fall sharply while birth rates remain high.
- Statement 2 is correct:
 - ✓ Stage 2 is defined by a rapid decline in death rate due to improvements in:
 - Public health
 - Sanitation
 - Medical facilities and vaccination
 - Food supply and nutrition
 - ✓ However, birth rates remain high because social and cultural practices change slowly.

Q 76. Ans. B

Explanation:

- Statement 1 is incorrect:
 - ✓ The Danube River is not the third longest river in Europe.
 - ✓ It is the second longest river in Europe, after the Volga River.
- Statement 2 is correct:
 - ✓ The Rhine River originates in the Swiss

Alps, specifically in the Graubünden region of Switzerland.

- ✓ It forms from the confluence of the Vorderrhein and Hinterrhein headstreams.
- Additional facts:
 - ✓ Volga River:
 - Longest river in Europe.
 - Flows entirely through Russia.
 - Empties into the Caspian Sea.
 - ✓ Danube River:
 - Second longest river in Europe.
 - Originates in the Black Forest (Germany).
 - Empties into the Black Sea.
 - ✓ Ural River:
 - Often considered the third longest in Europe.
 - Forms part of the boundary between Europe and Asia.

Q 77. Ans. D

Explanation:

- Formal Region (Uniform Region):
 - ✓ Area where a specific characteristic is uniform throughout.
 - ✓ Boundaries are clearly defined.
 - ✓ Typical examples:
 - Political regions: State of Bihar, Country of France
 - Physical regions: Sahara Desert
 - Cultural regions: Areas sharing a common language
- Functional Region (Nodal Region):
 - ✓ Organized around a central node or focal point.
 - ✓ Interaction decreases as distance from the center increases (distance decay).
 - ✓ Typical examples:
 - Delhi National Capital Region (NCR)
 - Market areas around cities
 - Transport or communication networks
- Vernacular Region (Perceptual Region):
 - ✓ Defined by people's perceptions, not strict boundaries.
 - ✓ Borders are not precisely fixed.
 - ✓ Typical examples:
 - Middle East
 - Hindi Heartland
 - American South

Q 78. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ The Panama Canal is about 82 km long.
 - ✓ It connects the Atlantic Ocean (via Caribbean Sea) with the Pacific Ocean, saving ships from the long route around Cape Horn.
- Statement 2 is incorrect:
 - ✓ The Panama Canal is not controlled by the United States.
 - ✓ Under the Torrijos–Carter Treaties (1977), full control was transferred to Panama on December 31, 1999.
 - ✓ It is now managed by the Panama Canal Authority.
- Statement 3 is correct:
 - ✓ The canal is built across the Isthmus of Panama, the narrow land bridge connecting North America and South America.

Q 79. Ans. A

Explanation:

- Golan Heights:
 - ✓ Located between Syria and Israel.
 - ✓ Captured by Israel from Syria during the 1967 Six-Day War.
 - ✓ Formally annexed by Israel in 1981, but the annexation is not recognised under international law (UN Security Council Resolution 497).
- Key Strategic Importance:
 - ✓ Military Advantage:
 - The Golan Heights is a high, rocky plateau, giving a strong strategic vantage point.
 - It overlooks northern Israel and parts of southern Syria.
 - ✓ Water Security:
 - Controls access to the Sea of Galilee, an important freshwater source for Israel.
 - ✓ Proximity to Damascus:
 - Located roughly 60 km from Damascus, making it militarily sensitive for Syria.
- Additional facts:
 - ✓ 1973 Yom Kippur War:
 - Syria attempted to retake the Golan Heights, but Israel retained control.

- ✓ UNDOF Presence:
 - The United Nations Disengagement Observer Force (UNDOF) maintains a buffer zone between Israeli and Syrian forces.
- ✓ Population Feature:
 - The region includes a significant Druze population, many of whom retain ties to Syria.

- It is an exclave of Russia (separated from mainland Russia but touching the sea).
- ✓ Baltic Region Strategy:
 - After NATO expansion and Nordic entries, the Baltic Sea region has gained major geopolitical importance.
- ✓ Suwałki Gap → Poland–Lithuania → Between Kaliningrad and Belarus.

Q 80. Ans. A

Explanation:

- Suwałki Gap (Suwałki Corridor):
 - ✓ Located along the border between Poland and Lithuania.
 - ✓ Length: About 65 km wide.
 - ✓ Strategic role: Separates Russia's Kaliningrad exclave from Belarus, a close Russian ally.
- Key strategic importance:
 - ✓ NATO Vulnerability:
 - The Suwałki Gap is the only land corridor linking the Baltic States (Estonia, Latvia, Lithuania) to the rest of NATO-controlled Europe.
 - If blocked, Baltic States could be isolated from NATO land support.
 - ✓ Russia–Belarus Factor:
 - Belarus maintains strong military alignment with Russia.
 - Joint military presence increases the strategic risk around this corridor.
 - ✓ Kaliningrad Significance:
 - Kaliningrad hosts advanced Russian missile and air-defense systems.
 - Acts as a forward military base in the Baltic region.
 - ✓ Germany–France:
 - Major boundary reference: Rhine River and historic Maginot Line fortifications.
 - ✓ Spain–Portugal:
 - Share the Iberian Peninsula.
 - Border called La Raya, among the oldest continuous borders in Europe.
 - ✓ Norway–Sweden:
 - Separated mainly by the Scandinavian Mountains (Scandes).
- Additional facts:
 - ✓ Kaliningrad:

Q 81. Ans. C

Explanation:

- Big Inch Pipeline:
 - ✓ Constructed in the United States (1942–1944) during World War II.
 - ✓ Primary purpose: Transport crude oil from East Texas oil fields to East Coast refineries (Philadelphia and New Jersey).
- Little Big Inch Pipeline:
 - ✓ Built alongside the Big Inch.
 - ✓ Used to transport refined petroleum products such as gasoline, kerosene, and heating oil.
- Key Concept Clarification:
 - ✓ Crude Oil: Transported through the Big Inch pipeline.
 - ✓ Refined Petroleum Products: Transported through the Little Big Inch pipeline.
 - ✓ Natural Gas: Transported through these pipelines only after World War II, when they were converted.
- Modern Pipeline References:
 - ✓ Nord Stream Pipelines: Natural gas pipelines linking Russia and Germany.
 - ✓ Druzhba Pipeline: One of the world's longest crude oil pipelines, supplying Europe.
 - ✓ Trans-Adriatic Pipeline (TAP): Carries natural gas from the Caspian region to Europe.

Q 82. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ Hunting is a high-risk, high-reward activity.
 - ✓ Food returns are uncertain; hunters may obtain large prey or return empty-handed.
- Statement 2 is correct:

- ✓ Gathering provides a stable and reliable food supply.
- ✓ Roots, fruits, nuts, and tubers offer consistent daily calories, forming the subsistence base of hunter-gatherer societies.
- Additional facts:
 - ✓ Dual subsistence strategy:
 - Hunter-gatherer societies survived by combining reliable gathering with occasional hunting success, reducing food insecurity.
 - ✓ Environmental exception:
 - In Arctic regions, such as among the Inuit, hunting and fishing dominate because plant resources are scarce.
 - ✓ Examples:
 - San (Bushmen): Kalahari Desert region.
 - Hadza: Tanzania region of East Africa.
 - Pygmies: Congo Basin forest region.
 - Sentinelese and Jarawas: Andaman Islands region of India.
 - ✓ Related Concept:
 - Optimal Foraging Theory states that people choose food sources that maximise energy gain while minimising effort and time.
- **Statement 2 is correct:** Both gorges and canyons are primarily formed by river erosion, where flowing water cuts downward into the land over long periods, often aided by tectonic uplift.
 - ✓ Formation mechanism:
 - Formed mainly through vertical river erosion (downcutting).
 - Tectonic uplift (rejuvenation) increases river gradient and velocity.
 - This accelerates deep erosion, leading to the formation of deep valleys.
 - Examples:
 - ✓ Major Himalayan Gorges (Antecedent Rivers):
 - Formed when rivers maintained their course during the Himalayan uplift.
 - Indus Gorge: near Nanga Parbat; among the deepest gorges (~5,200 m).
 - Brahmaputra (Dihang) Gorge: near Namcha Barwa.
 - Sutlej Gorge: near Shipki La.
 - ✓ Major canyons:
 - Grand Canyon (USA):
 - ◇ Formed by the Colorado River.
 - ◇ Considered the most famous canyon in the world.
 - Gandikota Canyon (Andhra Pradesh):
 - ◇ Formed by the Penna River.
 - ◇ Known as the “Grand Canyon of India.”
 - The Kosi River flows through the Chatra Gorge in the Mahabharat Range.
 - In West Champaran, rivers such as the Gandak and Masan form steep V-shaped valleys in the Someshwar Range, near Valmikinagar.

Q 83. Ans. C

Explanation:

- **Statement 1 is correct:** A gorge is typically narrower and steeper than a canyon, often having very steep or nearly vertical walls due to intense vertical river erosion.
 - ✓ Gorge:
 - A deep, narrow valley with very steep or vertical sides.
 - Usually almost equal in width at the top and bottom.
 - Commonly formed in hard, resistant rocks, such as granite.
 - Characterised by intense vertical downcutting by rivers.
 - ✓ Canyon:
 - A deep valley with step-like side slopes.
 - Generally wider at the top than at the bottom.
 - Commonly formed in horizontally bedded sedimentary rocks, where rock layers erode at different rates.

Q 84. Ans. D

Explanation:

- Assertion (A) is False:
 - ✓ El Niño is not associated with warming along the western coast of North America.
 - ✓ It is defined by abnormal warming of sea surface temperatures along the western coast of South America, especially near Peru and Ecuador.
- Reason (R) is True:
 - ✓ During El Niño, the normally strong easterly trade winds weaken or reverse.

- ✓ As a result, warm surface water shifts eastward toward South America, suppressing the usual cold-water upwelling along the coast.
- Key Concepts:
 - ✓ Normal Conditions (Walker Circulation):
 - ✓ Trade winds push warm water westward toward Indonesia and Australia.
 - ✓ Cold, nutrient-rich water rises (upwelling) along the Peru coast.
- El Niño Conditions:
 - ✓ Trade winds weaken.
 - ✓ Warm water shifts eastward toward South America.
 - ✓ Upwelling is suppressed, reducing marine productivity.
- Key facts:
 - ✓ Indian Monsoon:
 - El Niño: Generally leads to weaker monsoon and possible drought in India.
 - La Niña: Usually associated with stronger monsoon rainfall.
 - ✓ Fishing impact:
 - Suppression of upwelling reduces nutrient supply, causing decline in anchovy fish populations off Peru.

Q 85. Ans. A

Explanation:

- The Coriolis force arises due to the Earth's rotation, causing moving objects like winds and ocean currents to be deflected from their straight path.
- Its magnitude increases with latitude.
- It is zero at the Equator (0°) and maximum at the Poles (90°).
- Therefore, the Reason correctly explains the Assertion.
- Additional facts:
 - ✓ Ferrel's Law:
 - Northern Hemisphere: Deflection toward the right.
 - Southern Hemisphere: Deflection toward the left.
 - ✓ Cyclone Formation Rule:
 - Tropical cyclones do not form near the Equator because the Coriolis force is too weak to produce rotation.
 - ✓ Geostrophic Wind:

- At high altitudes, when Pressure Gradient Force balances Coriolis Force, winds flow parallel to isobars.
- ✓ Jet Streams:
 - Strong Coriolis force at higher latitudes helps maintain fast-moving jet streams in mid-latitudes and polar regions.

Q 86. Ans. C

Explanation:

- Fringing Reef:
 - ✓ Reef attached directly to coast with shallow or narrow lagoon.
- Barrier Reef:
 - ✓ Reef separated from land by a wide and deep lagoon, often navigable.
- Atoll:
 - ✓ Ring-shaped coral reef enclosing a central lagoon, usually without a central island.
- Patch Reef:
 - ✓ Small, isolated reef found on continental shelves or inside lagoons.
- Darwin's Subsidence Theory:
 - ✓ Fringing Reef: Forms first around a newly formed volcanic island.
 - ✓ Barrier Reef: Develops as the island slowly sinks, creating a wider lagoon.
 - ✓ Atoll: Forms when the island completely submerges, leaving only a coral ring.
- Key facts:
 - ✓ Global examples:
 - Great Barrier Reef (Australia): Largest coral reef system in the world.
 - Lakshadweep Islands (India): Classic example of atoll reefs.
 - Andaman and Nicobar Islands (India): Dominated by fringing reefs.
 - ✓ Coral bleaching:
 - Occurs when rising sea temperature forces corals to expel zooxanthellae algae, turning reefs white and weakening them.
 - ✓ Ideal Conditions for Coral Growth:
 - Warm water: About 20°C to 30°C.
 - Shallow depth: Usually less than 50 meters.
 - Clear, saline water with abundant sunlight.

Q 87. Ans. B

Explanation:

- Mass Movements (Mass Wasting)
 - ✓ Mass movement is the downslope movement of soil or rock under the force of gravity.
 - ✓ Movement occurs when shear stress exceeds shear strength of slope material.
 - ✓ Water increases movement by adding weight and reducing cohesion through increased pore water pressure. Examples of mass movement:
 - Mudflow:
 - ◇ Rapid mass movement due to heavy water saturation of soil, reducing internal friction and enabling easy particle movement.
 - ◇ Mudflows generally move along stream channels.
 - ◇ **Lahar** is a volcanic mudflow formed when volcanic ash mixes with snow, ice, or crater lake water.
 - Soil Creep:
 - ◇ Soil creep is the **slow** and continuous movement of soil down a slope.
 - ◇ It is mainly caused by freeze, thaw cycles and wet-dry expansion and contraction of soil.
 - Rock Fall:
 - ◇ Rock fall is the free fall of rock fragments from steep cliffs.
 - ◇ Continuous rock falls lead to the formation of **talus slopes**, also called **scree**, at the base of cliffs.
 - ✓ Exfoliation (weathering, not mass movement):
 - Exfoliation is a type of mechanical weathering in which rock **layers peel off** in sheets.
 - It is mainly caused by unloading and thermal expansion and contraction.
 - Exfoliation produces exfoliation domes, such as Half Dome in Yosemite National Park.

Q 88. Ans. C

Explanation:

- **Statement 1 is correct:** A tombolo is formed

through the deposition of sediments carried by longshore drift which moves materials along the coast.

- ✓ These materials accumulate where there is an indentation such as a bay or near an island.
- **Statement 2 is correct:** A tombolo is a special type of bar that connects an island to the mainland through a ridge of sand or shingle.
 - ✓ It represents a coastal depositional landform formed by marine processes.
 - ✓ Example:
 - A natural tombolo connects Ross Island and Smith Island (North Andaman, Diglipur).
- Spit (Sand Spit)
 - ✓ A spit is a ridge of sand attached to the mainland at one end, with the other end extending into the sea or bay.
 - ✓ A spit does not connect an island to the mainland.
 - ✓ Example:
 - The sand spit enclosing Chilika Lake (Odisha) restricts its connection with the Bay of Bengal.
- Offshore Bar
 - ✓ An offshore bar is a sand ridge formed parallel to the coastline.
 - ✓ It remains detached from the mainland.
- Barrier Island
 - ✓ A barrier island is a large offshore bar that remains permanently above high tide.
 - ✓ Barrier islands may support vegetation and settlements.
 - ✓ Example:
 - Sriharikota (Andhra Pradesh) is a barrier island separating Pulicat Lake from the Bay of Bengal.

Q 89. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ Retailing is the final stage in the supply chain.
 - ✓ It involves selling goods directly to consumers for personal or household use.
- Statement 2 is correct:
 - ✓ Wholesaling involves buying goods in bulk from manufacturers.

- ✓ Goods are sold to retailers or commercial users, not directly to consumers.
- ✓ Wholesalers usually operate through large warehouses, not retail shops.
- Key Facts:
 - ✓ Economic sector classification:
 - Primary Sector: Extraction of natural resources (agriculture, mining, fishing).
 - Secondary Sector: Manufacturing and construction.
 - Tertiary Sector: Services such as trade, transport, banking, retail.
 - Quaternary Sector: Knowledge-based services (IT, research, education).
 - Quinary Sector: High-level decision making (top executives, policy makers).
 - ✓ FDI in Retail:
 - Single-brand retail: 100 percent FDI allowed under automatic route.
 - Multi-brand retail: 51 percent FDI allowed with government approval.
 - ✓ E-Commerce Role:
 - Platforms like Amazon and Flipkart combine wholesale storage and retail sales, changing traditional supply chains.

Q 90. Ans. D

Explanation:

- Durand Line:
 - ✓ Boundary between Pakistan and Afghanistan.
 - ✓ Length: About 2,640 km.
 - ✓ Established in 1893.
 - ✓ Agreement between Sir Mortimer Durand (British India) and Amir Abdur Rahman Khan (Afghanistan).
 - ✓ Purpose: Define British India's sphere of influence and act as buffer during the Great Game.
- India–China:
 - ✓ The boundary includes McMahon Line (1914) in the eastern sector.
 - ✓ Present operational boundary called Line of Actual Control (LAC).
- Iran–Iraq:
 - ✓ Major boundary dispute linked to Shatt al-Arab waterway.
 - ✓ Defined under 1975 Algiers Agreement.

- Key facts:
 - ✓ Durand Line divides Pashtun and Baloch tribal regions, making it politically sensitive.
 - ✓ Afghanistan has not officially recognized the Durand Line as an international boundary.

Q 91. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ Kaliningrad is a Russian exclave separated from mainland Russia.
 - ✓ Formerly Königsberg, capital of East Prussia.
 - ✓ Captured by the Soviet Union in 1945 after World War II and renamed Kaliningrad.
 - ✓ Birthplace of philosopher Immanuel Kant.
- Statement 2 is correct:
 - ✓ Located on the Baltic Sea coast.
 - ✓ Borders Lithuania (north and east) and Poland (south).
 - ✓ Strategically placed between NATO and EU countries.
- Key facts:
 - ✓ Exclave vs Enclave:
 - ✓ Exclave: Territory separated from the main country (Kaliningrad of Russia).
 - ✓ Enclave: Territory surrounded by one country only.
 - Examples: Lesotho inside South Africa, Vatican inside Italy.
 - ✓ Suwalki Gap:
 - About 65 km wide land corridor between Poland and Lithuania.
 - It connects Belarus to the Kaliningrad region.
 - Strategically important for NATO and Russia.
 - ✓ Baltic Sea:
 - After Finland and Sweden joined NATO, most of the Baltic coastline became NATO-controlled.
 - Kaliningrad hosts Russia's Baltic Fleet.
 - Important as an ice-free port throughout the year.

Q 92. Ans. A

Explanation:

- Statement 1 is correct:

- ✓ Brain Drain means large-scale migration of highly skilled and educated people from developing to developed countries.
- ✓ Includes professionals like doctors, engineers, scientists seeking better salary, facilities, and lifestyle.
- Statement 2 is incorrect:
 - ✓ Migration of skilled people causes economic loss, not benefit, to the home country.
 - ✓ Developing countries invest heavily in education, and migration leads to loss of skilled manpower, innovation, and tax revenue.
- Key facts:
 - ✓ Push and Pull Factors:
 - Push factors: Low wages, poor infrastructure, limited research facilities.
 - Pull factors: Better opportunities, high living standards, advanced technology.
 - ✓ Reverse Brain Drain:
 - Return of skilled professionals to their home country.
 - Supports startup growth in cities like Bengaluru and Gurugram.
 - ✓ Remittance Impact:
 - Migrants send money back home (remittances).
 - India is the largest global recipient of remittances, exceeding 125 billion dollars annually.
- ✓ Most southern among the four cities.
- ✓ Located on Judean Mountains, west of Jordan Rift Valley.
- Key facts:
 - ✓ Levant Region:
 - Includes Syria, Lebanon, Jordan, Israel, Palestine, Cyprus.
 - Located along the Eastern Mediterranean coast.
 - ✓ Jordan Rift Valley:
 - Major part of Great Rift Valley system.
 - Contains Dead Sea, the lowest land point on Earth.
 - ✓ Important Regional Features:
 - Golan Heights: Strategic plateau between Damascus and Sea of Galilee.
 - Captured by Israel from Syria in 1967.
 - ✓ West to East Water Body Order (Middle East):
 - Mediterranean Sea → Sea of Galilee → Dead Sea → Red Sea → Persian Gulf

Q 93. Ans. A

Explanation:

- Beirut → Damascus → Amman → Jerusalem
- Beirut (Lebanon):
 - ✓ Latitude: About 33.89° N
 - ✓ Most northern among the four cities.
 - ✓ Located on the Mediterranean coast.
- Damascus (Syria):
 - ✓ Latitude: About 33.51° N
 - ✓ Located in southwestern Syria, not far north despite Syria's northern extent.
- Amman (Jordan):
 - ✓ Latitude: About 31.95° N
 - ✓ Situated on Transjordan Plateau, east of Jordan Rift Valley.
- Jerusalem (Israel/Palestine):
 - ✓ Latitude: About 31.77° N
- Mali:
 - ✓ Landlocked country in West Africa.
 - ✓ Located in the Sahel region south of the Sahara Desert.
- Niger:
 - ✓ Landlocked country in West Africa.
 - ✓ Part of the Sahel zone, prone to drought and desertification.
- Burkina Faso:
 - ✓ Landlocked country in West Africa.
 - ✓ Also located in the Sahel belt.
- Senegal:
 - ✓ Not landlocked.
 - ✓ Has a long Atlantic Ocean coastline.
 - ✓ Capital Dakar is a major port and lies on Cap-Vert Peninsula, the westernmost point of mainland Africa.
- Key facts:
 - ✓ Alliance of Sahel States (AES):
 - ✓ Formed by Mali, Niger, Burkina Faso after recent military coups.
 - ✓ These countries withdrew from ECOWAS and formed a mutual defense alliance.
- Sahel region:
 - ✓ Semi-arid belt between Sahara Desert (north) and Savanna (south).

- ✓ Highly vulnerable to drought and desertification.

● Great Green Wall Project:

- ✓ African Union initiative to control desert expansion.
- ✓ Planned 8,000 km vegetation belt from Senegal to Djibouti.

● Other major landlocked African countries:

- ✓ Chad, Central African Republic, South Sudan, Ethiopia, Uganda, Rwanda, Burundi, Zambia, Zimbabwe, Botswana.

Q 95. Ans. A

Explanation:

● Statement 1 is correct:

- ✓ Circular settlements develop around a central feature.
- ✓ Central features: lake, tank, village green, livestock enclosure.

● Statement 2 is correct:

- ✓ Star-shaped settlements form at nodal points where several roads meet.
- ✓ Houses spread along radiating roads, creating a star-like pattern.

● Statement 3 is correct:

- ✓ T-shaped settlements form at tri-junctions.
- ✓ Develop where a minor road meets a major road.
- ✓ Houses extend along three road arms.

● Statement 4 is incorrect:

- ✓ Cruciform settlements develop at four-way road intersections, not along river banks.
- ✓ Settlements along river banks are typically linear settlements.

● Key facts:

✓ Other Settlement Patterns:

- Linear Settlement: Develops along river banks, roads, railways, or coasts.
- Rectangular Settlement: Forms in plains, roads meet at right angles (grid pattern).
- Double Village: Develops on both sides of a river, usually near bridges or ferries.

✓ Settlement Distribution Types:

- Compact (Nucleated): Houses built close together, common in fertile plains like Gangetic plains.
- Dispersed (Isolated): Houses scattered, common in mountainous or forested regions.

Q 96. Ans. A

Explanation:

● Amazon River:

- ✓ Located in South America.
- ✓ Largest river in the world by water discharge.
- ✓ Has the largest drainage basin globally.
- ✓ Carries more water than the next seven largest rivers combined.

● Congo River:

- ✓ Located in Africa.
- ✓ The second-largest river in the world by discharge volume.
- ✓ The second-longest river in Africa after the Nile.
- ✓ Only major river to cross the Equator twice.
- ✓ The deepest river in the world.

● Yangtze River:

- ✓ Located in Asia (China).
- ✓ Longest river in Asia and third longest in the world.
- ✓ Longest river flowing entirely within one country.
- ✓ Three Gorges Dam is built on this river.

● Volga River:

- ✓ Located in Europe (Russia).
- ✓ Longest river in Europe.
- ✓ Flows entirely within Russia.
- ✓ Drains into the Caspian Sea, the world's largest inland lake.

● Key facts:

✓ Unique River Crossings:

- Congo River: Crosses the Equator twice.
- Mahi River (India): Crosses the Tropic of Cancer twice.

✓ Important Indian River Facts:

- Godavari: Longest river in Peninsular India, called Dakshin Ganga.

✓ Rivers Called "Sorrow":

- Kosi: Sorrow of Bihar.
- Damodar: Sorrow of Bengal.
- Mahanadi: Historically called the Sorrow of Odisha.
- Huang He (Yellow River): Sorrow of China.

Q 97. Ans. B

Explanation:

- Pair 1 is correct:

- ✓ Administrative towns function as centres of governance and administration.
- ✓ They serve as national or state headquarters.
- ✓ Examples: New Delhi, London, Patna, Chennai, Bhopal.
- Pair 2 is incorrect:
 - ✓ Transport cities are not limited to ports alone.
 - ✓ They include major railway junctions and inland transport hubs, not exclusively seaports.
 - ✓ Examples: Pt. Deen Dayal Upadhyaya Nagar (Mughalsarai), Itarsi, Chicago.
- Pair 3 is correct:
 - ✓ Mining towns develop in mineral-rich regions.
 - ✓ They support extraction and related activities.
 - ✓ Examples: Jharia (coal), Digboi (oil), Khetri (copper).
- Pair 4 is incorrect:
 - ✓ Garrison towns are built for military purposes, not trade or commerce.
 - ✓ They house troops, weapons, and military command centres.
- Key facts:
 - ✓ Functional Types of Towns (India):
 - Garrison Towns: Ambala, Jalandhar, Mhow, Danapur
 - Commercial Towns: Mumbai, Kolkata, Saharanpur
 - Industrial Towns: Jamshedpur, Bhilai, Durgapur
 - Educational Towns: Roorkee, Aligarh, Pilani
 - Religious Towns: Varanasi, Tirupati, Amritsar, Bodh Gaya
- Frost Point
 - ✓ When the dew point is below 0°C , water vapour directly changes into ice crystals.
 - ✓ This process forms frost (hoarfrost) and is called deposition.
- Dew forms under:
 - ✓ Clear skies, calm air, high humidity, and long nights.
- Latent Heat of Condensation
 - ✓ Condensation releases latent heat into the atmosphere.
 - ✓ This heat provides energy for thunderstorms and tropical cyclones.
- The Indo-Gangetic Plain, including Bihar, experiences dense winter fog (December–January).
- This is mainly radiation fog, caused by nocturnal cooling, lowering the temperature to the dew point.
- Freezing Point:
 - ✓ Freezing point of pure water is 0°C .
 - ✓ In the atmosphere, water droplets can remain liquid below 0°C up to about -40°C .
 - ✓ This phenomenon is called supercooling.
 - ✓ Freezing requires ice-nucleating particles.
- Boiling Point:
 - ✓ Boiling point of water is 100°C at sea level.
 - ✓ It decreases with an increase in altitude due to lower atmospheric pressure.
 - ✓ At high altitudes (Himalayas), water boils around $70-72^{\circ}\text{C}$.
- Melting Point:
 - ✓ Melting point of ice is 0°C .
 - ✓ Melting point is equal to the freezing point for pure water.
- Pressure Melting:
 - ✓ In glaciers, high pressure lowers the melting point of ice.
 - ✓ This is called pressure melting.
 - ✓ It causes basal melting, helping glaciers to move downslope.

Q 98. Ans. B

Explanation:

- Dew Point & Relative Humidity
 - ✓ Temperature at which air becomes saturated ($\text{RH} = 100\%$).
 - ✓ At this temperature, air cannot hold additional moisture.
 - ✓ Further cooling $\rightarrow$ condensation of water vapour into liquid.
 - ✓ Leads to the formation of dew, fog, or clouds, depending on conditions.

Q 99. Ans. A

Explanation:

- **Statement 1 is correct:** Seamounts are volcanic mountains that rise from the ocean floor with pointed summits and remain submerged without reaching the ocean surface.
 - ✓ Seamounts:

- Submarine volcanic mountains are formed by volcanic activity on the ocean floor.
- Characterised by pointed summits.
- Typically rise about 3,000 to 4,500 meters above the ocean floor.
- Common features of ocean basins are especially prevalent in the Pacific Ocean.
- Example:
 - ◊ The Emperor Seamount Chain in the Pacific Ocean is a classic example.
 - ◊ It represents an underwater extension of the Hawaiian Island chain.
- **Statement 2 is correct:** Guyots are flat-topped seamounts that were originally volcanic mountains, whose summits were levelled by wave erosion when near sea level and later submerged due to gradual subsidence.
 - ✓ Guyots (Tablemounts):
 - Also known as Tablemounts.
 - Initially formed as volcanic islands with pointed peaks.
 - Wave erosion flattened their tops when they were above or near sea level.
 - Subsequent subsidence of oceanic crust caused them to sink below sea level.
 - Hence, they show evidence of both erosion and submergence.
- Additional points:
 - ✓ Global Distribution:
 - More than 10,000 seamounts and guyots are estimated worldwide.
 - The Pacific Ocean contains the largest concentration of these features.
 - ✓ Economic Importance:
 - Slopes of seamounts and guyots contain cobalt-rich mineral crusts.
 - Important minerals include:
 - ◊ Cobalt
 - ◊ Nickel
 - ◊ Platinum
 - These minerals are valuable for modern industrial and battery technologies.
 - ✓ Ecological Importance:
 - Seamounts promote upwelling of nutrient-rich water.
 - This supports high marine productivity.
 - They act as feeding grounds for tuna, whales, and other migratory species.

India's Deep Sea Mineral Exploration

Pioneer investor status:

- India was the first country to receive an area for polymetallic nodule exploration in international waters.
- Designated as a "Pioneer Investor" by the International Seabed Authority (ISA) in 1987.
- Allocated about 75,000 sq km in the Central Indian Ocean Basin (CIOB).

Applications to ISA:

- India submitted two applications to the International Seabed Authority (ISA) for exploration in the Central Indian Ocean:

✓ Polymetallic Sulphides (PMS):

- Location: Carlsberg Ridge
- Area: 10,000 km²

✓ Cobalt-rich Ferromanganese Crusts:

- Location: Afanasy-Nikitin Seamount (ANS)
- Area: 3,000 km²

New PMS Contract:

- India signed a 15-year contract with the ISA for exclusive exploration rights for Polymetallic Sulphides (PMS) in the Carlsberg Ridge.
- India became the first country to hold two contracts with ISA for PMS exploration.
- India currently holds the largest PMS exploration area globally in the international seabed.
- India's two PMS contract regions include:
 - ✓ Central Indian Ridge and Southwest Indian Ridge
 - ✓ Carlsberg Ridge

Polymetallic Sulphides (PMS):

- Contain important minerals such as:

- ✓ Iron
- ✓ Copper
- ✓ Zinc
- ✓ Silver
- ✓ Gold
- ✓ Platinum

Institutional Framework:

- Exploration carried out under India's Deep Ocean Mission.
- Nodal Body:
 - ✓ National Centre for Polar and Ocean Research (NCPOR), Goa.
 - ✓ Operates under the Ministry of Earth Sciences (MoES).
- India hosted the 8th ISA Annual Contractors Meeting in Goa (18–20 September).

Q 100. Ans. A

Explanation:

- Statement 1 is correct:
 - ✓ Thermocline is a layer of rapid temperature decrease with depth.
 - ✓ Temperature changes faster here than in layers above or below.
- Statement 2 is correct:
 - ✓ Thermocline acts as a transition zone.
 - ✓ Above it: Surface mixed layer warmed by sunlight and wind.
 - ✓ Below it: Deep ocean layer, usually near freezing temperature.
- Statement 3 is incorrect:
 - ✓ Thermocline is almost absent in polar regions.
 - ✓ Surface water there is already very cold, similar to deep water.
 - ✓ This creates uniform temperature from surface to bottom.
- Key facts:
 - ✓ Major Oceanic "Clines":
 - Thermocline: Rapid change in temperature.
 - Halocline: Rapid change in salinity.
 - Pycnocline: Rapid change in density (due to temperature and salinity).
 - ✓ Isothermal Condition (Polar Regions):
 - Water temperature remains nearly uniform with depth.
 - Allows strong vertical mixing, making polar waters nutrient-rich.
 - ✓ Three Vertical Ocean Layers:
 - Epipelagic Zone: 0 to 200 m, warm and sunlit.
 - Mesopelagic Zone: 200 to 1000 m, contains thermocline.
 - Deep Ocean Zone: Below 1000 m, dark and near freezing.
- Labrador Current:
 - ✓ Cold current flowing southward from Arctic regions along eastern Canada.
 - ✓ Carries icebergs into North Atlantic shipping routes.
 - ✓ Met the Gulf Stream near Newfoundland, creating foggy fishing grounds.
- California Current:
 - ✓ Cold current flowing southward along western USA coast.
 - ✓ Keeps coastal regions like San Francisco cool and foggy.
- Peru Current (Humboldt Current):
 - ✓ Cold current flowing northward along western South America.
 - ✓ Creates strong upwelling, producing one of the richest fishing grounds in the world.
- Benguela Current:
 - ✓ Cold current flowing northward along southwestern Africa (Namibia, South Africa).
 - ✓ Brings cold water from the Southern Ocean into subtropical regions.
- Key facts:
 - ✓ Peru Current: Causes the Atacama Desert (Chile)
 - ✓ Benguela Current: Causes the Namib Desert (Namibia)
 - ✓ California Current: Associated with the Mojave and Sonoran Deserts
 - ✓ Fog and Fisheries Zones:
 - Fog and rich fisheries form where cold and warm currents meet.
 - Example: Labrador Current meets Gulf Stream at Grand Banks.
 - ✓ El Niño Effect:
 - Weakening of trade winds reduces cold Peru Current upwelling.
 - Warm water replaces cold water, reducing nutrients and harming fisheries.

Q 101. Ans. B

Explanation:

- Labrador Current: Eastern coast of North America
- California Current: Western coast of North America
- Peru Current: Western coast of South America
- Benguela Current: Western coast of southern Africa

Q 102. Ans. B

Explanation:

- Assertion (A) is true:
 - ✓ Tropical cyclones rarely form within 5° latitude of the Equator.
 - ✓ Even though ocean waters are very warm there, cyclones do not develop in this zone.

- Reason (R) is true:
 - ✓ Tropical cyclones are powered by the latent heat of condensation.
 - ✓ Warm, moist air rises, condenses into clouds, and releases heat.
 - ✓ This released heat intensifies rising air and strengthens the cyclone.
- Why (R) is not the correct explanation of (A):
 - ✓ The absence of cyclones near the Equator is due to the lack of the Coriolis Force, not the lack of heat.
 - ✓ Coriolis Force is zero at the Equator, so the rotating motion required for cyclone formation cannot develop.

Q 103. Ans. B

Explanation:

- Brickfielder:
 - ✓ Hot, dry, dusty wind of Australia.
 - ✓ Blows from the desert interior to the southern coasts during summer.
 - ✓ Often carries thick dust clouds.
- Chinook:
 - ✓ Warm, dry descending wind on the eastern slopes of the Rocky Mountains (USA, Canada).
 - ✓ Called “Snow Eater” because it melts snow rapidly in winter.
- Mistral:
 - ✓ Strong, cold, dry wind of France.
 - ✓ Blows through the Rhône Valley into the Gulf of Lyon (Mediterranean).
- Loo:
 - ✓ Very hot, dry summer wind of North India.
 - ✓ Blows across the Indo-Gangetic Plain during May–June.
 - ✓ Causes severe heat waves.
- Key facts:
 - ✓ Foehn Effect:
 - Warm, dry wind forms when moist air rises, loses moisture, and then descends and warms rapidly.
 - Chinook (Rockies) and Foehn (Alps) are classic examples.
 - ✓ Other Important Local Winds:
 - Sirocco: Hot, dusty wind from the Sahara to the Mediterranean; causes dust-laden “blood rain” in southern Europe.
 - Harmattan: Dry, dusty wind from the Sahara to West Africa; called “The Doctor.”

Q 104. Ans. A

Explanation:

- Alluvial Fan:
 - ✓ Formed when a fast mountain stream enters a plain, causing sudden loss of velocity.
 - ✓ Deposits coarse sediments in a fan-shaped pattern at foothills.
 - ✓ Example: Kosi megafan in North Bihar.
- Delta:
 - ✓ Formed where a river enters a sea or lake and velocity drops.
 - ✓ Deposits fine sediments in a triangular shape (like Greek letter Delta).
 - ✓ Example: Ganga–Brahmaputra (Sundarbans) Delta, the largest delta in the world.
- Natural Levee:
 - ✓ Formed during floods when rivers overflow banks.
 - ✓ Heavier sediments deposit close to the channel, creating raised embankments along river sides.
 - ✓ Levee breaches can cause severe flooding, especially in Bihar plains.
- Point Bar:
 - ✓ Depositional feature formed on the inner convex bank of a river meander.
 - ✓ Occurs where water flow is slowest.
- Cut Bank:
 - ✓ Forms on the outer concave bank of a meander.
 - ✓ Zone of maximum erosion due to faster flow.
- Key facts:
 - ✓ Kosi River forms a massive alluvial fan, causing frequent course shifts called “Sorrow of Bihar.”
- Oxbow Lakes:
 - ✓ Formed when meander loops are cut off from the main river.
 - ✓ Known locally in Bihar as Mauns or Chauras.
 - ✓ Example: Kanwar Jheel (Begusarai), Asia’s largest freshwater oxbow lake.

Q 105. Ans. A**Explanation:**

- Crust:
 - ✓ Outermost and thinnest layer of Earth.
 - ✓ Thickness: About 5 km under oceans to about 70 km under mountains.
- Mantle:
 - ✓ Thickest Earth layer beneath the crust.
 - ✓ Extends up to 2900 km depth.
 - ✓ Forms about 84 percent of Earth's volume.
- Outer Core:
 - ✓ Composed mainly of liquid iron and nickel.
 - ✓ Movement of liquid metal generates Earth's magnetic field.
- Inner Core:
 - ✓ Composed of solid iron and nickel.
 - ✓ Remains solid despite temperatures above 5400°C due to very high pressure.

Q 106. Ans. B**Explanation:**

- Assertion (A) is true:
 - ✓ Crops grown on middle slopes are safer from frost than those at the valley bottom.
 - ✓ Valley bottoms become colder due to the accumulation of cold air, making them frost-prone.
- Reason (R) is false:
 - ✓ At night, cold air, not warm air, flows downslope.
 - ✓ Cold air is denser and heavier, so it settles in valleys.
- Key facts:
 - ✓ Frost Pocket Effect:
 - During clear nights, slopes lose heat quickly.
 - Cold air moves downward (katabatic flow) and collects in valleys.
 - Valley bottoms become frost pockets, while middle slopes form a warmer thermal belt.
 - ✓ Temperature Inversion:
 - Occurs when cold air remains near the surface, and warmer air lies above it.
 - Opposite of the normal condition where temperature decreases with height.
 - ✓ Slope Winds:
 - Katabatic Winds: Cold air flows down slope at night.

- Anabatic Winds: Warm air moves up the slope during the daytime.

Q 107. Ans. B**Explanation:**

- Assertion (A) is true:
 - ✓ In 1912, Alfred Wegener proposed the Continental Drift Theory.
 - ✓ He stated that all continents were once united into a single supercontinent called Pangaea (meaning "all earth").
- Reason (R) is true:
 - ✓ Identical fossils found on continents now separated by oceans supported Wegener's idea.
 - ✓ This fossil similarity proved that continents were once connected.
- Why (R) is not the correct explanation of (A):
 - ✓ Fossil distribution supports the theory as evidence, but it does not explain the mechanism of continental movement or breakup.
- Key Facts:
 - ✓ Major fossil evidence:
 - Mesosaurus: Freshwater reptile fossils found in Africa and South America, proving continents were once joined.
 - Glossopteris: Plant fossils found in India, South America, Africa, Antarctica, and Australia.
 - Other Supporting Evidence:
 - ◇ Jigsaw Fit: Coastlines of South America and Africa match closely.
 - ✓ Major geological terms:
 - Panthalassa: The single ocean surrounding Pangaea.
 - Laurasia: Northern landmass (North America, Europe, Asia).
 - Gondwanaland: Southern landmass (South America, Africa, India, Antarctica, Australia).
 - Tethys Sea: Sea between Laurasia and Gondwanaland; its sediments later formed the Himalayas.
 - ✓ Limitations of Wegener's theory:
 - Wegener could not explain the driving force of continental movement.
 - The actual mechanism was later explained by Plate Tectonics and mantle convection currents.

Q 108. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ Ocean temperature decreases from the Equator to the Poles.
 - ✓ Equator receives direct solar rays, while the poles receive slanting rays.
 - ✓ Typical temperatures: Equator about 22°C to 26°C, Polar regions near freezing.
- Statement 2 is incorrect:
 - ✓ Northern Hemisphere oceans are warmer (about 19°C) than Southern Hemisphere oceans (about 16°C).
 - ✓ Reason: The larger landmass in the Northern Hemisphere heats faster and warms nearby oceans.
- Statement 3 is incorrect:
 - ✓ Offshore winds move warm surface water away from the coast.
 - ✓ This causes upwelling, where cold deep water rises, reducing coastal temperature.
 - ✓ Warm water accumulates near coasts mainly due to onshore winds, not offshore winds.
- Statement 4 is correct:
 - ✓ Enclosed or semi-enclosed seas in tropical regions record very high temperatures.
 - ✓ Limited mixing with the open ocean allows water to heat more rapidly.
- Key facts:
 - ✓ Highest Ocean Temperatures:
 - Recorded slightly north of the Equator, especially in the Red Sea and the Persian Gulf, often exceeding 30°C.
 - ✓ Thermal Equator:
 - Line joining places of the highest average temperature.
 - Shifts slightly north of the Equator due to the larger Northern Hemisphere landmass.
 - ✓ Upwelling Importance:
 - Brings nutrient-rich cold water to the surface.
 - Creates major fishing grounds such as Peru coast fisheries.

Q 109. Ans. A

Explanation:

- Claimants to territories within the Spratly

Islands region:

- ✓ China, Taiwan, Vietnam, Philippines, Malaysia, Brunei
- ✓ Malaysia claims the southern Spratly Islands within its Exclusive Economic Zone (EEZ)
- ✓ Brunei:
 - Claims the Louisa Reef within its EEZ
 - Least assertive but legally involved
- ✓ Vietnam claims the entire Spratly and Paracel Islands based on historical grounds
- ✓ Indonesia:
 - Not a claimant in the Spratly dispute
 - Conflict with China near the Natuna Islands due to overlap with China's claims
- ✓ Nine-Dash Line:
 - This is the boundary used by China to claim roughly 90% of the South China Sea.
- ✓ 2016 Hague Ruling:
 - The Permanent Court of Arbitration rejected China's claims
 - Declared the Nine-Dash Line invalid under UNCLOS
 - China rejected the ruling
- ✓ Other Disputes:
 - Paracel Islands: China, Taiwan, Vietnam
 - Scarborough Shoal: China and Philippines

Q 110. Ans. A

Explanation:

- Assertion (A) is correct:
 - ✓ Early Earth's primordial and secondary atmospheres were reducing in nature.
 - ✓ They mainly contained water vapour, nitrogen, carbon dioxide, methane, and ammonia.
 - ✓ Free oxygen was absent until the Great Oxygenation Event caused by cyanobacteria.
- Reason (R) is correct and explains (A):
 - ✓ Strong ultraviolet radiation splits water vapour into hydrogen and oxygen (photolysis).
 - ✓ Hydrogen escaped into space, while oxygen reacted with gases like methane and ammonia and with surface minerals such as iron.

- ✓ This prevented free oxygen from accumulating, keeping the atmosphere reducing.
- Key facts:
 - ✓ Stages of Atmospheric Evolution:
 - Primordial Atmosphere Loss: Early hydrogen and helium were removed by solar winds.
 - Degassing Stage: Volcanoes release carbon dioxide, water vapour, and nitrogen.
 - Biotic Stage: Cyanobacteria released oxygen about 2.4 billion years ago.
 - ✓ Reducing vs Oxidising Atmosphere:
 - Early atmosphere: Reducing, oxygen-poor.
 - Modern atmosphere: Oxidising, oxygen-rich.
 - ✓ Origin of Life Link:
 - Lack of oxygen supported chemical evolution (Oparin–Haldane theory).
 - The ozone layer formed only after oxygen accumulated, protecting Earth from UV radiation.
- ✓ Major chemical industries are concentrated in eastern coastal regions.
- Key facts:
 - ✓ Major Industrial Regions (World):
 - Ruhr Valley (Germany): Major coal, iron, steel, and chemical region of Europe.
 - Keihin Region (Tokyo–Yokohama): Major industrial zone of Japan.
 - Hanshin Region (Osaka–Kobe): Important manufacturing and shipbuilding hub.
 - Rust Belt (USA): Former steel and automobile hub around the Great Lakes.
 - ✓ Major Industrial Regions (India):
 - Mumbai–Pune: Major automobile and chemical hub.
 - Hugli Region (West Bengal): Traditional jute textile centre.
 - Ahmedabad–Vadodara: Major textile and petrochemical region.
 - Bengaluru–Tamil Nadu Corridor: Leading electronics and aerospace region.

Q 111. Ans. A

Explanation:

- Automobile: Germany
 - ✓ Germany is the birthplace of the modern automobile.
 - ✓ Major companies include Volkswagen, BMW, and Mercedes-Benz.
 - ✓ Automobile industry forms the core of German manufacturing.
- Shipbuilding: Japan
 - ✓ Japan became the world leader in shipbuilding after World War II.
 - ✓ Still a major hub for high-technology ship construction, despite competition from China and South Korea.
- Textile: Southern United States
 - ✓ Textile industry shifted from New England to the Piedmont region (North Carolina, South Carolina, Georgia).
 - ✓ The shift occurred closer to cotton-growing regions.
- Chemical: China
 - ✓ China has been the world's largest producer of chemicals since around 2010.

Q 112. Ans. D

Explanation:

- Assertion (A) is incorrect:
 - ✓ Loss of agricultural land due to urban sprawl is largely irreversible.
 - ✓ Once land is covered with concrete, asphalt, and buildings, soil structure and fertility are permanently damaged, making restoration to productive farmland impractical.
- Reason (R) is correct:
 - ✓ Urbanisation involves deep excavation, soil compaction, and pollution.
 - ✓ Topsoil formation is extremely slow, taking 500 to 1,000 years to form one inch, so destroyed fertile soil cannot be quickly replaced.
- Key facts:
 - ✓ Land Use Trend:
 - Land put to non-agricultural uses is one of the fastest-growing land-use categories.
 - Bihar has about 60 percent Net Sown Area, but urban expansion around Patna, Muzaffarpur, and Gaya is reducing fertile alluvial farmland.

- ✓ Urban expansion over farmland contributes to the Urban Heat Island effect, as concrete surfaces absorb more heat than vegetated land.

Q 113. Ans. B

Explanation:

- Statement 1 is true:
 - ✓ Gravity Model is based on Newton's Law of Gravitation.
 - ✓ Migration flow is directly proportional to the population size of the two places.
 - ✓ Larger cities or regions attract more migrants.
- Statement 2 is true:
 - ✓ Migration follows the Distance Decay principle.
 - ✓ Migration is inversely proportional to distance.
 - ✓ Greater distance increases cost, effort, and uncertainty, reducing migration.
- Statement 3 is false:
 - ✓ Gravity Model works at a macro level (regions or countries).
 - ✓ It does not explain individual motives like age or personal push-pull factors.
- Statement 4 is true:
 - ✓ Gravity Model is used for international migration, trade flows, and border studies, not just internal migration.
- Key facts:
 - ✓ Ravenstein's Laws (1885):
 - Most migrants move short distances.
 - Long-distance migrants usually move to major economic centres.
 - ✓ Push-Pull Theory (Everett Lee):
 - Push factors: unemployment, poverty, disasters.
 - Pull factors: jobs, safety, better wages.

Q 114. Ans. C

Explanation:

- Assertion (A) is correct:
 - ✓ In developed countries, the share of the secondary sector has declined.
 - ✓ These economies have become post-industrial, with dominance of tertiary (services) and quaternary (knowledge-based) sectors.

- Reason (R) is incorrect:

- ✓ Manufacturing itself is the secondary sector.
- ✓ If a country shifts to manufacturing, the secondary sector would increase, not decline.
- ✓ Developed countries have actually shifted away from manufacturing toward services.

- Key facts:

- ✓ India's pattern:
- ✓ India moved largely from the primary to the tertiary sector without a long industrial phase.
- ✓ The tertiary sector contributes over 50 percent of GDP, but the primary sector employs about 45 percent workforce, creating a job mismatch.
- ✓ Other worker categories:
 - Pink-collar: Service jobs traditionally held by women (teaching, nursing).

Q 115. Ans. D

Explanation:

- Statement 1 is correct:
 - ✓ Commercial livestock rearing is capital-intensive.
 - ✓ Requires heavy investment in ranches, machinery, refrigerated transport, veterinary services, and breed improvement.
- Statement 2 is correct:
 - ✓ It is a specialised activity.
 - ✓ Ranches usually focus on one animal type such as cattle, sheep, goats, or horses to ensure efficiency and uniform products.
- Statement 3 is incorrect:
 - ✓ Commercial livestock rearing is practiced on large permanent ranches, not small farms.
 - ✓ Uses modern scientific methods, not traditional subsistence techniques.
- Statement 4 is correct:
 - ✓ Commercial livestock rearing dominates temperate grasslands, which provide vast grazing lands suitable for large-scale ranching.

Q 116. Ans. C

Explanation:

- Peru:
 - ✓ Second largest producer of Copper after Chile.

- ✓ Among top three producers of Silver after Mexico and Zinc after China.
- Indonesia:
 - ✓ World's largest producer of Nickel.
 - ✓ Has banned export of raw nickel ore to promote electric vehicle battery manufacturing within the country.
- Brazil:
 - ✓ Home to Carajás Mine, the largest iron ore mine in the world.
 - ✓ Second largest producer of Iron Ore after Australia.
 - ✓ Also a major producer of Bauxite, the ore used for aluminium.
- Russia:
 - ✓ Major global mining country with vast reserves in Siberia.
 - ✓ Norilsk Nickel produces about 40 percent of the world's Palladium.
 - ✓ Alrosa is the world's largest producer of rough diamonds by volume.
- Uranium Producers:
 - ✓ World:
 - Kazakhstan is the largest producer of Uranium.
 - Canada is the second largest producer.
 - Australia is the third largest producer.
 - ✓ India:
 - Jaduguda Mine in Jharkhand was the first uranium mine in India.
 - Tummalapalle Mine in Andhra Pradesh holds major recent uranium reserves.
- Major Seismic Belts (Big Three):
 - ✓ Circum-Pacific Belt (Ring of Fire):
 - Accounts for about 80% of major earthquakes.
 - Along the margins of the Pacific Ocean (Americas, Japan, Philippines, New Zealand).
 - ✓ Alpine-Himalayan Belt:
 - Accounts for about 15% of earthquakes.
 - Extends from the Mediterranean region to the Himalayas.
 - Most important belt affecting India.
 - ✓ Mid-Atlantic Ridge:
 - A divergent plate boundary in the Atlantic Ocean.
 - Produces mostly undersea earthquakes.
- India's Seismic Zones:
 - ✓ India is divided into Zones II, III, IV, and V (by BIS).
 - ✓ Zone V: Highest earthquake risk.
 - ✓ North Bihar districts near the Nepal border (Madhubani, Supaul, Araria, Sitamarhi) fall in Zone V due to the Himalayan plate collision zone.
- Important Terms:
 - ✓ Focus (Hypocenter): Point inside the Earth where the earthquake starts.
 - ✓ Epicentre: Point on Earth's surface directly above the focus.

Q 118. Ans. C

Explanation:

- S-wave shadow zone is much larger than the P-wave shadow zone.
- ✓ P-wave Shadow Zone:
 - Occurs between 105° and 145° from the earthquake epicentre.
 - Cause: Refraction (bending) of P-waves when they enter the liquid outer core.
- ✓ S-wave Shadow Zone:
 - Begins beyond 105° from the epicentre and extends across a very large area of Earth.
 - Cause: S-waves cannot travel through liquids, so they are blocked by the liquid outer core.
- ✓ Shadow zones helped scientists prove that:
 - Earth has a liquid outer core.
 - Earth has a solid inner core.

Q 117. Ans. A

Explanation:

- Assertion (A) is correct:
 - ✓ Earthquakes occur mainly in narrow belts, not randomly across the Earth.
 - ✓ These belts are called seismic belts.
- Reason (R) is correct and explains (A):
 - ✓ Earth's lithosphere is divided into tectonic plates.
 - ✓ Plates move slowly over the semi-fluid asthenosphere.
 - ✓ At plate boundaries, plates get stuck, stress builds, and the sudden release of energy causes earthquakes.
 - ✓ Therefore, plate boundaries form seismic belts.

- Key Wave Properties:
 - ✓ P-waves (Primary Waves):
 - Type: Longitudinal waves.
 - Fastest seismic waves.
 - Travel through solids, liquids, and gases.
 - ✓ S-waves (Secondary Waves):
 - Type: Transverse waves.
 - Slower than P-waves.
 - Travel only through solids.
 - ✓ Surface Waves:
 - Types: L-waves and Rayleigh waves.
 - Travel along Earth's surface.
 - Most destructive during earthquakes.
- Major Earth Interior Boundaries:
 - ✓ Moho Discontinuity: Crust–Mantle boundary.
 - ✓ Gutenberg Discontinuity: Mantle–Outer Core boundary.
 - ✓ Lehmann Discontinuity: Outer Core–Inner Core boundary.
- Area with many drumlins clustered together.
- ✓ Depositional vs Erosional Glacial Landforms:
 - Depositional: Terminal moraine, Esker, Drumlin, Outwash plain.
 - Erosional: Cirque, Horn, Arête, U-shaped valley.
- ✓ Fjord:
 - A deep U-shaped glacial valley flooded by seawater.
 - Common along Norway's coastline.

Q 120. Ans. A

Explanation:

- Statement 1 is correct:
 - ✓ Terrestrial planets (Mercury, Venus, Earth, Mars) formed close to the Sun.
 - ✓ High temperature near the Sun prevented light gases (hydrogen, helium) from condensing.
 - ✓ Only heavy rocky materials condensed, forming rocky, dense planets.
- Statement 2 is incorrect:
 - ✓ Solar winds were strongest near the Sun, not far from it.
 - ✓ These strong winds stripped gases from terrestrial planets.
 - ✓ Jovian planets (Jupiter, Saturn, Uranus, Neptune) formed far from the Sun, where weaker solar winds allowed them to retain thick gaseous atmospheres.
- Key facts:
 - ✓ Planetary Division:
 - Asteroid Belt (between Mars and Jupiter) separates:
 - ◇ Terrestrial Planets (inner)
 - ◇ Jovian Planets (outer)
 - ✓ Gravity Factor:
 - Terrestrial planets: Small size, weaker gravity, unable to retain gases.
 - Jovian planets: Large size, strong gravity, retained thick atmospheres.
 - ✓ Planet Types:
 - Terrestrial Planets: Rocky, solid surface, high density.
 - Jovian Planets: Gas giants or ice giants, mostly gaseous, low density.
 - ✓ Density Extremes:

Q 119. Ans. A

Explanation:

- Terminal Moraine: Ridge deposited at the glacier's end
 - ✓ Formed at the snout (end) of a glacier.
 - ✓ Marks the maximum advance of the glacier.
 - ✓ Made of unstratified till (mixed debris).
- Esker: Sinuous ridge formed by subglacial streams
 - ✓ Winding ridge of sand and gravel.
 - ✓ Formed by meltwater streams flowing beneath glacier ice.
 - ✓ A glacio-fluvial landform.
- Drumlin: Oval hill aligned with ice movement
 - ✓ Smooth, elongated hill shaped like an inverted spoon.
 - ✓ Blunt end (stoss): Faces the direction from which the ice came.
 - ✓ Gentle slope (lee): Points in the direction of ice movement.
- Outwash Plain: Glacio-fluvial deposits beyond glacial limits
 - ✓ Broad, gently sloping plain.
 - ✓ Formed by meltwater depositing sorted sand and gravel beyond the glacier front.
- Key facts:
 - ✓ Basket of Eggs Topography:

- Earth: Densest planet ($\sim 5.51 \text{ g/cm}^3$).
- Saturn: Least dense ($\sim 0.68 \text{ g/cm}^3$); theoretically can float on water.

Q 121. Ans. C

Explanation:

- Assertion (A) is correct:
 - ✓ Ozone layer is located in the stratosphere (15–30 km).
 - ✓ It absorbs 97–99% of harmful UV-B and UV-C radiation.
 - ✓ Protects life by preventing skin cancer and DNA damage.
- Reason (R) is incorrect:
 - ✓ Stratospheric warming occurs because ozone absorbs incoming shortwave UV radiation, not outgoing infrared radiation.
 - ✓ Ozone acts as a harmful greenhouse gas mainly when present in the troposphere, where it behaves as a pollutant.
- Good vs Bad Ozone:
 - ✓ Good Ozone:
 - Located in the stratosphere.
 - Formed naturally from UV radiation acting on oxygen (O_2).
 - Protects Earth from harmful ultraviolet radiation.
 - ✓ Bad Ozone:
 - Located in the troposphere.
 - Formed from vehicle emissions and sunlight.
 - Major component of photochemical smog; harmful to lungs and climate.
- Key facts:
 - ✓ Ozone layer thickness measured in Dobson Units (DU).
 - ✓ Main Causes of Ozone Depletion:
 - CFCs, HCFCs, and Halons release chlorine that destroys O_3 molecules.
 - ✓ Major International Agreements:
 - Montreal Protocol (1987): Phased out ozone-depleting substances.
 - Kigali Amendment: Targets the reduction of HFCs (strong greenhouse gases).
 - ✓ Ozone hole location:
 - Most severe over Antarctica during September to November (Southern Hemisphere spring).

Q 122. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ Asteroids: Rocky, airless remnants from early solar system formation.
 - ✓ Most asteroids orbit in the Asteroid Belt, located between Mars and Jupiter.
- Statement 2 is incorrect:
 - ✓ Comet Tail Direction: Tail does not point toward the Sun.
 - ✓ Solar wind and radiation push the tail away from the Sun, regardless of the comet's direction.
- Statement 3 is correct:
 - ✓ Terminology based on location:
 - ✓ Meteoroid: Small rock moving in space.
 - ✓ Meteor: Rock enters Earth's atmosphere and burns ("shooting star").
 - ✓ Meteorite: Fragment that reaches Earth's surface.
- Key facts:
 - ✓ Example:
 - Lonar Lake (Maharashtra): Formed by a meteorite impact during the Pleistocene Epoch.
 - Only known hyper-velocity impact crater in basalt rock.
 - ✓ Famous periodic comet:
 - Halley's Comet: Appears every 75–76 years.
 - Last appearance: 1986
 - Next appearance: 2061
 - ✓ Asteroid Belt:
 - Ceres: Largest body in the asteroid belt.
 - Classified as a Dwarf Planet due to its spherical shape.

Q 123. Ans. A

Explanation:

- Assertion (A) is correct:
 - ✓ Extrusive igneous rocks (e.g., Basalt) have a fine-grained structure.
 - ✓ Intrusive igneous rocks (e.g., Granite) have coarse-grained structure.
- Reason (R) is correct and explains (A):
 - ✓ Crystal size depends on the cooling time of magma/lava.
 - ✓ Intrusive rocks: Magma cools slowly underground, allowing large crystals to form.

- ✓ Extrusive rocks: Lava cools rapidly at the surface, forming small crystals.
- Key facts:
 - ✓ Glassy Rock:
 - If cooling is extremely rapid, no crystals form.
 - Example: Obsidian (volcanic glass).
 - ✓ Vesicular Rock:
 - Gas-rich lava forms rocks with air bubbles.
 - Example: Pumice, which can float on water.
 - ✓ Intrusive Landforms:
 - Batholith: Very large underground magma body.
 - Sill: Horizontal intrusion.
 - Dyke: Vertical intrusion.

Q 124. Ans. B

Explanation:

- Venus is the hottest planet with an average temperature of about 475°C.
 - ✓ Reason: Runaway greenhouse effect.
 - ✓ Atmosphere: About 96% Carbon Dioxide (CO₂) with sulfuric acid clouds.
 - ✓ Thick atmosphere traps heat and prevents it from escaping.
- Satellites:
 - ✓ Venus: No natural satellites (moons).
 - ✓ Mercury: No natural satellites.
- Other facts:
 - ✓ Mercury:
 - Closest planet to the Sun.
 - Has no thick atmosphere, so heat escapes quickly.
 - Experiences extreme temperature range (very hot days, very cold nights).
 - ✓ Mars:
 - Called the Red Planet (iron oxide surface).
 - Thin, cold atmosphere.
 - Moons: Phobos and Deimos.
 - ✓ Uranus:
 - An Ice Giant.
 - Coldest planetary atmosphere (~-224°C).
 - ✓ Has 27 known moons (e.g., Titania, Oberon).
- Key facts:

- ✓ Venus Titles:
 - Morning Star and Evening Star: Brightest planet visible from Earth after the Moon.
 - Earth's Twin: Similar size, mass, and density to Earth.
- ✓ Retrograde Rotation:
 - Most planets rotate west to east.
 - Venus and Uranus rotate east to west (retrograde).
- ✓ Longest Day Fact:
 - 1 day on Venus: About 243 Earth days.
 - 1 year on Venus: About 225 Earth days.
 - Thus, a day on Venus is longer than its year.

Q 125. Ans. A

Explanation:

- Strait of Malacca:
 - ✓ The Strait of Malacca separates the Malay Peninsula from the Indonesian island of Sumatra.
 - ✓ It connects the Andaman Sea (Indian Ocean) with the South China Sea (Pacific Ocean).
 - ✓ Length of the strait is about 800 km, making it one of the longest straits in the world.
 - ✓ At its narrowest point (Phillips Channel near Singapore), the width is about 2.8 km.
 - ✓ It is one of the busiest maritime chokepoints, handling about 80,000 ships annually.
 - ✓ Nearly 25–30% of global maritime trade passes through this strait.
 - ✓ Malacca Dilemma
 - The Malacca Dilemma refers to China's dependence on the Strait of Malacca.
 - Nearly 80% of China's crude oil imports pass through this strait, creating strategic vulnerability.
- ✓ Historical context:
 - During the Pala Empire, Buddhist scholars from Nalanda and Vikramashila, including Atisha Dipankara, travelled through this maritime route to reach the Srivijaya Empire (Sumatra).
 - Under India's Act East Policy, exports from eastern ports such as Kolkata and Haldia move through the Strait of Malacca to reach ASEAN and East Asian countries.

- Strait of Hormuz:
 - ✓ The Strait of Hormuz connects the Persian Gulf with the Gulf of Oman and the Arabian Sea.
 - ✓ It separates Iran (north) from Oman and the UAE (south).
 - ✓ About 21 million barrels of oil per day pass through this strait.
 - ✓ It handles nearly 20% of the global petroleum trade.
- Strait of Gibraltar:
 - ✓ The Strait of Gibraltar connects the Atlantic Ocean with the Mediterranean Sea.
 - ✓ It separates Spain (Europe) from Morocco (Africa).
 - ✓ Its narrowest width is about 14.3 km.
 - ✓ It serves as the western gateway to the Mediterranean Sea.
- Sunda Strait:
 - ✓ The Sunda Strait connects the Indian Ocean with the Java Sea.
 - ✓ It lies between the Indonesian islands of Sumatra and Java.
 - ✓ It is an alternative route to the Strait of Malacca.
 - ✓ The strait contains the Krakatoa volcano, making it tectonically active.

Q 126. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ June 21: Summer Solstice (Northern Hemisphere)
 - ✓ Northern Hemisphere tilts towards the Sun.
 - ✓ Sun's direct rays fall on the Tropic of Cancer (23.5° N).
 - ✓ Results: Longest day and shortest night in the Northern Hemisphere.
- Statement 2 is incorrect:
 - ✓ December 22: Winter Solstice (Northern Hemisphere)
 - ✓ Sun's direct rays fall on the Tropic of Capricorn (23.5° S), not Cancer.
 - ✓ The longest day occurs in the Southern Hemisphere, the shortest in the Northern.
- Statement 3 is correct:
 - ✓ March 21 and September 23: Equinoxes
 - ✓ Sun's direct rays fall on the Equator.

- ✓ Day and night are equal (12 hours each) everywhere.

● Statement 4 is incorrect:

- ✓ Earth's axis remains tilted at 23.5° during the equinox.
- ✓ Axis is neither tilted towards nor away from the Sun.
- ✓ Day-night boundary (terminator) passes through both poles.

● Key facts:

- ✓ Perihelion and Aphelion:
 - Perihelion: Earth is closest to the Sun around January 3.
- ✓ Aphelion: Earth farthest from the Sun around July 4.
 - Seasons occur due to axial tilt, not distance from the Sun.

Q 127. Ans. C

Explanation:

- If Earth's axis had 0° tilt, there would be no seasons.
 - ✓ The sun would remain over the Equator year-round.
 - ✓ All places would receive 12 hours a day and 12 hours a night daily.
 - ✓ Seasons occur due to Earth's axial tilt (23.5°), not a perpendicular axis.
- Earth's Axis Tilt:
 - ✓ Tilt angle: 23.5° from vertical.
 - ✓ Angle with orbital plane: 66.5° .
- Heat Distribution:
 - ✓ Sun's rays fall vertically near tropics → concentrated heat.
 - ✓ At poles, rays fall slanting → heat spreads over a larger area → colder regions.
- Northern Hemisphere Summer:
 - ✓ Occurs when the Northern Hemisphere tilts toward the Sun (around June 21).
 - ✓ Results in longer days and more direct sunlight.
- Key events:
 - ✓ Solstices:
 - June 21: Summer Solstice (Northern Hemisphere); Sun overhead at Tropic of Cancer; longest day in the north.
 - December 22: Winter Solstice (Northern Hemisphere); Sun overhead at Tropic of Capricorn; longest day in the south.

- ✓ Equinoxes:
 - March 21: Vernal Equinox
 - September 23: Autumnal Equinox
 - Sun overhead at Equator; day and night are equal worldwide.
- ✓ Special phenomena:
 - Midnight Sun:
 - Regions beyond 66.5° N (Arctic Circle) experience 24-hour daylight during summer.
- Planetary fact:
 - ✓ Uranus: Axial tilt about 98°; experiences extreme seasons, with ~42 years of continuous daylight or darkness at the poles.
- This provides Israel and Jordan indirect access to the Red Sea.
- Bab-el-Mandeb Strait
 - ✓ The Red Sea connects to the Gulf of Aden and the Indian Ocean through the Bab-el-Mandeb Strait.
 - ✓ Bab-el-Mandeb means “Gate of Tears”.
- Oceanographic & Tectonic Features
 - ✓ The Red Sea lies along a rift zone between the African Plate and the Arabian Plate.
 - ✓ It is a divergent plate boundary.
- The Red Sea has high salinity (~40 ppt) due to:
 - ✓ High evaporation
 - ✓ Very few rivers drain into it
 - ✓ Limited water exchange

Q 128. Ans. D

Explanation:

- Oman:
 - ✓ Oman is located on the southeastern coast of the Arabian Peninsula.
 - ✓ Oman does not border the Red Sea.
 - ✓ Maritime Borders:
 - Bounded by the Arabian Sea to the southeast.
 - Bounded by the Gulf of Oman to the northeast.
 - ✓ Geopolitical Importance:
 - Oman shares control of the Strait of Hormuz with Iran.
 - The Strait of Hormuz is a major global oil transit chokepoint, linking the Persian Gulf to the open ocean.
- Countries bordering the Red Sea:
 - ✓ The Red Sea is bordered by six countries:
 - Djibouti
 - Eritrea
 - Sudan
 - Saudi Arabia
 - Egypt
 - Yemen
 - ✓ (Useful memory: DESSEY → Djibouti, Eritrea, Sudan, Saudi Arabia, Egypt, Yemen)
- Northern Extensions of the Red Sea
 - ✓ At its northern end, the Red Sea divides into:
 - ✓ Gulf of Suez — lies entirely within Egypt.
 - ✓ Gulf of Aqaba — bordered by Israel and Jordan.
- Sudan:
 - ✓ Sudan has a long coastline along the western Red Sea.
 - ✓ Port Sudan is its main seaport and major trade hub.
- Jordan:
 - ✓ Jordan has a short coastline (~26 km) along the Gulf of Aqaba.
 - ✓ Port of Aqaba is Jordan’s only seaport.
- Red Sea & Ancient India:
 - ✓ The Red Sea served as a major trade route between the Roman Empire and ancient India.
 - ✓ The Periplus of the Erythraean Sea (1st century CE) describes the trade of goods from Magadha (modern Bihar).
 - Major exports included:
 - Gangetic spikenard (aromatic oil)
 - Fine muslin cloth
 - ✓ These goods moved via inland rivers to ports, then through the Arabian Sea and Red Sea to Roman Egypt.

Q 129. Ans. D

Explanation:

- The Earth rotates 360 degrees of longitude in

24 hours. Therefore, 15 degrees of longitude corresponds to a time difference of 1 hour.

- The longitudinal difference between the Greenwich Meridian (0 degrees) and 60 degrees West longitude is 60 degrees.
- Time difference = 60 divided by 15 = 4 hours.
- Locations situated to the west of the Greenwich Meridian experience earlier local time. Therefore, 4 hours must be subtracted from Greenwich time.
- Greenwich time = 10:00 a.m.
- Local time at 60 degrees West = 10:00 a.m. minus 4 hours = 6:00 a.m.

Q 130. Ans. D

Explanation.

- Assertion (A) is false:
 - ✓ Wind velocity increases, not decreases, when the pressure gradient becomes steeper.
 - ✓ Steeper pressure gradient → faster movement of air → stronger winds.
- Reason (R) is correct:
 - ✓ Isobars are lines joining places of equal atmospheric pressure.
 - ✓ Closely spaced isobars indicate a steep pressure gradient, which produces strong winds.
 - ✓ Widely spaced isobars indicate a gentle pressure gradient and light winds.
- Key facts:
 - ✓ Basic Wind Rule:
 - Air moves from high-pressure areas to low-pressure areas.
 - ✓ Geostrophic Wind:
 - At higher altitudes, the Pressure Gradient Force balances the Coriolis Force.
 - Wind then flows parallel to isobars.
 - ✓ Important Isoline Terms:
 - Isobar: Equal pressure
 - Isotherm: Equal temperature
 - Isohyet: Equal rainfall
 - Isobath: Equal ocean depth
 - Isohaline: Equal salinity

Q 131. Ans. A

Explanation.

- Countries not crossed by the Equator:

- ✓ Venezuela: Lies north of the Equator.
- ✓ Rwanda: Lies south of the Equator.
- ✓ Malaysia: Lies north of the Equator.

- Countries crossed by the Equator:

- ✓ Equator passes through 13 countries, 3 continents, and 3 major oceans.

- ✓ Country List (West to East):

- South America:

- ◇ Ecuador, Colombia, Brazil

- Africa:

- ◇ São Tomé & Príncipe, Gabon, Republic of the Congo, Democratic Republic of the Congo, Uganda, Kenya, Somalia

- Asia-Pacific:

- ◇ Maldives, Indonesia, Kiribati

- Major water bodies crossed:

- ✓ Atlantic Ocean, Indian Ocean, Pacific Ocean.

- Key facts:

- ✓ Congo River is the only major river that crosses the Equator twice.

- ✓ The equator does not pass through India.

- ✓ Southernmost point of India: Indira Point (6°45'N), about 876 km north of the Equator.

- ✓ Quito (Ecuador): The closest capital city to the Equator and among the highest capital cities globally.

- ✓ Lake Victoria: The largest tropical lake in the world and is intersected by the Equator.

Q 132. Ans. B

Explanation:

- Exfoliation (mechanical weathering):

- ✓ Exfoliation is a type of mechanical weathering in which outer rock layers peel off in curved sheets (like onion layers).

- ✓ It is mainly caused by unloading (pressure release) and thermal expansion and contraction.

- ✓ Exfoliation is most common in massive unstratified rocks, especially **granite and gneiss**.

- ✓ Examples:

- Important exfoliation domes occur in the Peninsular Plateau.

- Notable examples include Bhongir Dome (Telangana) and the Ranchi Plateau (Jharkhand).

- **Mandar Hill (Banka):** A large monolithic granite hill, about 700 ft high, showing dome-like exfoliation features.
- **Barabar Hills (Jehanabad):**
 - ◇ Composed of granite and gneiss.
 - ◇ Rounded hill forms result from exfoliation and spheroidal weathering.
 - ◇ Famous for the Mauryan-period rock-cut caves.
- ✓ Chemical weathering:
 - Chemical weathering is the decomposition of rocks through chemical reactions.
 - Major processes include:
 - ◇ Solution, Carbonation, Hydration, and Oxidation–Reduction.
 - ◇ Chemical weathering is most active in **hot and humid** tropical climates.
 - Example:
 - ◇ Carbonation of limestone leads to **Karst topography**, forming caves, sinkholes, stalactites, etc.
- ✓ Glacial erosion:
 - Glacial erosion is caused by the movement of glaciers.
 - Main processes include Plucking and Abrasion.
 - Important glacial landforms include:
 - ◇ U-shaped valleys, Cirques, and Horns.
- ✓ Fluvial action (running water):
 - Fluvial action refers to the work of rivers through erosion, transportation, and deposition.
 - Major fluvial landforms include:
 - ◇ V-shaped valleys, Gorges, Canyons, Meanders, Oxbow lakes, and Deltas.
 - ◇ Fluvial action is the dominant geomorphic process in **Bihar**, shaping the Indo-Gangetic Plain.
 - Major rivers responsible include:
 - ◇ Ganga, Kosi, Gandak, and Son.
 - ◇ These rivers deposit large amounts of alluvium, forming fertile plains.
- ✓ Earth rotates from west to east, so the Eastern Hemisphere is ahead of the Western Hemisphere.
- ✓ Crossing the International Date Line (IDL) from West to East (Japan → Hawaii) means gaining one day (repeat a calendar day).
- ✓ Rule:
 - West → East: Gain one day
 - East → West: Lose one day
- Reason (R) is correct:
 - ✓ The IDL is not a straight 180° line.
 - ✓ It zigzags across the Pacific Ocean so that nearby islands and countries remain on the same calendar date.
- Why (R) is not the correct explanation of (A):
 - ✓ Gain or loss of a day occurs due to Earth's rotation (360°) and time-zone calculation, not because of the zigzag path of the IDL.
- Key facts:
 - ✓ Major IDL Deviation:
 - Occurs near the Bering Strait, separating Russia (Siberia) and the USA (Alaska) into different calendar dates.
 - ✓ Kiribati Adjustment (1995):
 - Kiribati shifted IDL eastward to keep all islands on the same date.
 - It becomes the first country to celebrate the New Year.
 - ✓ Longitude-Time Rule:
 - 1° longitude = 4 minutes
 - 15° longitude = 1 hour
 - ✓ India Time Standard:
 - Indian Standard Time (IST): Based on 82.5° E longitude (near Mirzapur, Uttar Pradesh).
 - IST = GMT + 5 hours 30 minutes.

Q 134. Ans. A

Explanation.

- London: Thames
 - ✓ The River Thames flows through London and empties into the North Sea.
 - ✓ A major tidal and navigable river.
- Paris: Seine
 - ✓ The River Seine flows through Paris.
 - ✓ Empties into the English Channel at Le Havre.
- Cairo: Nile
 - ✓ The River Nile flows through Cairo.

Q 133. Ans. B

Explanation.

- Assertion (A) is correct:

- ✓ Forms the fertile Nile Delta before entering the Mediterranean Sea.
- New York: Hudson
 - ✓ The Hudson River flows through New York State.
 - ✓ Empties into the Atlantic Ocean at New York Harbour.
- Key facts:
 - ✓ Danube River (Europe):
 - The second-longest river in Europe.
 - Flows through 10 countries.
 - Major capitals on the Danube: Vienna, Bratislava, Budapest, Belgrade.
 - ✓ Important global city on a river:
 - Rome: Tiber River
 - Washington D.C.: Potomac River
 - Lahore: Ravi River
 - ✓ Some Indian city on a river:
 - Lucknow: Gomti River
 - Jabalpur: Narmada River (Dhuandhar Falls)
 - Surat: Tapi River
 - Kota: Chambal River
 - Hyderabad: Musi River

Q 135. Ans. C

Explanation:

- Assertion (A) is correct:
 - ✓ Kerch Strait: Connects the Sea of Azov to Black Sea.
 - ✓ Highly strategic for Russia and Ukraine, especially for access to the Mariupol port.
 - ✓ Separates Crimean Peninsula (west) from the Taman Peninsula (Russia, east).
- Reason (R) is incorrect:
 - ✓ The Kerch Strait does not connect Black Sea to the Mediterranean Sea.
 - ✓ The Black Sea–Mediterranean route is through the Turkish Straits, not the Kerch Strait.
- Correct Mediterranean Route:
 - ✓ Bosphorus Strait: Black Sea → Sea of Marmara
 - ✓ Dardanelles Strait: Sea of Marmara → Aegean Sea → Mediterranean Sea
- Key facts:
 - ✓ Montreux Convention (1936):
 - Gives Turkey control over the Bosphorus and Dardanelles.

- Allows regulation of warship movement, used during the Russia–Ukraine conflict.
- ✓ Other Important Strategic Straits:
 - Bab-el-Mandeb: Red Sea → Gulf of Aden; between Yemen and Djibouti.
 - Strait of Hormuz: Persian Gulf ↔ Gulf of Oman; between Iran and Oman/UAE; world's major oil chokepoint.

Q 136. Ans. B

Explanation:

- South America: Aconcagua
 - ✓ Located in the Andes Mountains (Argentina).
 - ✓ Height: 6,960 m.
 - ✓ Highest peak in the Southern Hemisphere, the Western Hemisphere, and outside Asia.
- Africa: Mount Kilimanjaro
 - ✓ Located in Tanzania.
 - ✓ Highest peak in Africa.
 - ✓ Highest free-standing mountain (not part of a long mountain chain).
- Europe: Mount Elbrus
 - ✓ Located in the Caucasus Mountains (Russia).
 - ✓ A dormant volcano marking the boundary between Europe and Asia.
- North America: Denali (Mount McKinley)
 - ✓ Located in the Alaska Range (USA).
 - ✓ Known for the greatest base-to-peak rise on land.
- Complete Seven Summits List:
 - ✓ Asia: Mount Everest
 - ✓ Africa: Mount Kilimanjaro
 - ✓ Europe: Mount Elbrus
 - ✓ North America: Denali
 - ✓ South America: Aconcagua
 - ✓ Antarctica: Vinson Massif
 - ✓ Australia: Mount Kosciuszko
- Key facts:
 - ✓ Longest vs Highest Mountain Range:
 - ✓ Andes: Longest continental mountain range.
 - ✓ Himalayas: The highest mountain range.
- Mountain types:
 - ✓ Young Fold Mountains: Himalayas, Andes, Rockies, Alps
 - ✓ Old Fold Mountains: Aravalli (India), Ural (Russia)

Q 137. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ Sea Surface Temperature (SST) must be above 27°C.
 - ✓ Warm water supplies moisture and latent heat, which powers cyclone development.
- Statement 2 is correct:
 - ✓ Coriolis force is essential to generate the cyclonic rotation (vortex).
- Statement 3 is incorrect:
 - ✓ Tropical cyclones do not form near the Equator (0°–5° N/S) because the Coriolis force is nearly zero, so air cannot develop the required rotational spin.
 - ✓ They require high temperature, so temperature is not the limiting factor.
- Statement 4 is correct:
 - ✓ Cyclones weaken after landfall because:
 - The supply of warm, moist ocean air stops.
 - Surface friction increases over land.
- Key facts:
 - ✓ Cyclone Eye:
 - The central part of a cyclone is characterised by calm weather, sinking air, and the lowest pressure.
 - ✓ Cyclone Rotation:

■ Northern	Hemisphere:
Counter-clockwise	
■ Southern Hemisphere:	Clockwise
 - ✓ Regional Names of Tropical Cyclones:
 - Hurricanes: Atlantic Ocean and Eastern Pacific
 - Typhoons: Western Pacific
 - Willy-Willies: Western Australia
 - Cyclones: Indian Ocean

Q 138. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ Atacama Desert: Located in northern Chile.
 - ✓ Known as the driest non-polar desert in the world.
- Statement 2 is incorrect:
 - ✓ Kalahari Desert: Not confined to South Africa.
 - ✓ Extends across Botswana (major share), Namibia, and South Africa.

- ✓ It is a semi-desert, not a true desert in all parts.

- Statement 3 is correct:

- ✓ Great Victoria Desert: The largest desert in Australia.
- ✓ Spreads across Western Australia and South Australia.

- Statement 4 is correct:

- ✓ Gobi Desert: A major cold winter desert.
- ✓ Located in southern Mongolia and northern China.
- ✓ Formed due to the rain shadow of the Himalayas.

- Additional information:

- ✓ Cold Ocean Current Influence (West-Coast Deserts):
 - Atacama Desert: Influenced by the Peru (Humboldt) Current.
 - Namib and Kalahari: Influenced by the Benguela Current.
 - Mojave and Sonoran: Influenced by the California Current.
- ✓ Australia Deserts:
 - Australia is the driest inhabited continent.
 - Major deserts include: Great Victoria, Great Sandy, Tanami, Simpson, and Gibson.
- ✓ Tribes:
 - Kalahari Desert: Home of the San (Bushmen) hunter-gatherer tribe.
- ✓ India example:
 - Ladakh: A cold desert formed due to the rain shadow of the Himalayas, similar to the Gobi Desert.

Q 139. Ans. B

Explanation:

- Assertion (A) is correct:

- ✓ There is an inverse relationship between female literacy and Total Fertility Rate (TFR).

- ✓ Educated women:

- Marry at a later age
- Have better awareness of family planning
- Participate more in the workforce
- Result: Lower fertility levels.

- Reason (R) is correct:

- Infant Mortality Rate (IMR) varies across Indian states.

- Higher IMR: Madhya Pradesh (historically high).
- Lower IMR: Kerala and Goa (very low, single-digit IMR).
- Why (R) is not the correct explanation of (A):
 - ✓ Variation in IMR across states is a separate demographic fact.
 - ✓ Female literacy reduces fertility due to social and economic changes, not because IMR varies geographically.
- Other facts:
 - ✓ Replacement Level Fertility: 2.1 children per woman.
 - ✓ India TFR: 2.0 (NFHS-5), indicating movement toward stabilisation.

Q 140. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ Lake Baikal (Russia):
 - World's deepest lake (~1,620 m).
 - Oldest lake (~25 million years).
 - Holds about 20% of the world's unfrozen freshwater.
 - Largest freshwater lake by volume.
- Statement 2 is correct:
 - ✓ Lake Superior (USA–Canada):
 - Largest freshwater lake by surface area.
 - The Caspian Sea is larger in area but is saltwater, so not counted among freshwater lakes.
- Statement 3 is incorrect:
 - ✓ Lake Victoria is the main source of the White Nile, not the Blue Nile.
 - ✓ Blue Nile originates from Lake Tana (Ethiopian Highlands).
- Key supporting facts:
 - ✓ Nile River Fact:
 - White Nile and the Blue Nile meet at Khartoum (Sudan) to form the main Nile River.
 - ✓ Lake Victoria Facts:
 - Second-largest freshwater lake by surface area.
 - Equator passes through Lake Victoria.
- Indian example:
 - ✓ Wular Lake (Jammu & Kashmir): Largest freshwater lake in India.
 - ✓ Formed by tectonic activity; fed by the

Jhelum River.

- Great Lakes:
 - ✓ HOMES: Huron, Ontario, Michigan, Erie, Superior
 - ✓ Only Lake Michigan lies entirely within the USA; others are shared with Canada.

Q 141. Ans. B

Explanation:

- Statement 1 is correct:
 - ✓ Isthmus of Panama: Connects North America and South America.
 - ✓ It separates the Caribbean Sea (Atlantic Ocean) from the Pacific Ocean.
- Statement 2 is incorrect:
 - ✓ Isthmus of Suez: Connects Africa and Asia (Sinai Peninsula), not Europe.
 - ✓ It separates the Mediterranean Sea and the Red Sea.
- Statement 3 is correct:
 - ✓ Kra Isthmus: Located mainly in Thailand.
 - ✓ It forms the narrowest part of the Malay Peninsula, connecting it to mainland Asia.
- Statement 4 is incorrect:
 - ✓ Isthmus of Corinth: Connects the Peloponnese Peninsula to mainland Greece, not Italy.
 - ✓ Italy is connected to mainland Europe through the Alps, not an isthmus.
- Key Concepts:
 - ✓ Isthmus: Connects land, separates water bodies.
 - ✓ Strait: Connects water bodies, separates land.
- Important Strait:
 - ✓ Strait of Gibraltar: Connects the Atlantic Ocean and the Mediterranean Sea; separates Spain and Morocco.
 - ✓ Strait of Malacca: Connects Andaman Sea and South China Sea; located south of Kra Isthmus.
- Canal Link:
 - ✓ Suez Canal: Cut through the Isthmus of Suez (opened 1869).
 - ✓ Panama Canal: Cut through the Isthmus of Panama (opened 1914).

Q 142. Ans. B**Explanation:**

- Assertion (A) is correct:
 - ✓ Global population is rapidly ageing.
 - ✓ Population aged 65+ years is projected to rise from 10% (2022) to 16% (2050) (UN estimate).
- Reason (R) is correct:
 - ✓ Infant Mortality Rate (IMR) and Child Mortality Rate (CMR) have declined globally due to:
 - Vaccination expansion
 - Institutional deliveries
 - Improved maternal and child healthcare
- Why (R) is not the correct explanation of (A):
 - ✓ Lower infant mortality initially increases the young population, widening the base of the population pyramid.
 - ✓ Population ageing mainly occurs due to:
 - Declining fertility rates (fewer births)
 - Increased life expectancy (better adult and elderly healthcare)
- Related facts from Census 2011:
 - ✓ Elderly population (60+): 8.6%
 - ✓ Kerala: Highest elderly proportion
 - ✓ Bihar: Among the lowest elderly share due to high birth rate
- Dependency Ratio:
 - ✓ Ratio of dependents (0–14 years and 60+ years) to working-age population (15–59 years).
 - ✓ An ageing population increases the old-age dependency ratio, creating economic pressure (seen in Europe and China).

Q 143. Ans. B**Explanation:**

- Nomadic Herding: Saharan Africa
 - ✓ Primitive subsistence system: movement with livestock in arid and semi-arid regions such as the Sahara, the Arabian Peninsula, Central Asia, and the tundra (reindeer herding).
- Intensive Subsistence: Monsoon Asia
 - ✓ Found in India, China, and Southeast Asia.
 - ✓ Small landholdings, high labour, high yield per area.
 - ✓ Types: Wet-paddy (rice) and non-paddy (wheat, soybean).

- Commercial Grain Farming: Interior North America, Russia, Australia
 - ✓ Practised in semi-arid mid-latitudes.
 - ✓ Large farms, high mechanisation, low yield per acre, but high yield per worker.
- Specialised Horticulture: Northwestern Europe
 - ✓ High-value crops: vegetables, fruits, flowers for urban markets.
 - ✓ Capital and technology-intensive.
 - ✓ Also called Truck Farming.
- Other important agricultural types:
 - ✓ Mediterranean Agriculture: Famous for grapes (viticulture) and citrus fruits.
 - ✓ Plantation Agriculture: Large colonial estates; Fazendas are coffee plantations in Brazil.

Q 144. Ans. C**Explanation:**

- Statement 1 is correct:
 - ✓ Stage 3 (Late Expanding Stage): Birth rate declines, death rate remains low.
 - ✓ Gap between births and deaths narrows, so population growth slows after the Stage 2 explosion.
- Statement 2 is correct:
 - ✓ Falling birth rate in Stage 3 due to:
 - Urbanisation increases.
 - Shift from an agrarian to an industrial economy (children become an economic cost).
 - Infant mortality declines.
 - Female literacy and empowerment increase.
- Other DTT Stages:
 - ✓ Stage 1: High Stationary
 - High birth rate + High death rate.
 - Very slow population growth.
 - Typical of pre-industrial societies.
 - ✓ Stage 2: Early Expanding
 - High birth rate + Rapid fall in death rate.
 - Leads to a population explosion.
 - ✓ Stage 4: Low Stationary
 - Low birth rate + Low death rate.
 - Population stabilises or declines.
 - Examples: Japan, Germany.
 - ✓ Key India Facts:
 - India: Currently in Stage 3; TFR about 2.0 (NFHS-5).

- Bihar: In late Stage 2 to early Stage 3; highest TFR 2.98 (NFHS-5).

- Related concept:

- ✓ Demographic Dividend: Occurs in Stage 3 when falling births reduce the dependency ratio and increase the working-age population (15–59 years).

Q 145. Ans. B

Explanation:

- Countries not crossed:
 - ✓ Pakistan: Entirely north of the Tropic of Cancer.
 - ✓ Yemen, Thailand: Entirely south of the Tropic of Cancer.
- Countries crossed by the Tropic of Cancer:
 - ✓ Passes through 16 countries, 3 continents, and 6 water bodies.
- West to East order:
 - ✓ Mexico, Bahamas
 - ✓ Western Sahara, Mauritania, Mali, Algeria, Niger, Libya, Egypt
 - ✓ Saudi Arabia, UAE, Oman, India, Bangladesh, Myanmar, China, Taiwan
- Path through states (West to East):
 - ✓ Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram
- Other facts:
 - ✓ Mahi river: The only river in India crossing Tropic of Cancer twice.
 - ✓ Congo river: Crosses the Equator twice.
 - ✓ Limpopo River: Crosses the Tropic of Capricorn twice.
- Capital link:
 - ✓ Ranchi: State capital located on/closest to the Tropic of Cancer.

Q 146. Ans. B

Explanation:

- Statement 1 is incorrect:
 - ✓ The Ring of Fire is not a complete circle.
 - ✓ It is a horseshoe-shaped belt about 40,000 km long.
 - ✓ Extends from New Zealand → East Asia → Aleutian Islands → the western coasts of North and South America.
 - ✓ Southern Pacific boundary near Antarctica

lacks continuous subduction zones, so the ring is incomplete.

- Statement 2 is correct:

- ✓ The Ring of Fire is mainly formed along convergent plate boundaries.
- ✓ Here, dense oceanic plates subduct beneath lighter plates, leading to volcanic and earthquake activity.

- Statement 3 is correct:

- ✓ Ring of Fire contains about 75% of the world's active volcanoes.
- ✓ About 90% of global earthquakes occur in this belt.

- Key features:

- ✓ Island Arcs:
 - Formed due to oceanic–oceanic subduction.
 - Examples include the Aleutian Islands, Kuril Islands, Japan, Philippines.
- ✓ Wadati–Benioff Zone:
 - Inclined zone of deep-focus earthquakes along the subducting plate.
- ✓ Tsunami Generation:
 - Large megathrust earthquakes in subduction zones generate major tsunamis.
 - Examples include the 2004 Indian Ocean tsunami and the 2011 Japan (Tohoku) tsunami.

Q 147. Ans. B

Explanation:

- Basalt:
 - ✓ Basalt is a fine-grained extrusive igneous rock.
 - ✓ Formed by the rapid cooling of lava at the Earth's surface.
- Quartzite:
 - ✓ Quartzite is a non-foliated metamorphic rock.
 - ✓ Formed when quartz sandstone undergoes heat and pressure.
- Sandstone:
 - ✓ Sandstone is a sedimentary (clastic) rock.
 - ✓ Formed by compaction and cementation of sand grains, mainly quartz.
- Marble:
 - ✓ Marble is a metamorphic rock.
 - ✓ Formed from limestone or dolomite under heat and pressure.

- Metamorphic conversion list:
 - ✓ Limestone → Marble
 - ✓ Sandstone → Quartzite
 - ✓ Shale/Clay → Slate → Phyllite → Schist
 - ✓ Granite → Gneiss
 - ✓ Coal → Graphite or Diamond
- Fossil rule:
 - ✓ Fossils are found only in sedimentary rocks.
 - ✓ Igneous and metamorphic processes destroy organic remains due to heat and pressure.
- Indian geology:
 - ✓ Vindhyan rock system is known for large deposits of red sandstone.
 - ✓ Famous monuments such as the Red Fort, Agra Fort, and Sanchi Stupa are built using this stone.

Q 148. Ans. A

Explanation:

- Statement 1 is correct:
 - ✓ Intrusive (plutonic) igneous rocks form deep inside the Earth.
 - ✓ Magma cools slowly due to insulation by surrounding rocks.
 - ✓ Slow cooling forms large, well-developed crystals (phaneritic texture).
 - ✓ Example: Granite.
- Statement 2 is incorrect:
 - ✓ Extrusive igneous rocks form from lava cooling rapidly on the surface.
 - ✓ Rapid cooling prevents the growth of large crystals.
 - ✓ These rocks have very small crystals or a glassy texture (aphanitic texture).
 - ✓ Examples: Basalt and Obsidian.
- Key examples:
 - ✓ Granite: Typical intrusive igneous rock with large crystals.
 - ✓ Basalt: Common extrusive igneous rock.
 - ✓ Obsidian: Volcanic glass with no visible crystals.
- Indian example:
 - ✓ Deccan Plateau is mainly made of basalt formed by fissure eruptions.
 - ✓ Weathering of basalt forms black cotton soil (Regur), important for cotton cultivation.
- Chemical Classification:
 - ✓ Acidic Igneous Rocks:
 - Silica content is more than 65%.
 - Light-colored and lower density.
 - Example: Granite.
 - ✓ Basic Igneous Rocks:
 - Silica content is less than 55%.
 - Dark-colored and higher density.
 - Examples: Basalt, Gabbro.
 - ✓ Hypabyssal Rocks:
 - Form just below the surface in cracks such as dykes and sills.
 - Example: Dolerite.

Q 149. Ans. C

Explanation:

- Mt. Cotopaxi: Ecuador
 - ✓ Active stratovolcano in the Andes Mountains.
 - ✓ Part of the Pacific Ring of Fire.
 - ✓ One of the most active volcanoes in the world.
- Mt. Etna: Italy
 - ✓ Located on Sicily Island.
 - ✓ Highest and most active volcano in Europe.
 - ✓ Formed due to the African Plate subducting under the Eurasian Plate.
- Mt. Fuji: Japan
 - ✓ Highest peak of Japan is located on Honshu Island.
 - ✓ Active stratovolcano; last eruption in 1707.
 - ✓ Located at a triple junction of the Amurian, Okhotsk, and Philippine Sea plates.
- Mt. Kilimanjaro: Tanzania
 - ✓ The highest peak in Africa.
 - ✓ Dormant stratovolcano with three cones: Kibo, Mawenzi, and Shira.
 - ✓ Famous for its snow-capped summit near the equator.
- Other volcano facts:
 - ✓ Mt. Stromboli: Italy
 - Known as the “Lighthouse of the Mediterranean” due to frequent glowing eruptions.
 - ✓ Ojos del Salado: Chile and Argentina
 - Highest active volcano in the world.
- Indian facts:
 - ✓ Barren Island: Andaman Sea
 - India’s only active volcano.
 - ✓ Narcondam Island: Andaman Sea
 - Classified as a dormant volcano.

- ✓ Plate Tectonics Fact
 - Australia mainland: No active volcanoes because it lies within the Indo-Australian Plate, away from active plate boundaries.

Q 150. Ans. B

Explanation:

- Statement 1 is incorrect:
 - ✓ SIAL (Silica + Aluminium) forms the continental crust, not the mantle.
 - ✓ It is discontinuous and largely absent beneath ocean floors.
- Statement 2 is incorrect:
 - ✓ SIMA (Silica + Magnesium) forms the oceanic crust, not the mantle.
 - ✓ It is denser ($2.9-3.0 \text{ g/cm}^3$) than SIAL (about 2.7 g/cm^3).
 - ✓ Lighter SIAL continents float over denser SIMA.
- Statement 3 is correct:
 - ✓ NIFE (Nickel + Iron) forms the Earth's core.
 - ✓ Core density ranges from 11 to 13 g/cm^3 .
 - ✓ Movement of liquid iron in the outer core generates Earth's magnetic field (Geodynamo effect).
- Key discontinuities:
 - ✓ Conrad Discontinuity: Between SIAL and SIMA layers within the crust.
 - ✓ Mohorovičić (Moho) Discontinuity: Between crust and mantle.
- Element abundance:
 - ✓ Most abundant element in the whole Earth: Iron (about 35%).
 - ✓ Most abundant element in Earth's crust: Oxygen (about 46%), followed by Silicon (about 27%).

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